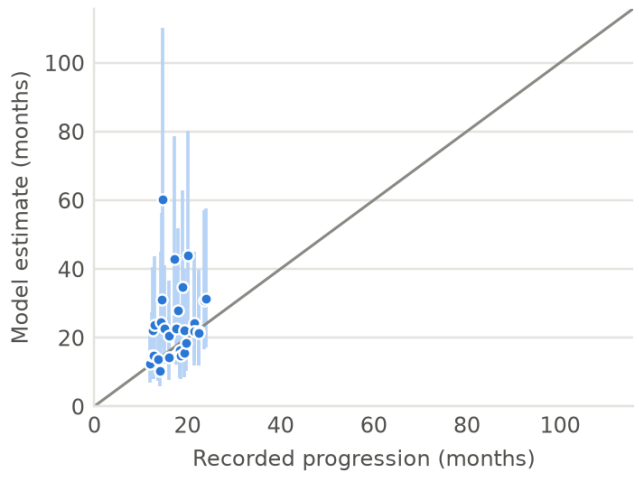
Group B · Progression in year 2 recorded progression-free survival: 12–24 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
30	**86**	**17.8 mo**	**63%**
2.8% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 12.0–24.0 mo	estimate in this window · median |error| 4 mo



Most frequent path end-points in this group	Patients
SUMOylation of DNA damage response and repair proteins	5
breast ductal adenocarcinoma	5
Neutrophil degranulation	3
Striated Muscle Contraction	3
PKMTs methylate histone lysines	2
Meiotic recombination	2
Sema3A PAK dependent Axon repulsion	2
Olfactory Signaling Pathway	2

Rules firing most often in this group	HR	Patients
F5-R1: M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	0.32	6
F5-R2: M stage = M0 AND 26.5 ≤ Age < 83.5	0.35	6
F3-R7: M stage = M0 AND 40.5 ≤ Age < 86.5	0.37	5
F4-R4: 26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	0.38	5
F4-R5: M stage = M0 AND Age ≥ 34.5	0.47	4
F4-R3: Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	0.36	3

Most common driver alterations

TP53 mutated (43%), MYC amplified (27%), PIK3CA mutated (23%), ERBB2 amplified (17%), KMT2C mutated (17%), CCND1 amplified (7%), RUNX1 mutated (7%), FOXA1 mutated (7%), GATA3 mutated (7%), BRCA1 mutated (7%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-A8-A08X	43	III	Her2	12.0 mo	12 mo	✓	91	B_progression_12-24m/TCGA-A8-A08X.pdf
TCGA-D8-A1XC	85	III	LumA	12.4 mo	15 mo	✓	91	B_progression_12-24m/TCGA-D8-A1XC.pdf
TCGA-A2-A0CM	40	II	Basal	12.6 mo	22 mo	✓	55	B_progression_12-24m/TCGA-A2-A0CM.pdf
TCGA-C8-A12Q	78	III	Her2	12.7 mo	15 mo	✓	97	B_progression_12-24m/TCGA-C8-A12Q.pdf
TCGA-BH-A1EY	79	II	LumA	13.0 mo	24 mo	✓	25	B_progression_12-24m/TCGA-BH-A1EY.pdf
TCGA-A7-A425	70	III	LumA	13.8 mo	14 mo	✓	89	B_progression_12-24m/TCGA-A7-A425.pdf
TCGA-D8-A1JA	60	missing	Her2	14.2 mo	10 mo	✗	15	B_progression_12-24m/TCGA-D8-A1JA.pdf
TCGA-AR-A2LH	55	III	Normal	14.3 mo	25 mo	✗	86	B_progression_12-24m/TCGA-AR-A2LH.pdf
TCGA-E9-A1N6	52	II	missing	14.6 mo	31 mo	✗	70	B_progression_12-24m/TCGA-E9-A1N6.pdf
TCGA-B6-A0WW	58	X	LumB	14.7 mo	60 mo	✗	93	B_progression_12-24m/TCGA-B6-A0WW.pdf
TCGA-B6-A409	44	III	Basal	15.0 mo	23 mo	✓	85	B_progression_12-24m/TCGA-B6-A409.pdf
TCGA-B6-A0IK	63	III	Her2	16.1 mo	14 mo	✓	94	B_progression_12-24m/TCGA-B6-A0IK.pdf
TCGA-B6-A0X1	48	II	Basal	16.1 mo	21 mo	✓	98	B_progression_12-24m/TCGA-B6-A0X1.pdf
TCGA-E9-A2JS	72	II	LumB	17.2 mo	43 mo	✗	37	B_progression_12-24m/TCGA-E9-A2JS.pdf
TCGA-GM-A2D9	69	I	LumA	17.6 mo	23 mo	✓	5	B_progression_12-24m/TCGA-GM-A2D9.pdf
TCGA-AR-A0TY	54	II	LumB	17.9 mo	28 mo	✗	30	B_progression_12-24m/TCGA-AR-A0TY.pdf
TCGA-A7-A13E	62	II	Basal	18.3 mo	15 mo	✓	51	B_progression_12-24m/TCGA-A7-A13E.pdf
TCGA-A2-A0YJ	39	III	Basal	18.4 mo	16 mo	✓	91	B_progression_12-24m/TCGA-A2-A0YJ.pdf
TCGA-B6-A0WX	40	III	Basal	18.5 mo	15 mo	✓	93	B_progression_12-24m/TCGA-B6-A0WX.pdf
TCGA-A2-A3XU	35	II	Basal	18.9 mo	35 mo	✗	52	B_progression_12-24m/TCGA-A2-A3XU.pdf
TCGA-A7-A13G	79	II	LumA	19.3 mo	22 mo	✓	32	B_progression_12-24m/TCGA-A7-A13G.pdf
TCGA-AR-A1AI	47	II	Basal	19.3 mo	15 mo	✓	29	B_progression_12-24m/TCGA-AR-A1AI.pdf
TCGA-E2-A2P5	78	III	missing	19.6 mo	18 mo	✓	94	B_progression_12-24m/TCGA-E2-A2P5.pdf
TCGA-BH-A18J	56	IV	LumA	20.1 mo	44 mo	✗	99	B_progression_12-24m/TCGA-BH-A18J.pdf
TCGA-A2-A0SW	82	IV	LumB	21.5 mo	24 mo	✗	99	B_progression_12-24m/TCGA-A2-A0SW.pdf
TCGA-AC-A2B8	84	II	LumA	21.5 mo	22 mo	✓	29	B_progression_12-24m/TCGA-AC-A2B8.pdf
TCGA-LQ-A4E4	73	III	missing	22.4 mo	22 mo	✓	92	B_progression_12-24m/TCGA-LQ-A4E4.pdf
TCGA-A2-A3XS	62	III	missing	22.4 mo	21 mo	✓	94	B_progression_12-24m/TCGA-A2-A3XS.pdf
TCGA-B6-A0X4	62	II	LumA	23.5 mo	31 mo	✗	42	B_progression_12-24m/TCGA-B6-A0X4.pdf
TCGA-EW-A1P7	59	II	Normal	24.0 mo	31 mo	✗	59	B_progression_12-24m/TCGA-EW-A1P7.pdf

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**43**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**Her2**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T4D**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A8-A08X-01** · Aneuploidy score: **23** · TMB (nonsyn., /Mb): **1.1** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 32%

Model-estimated progression time

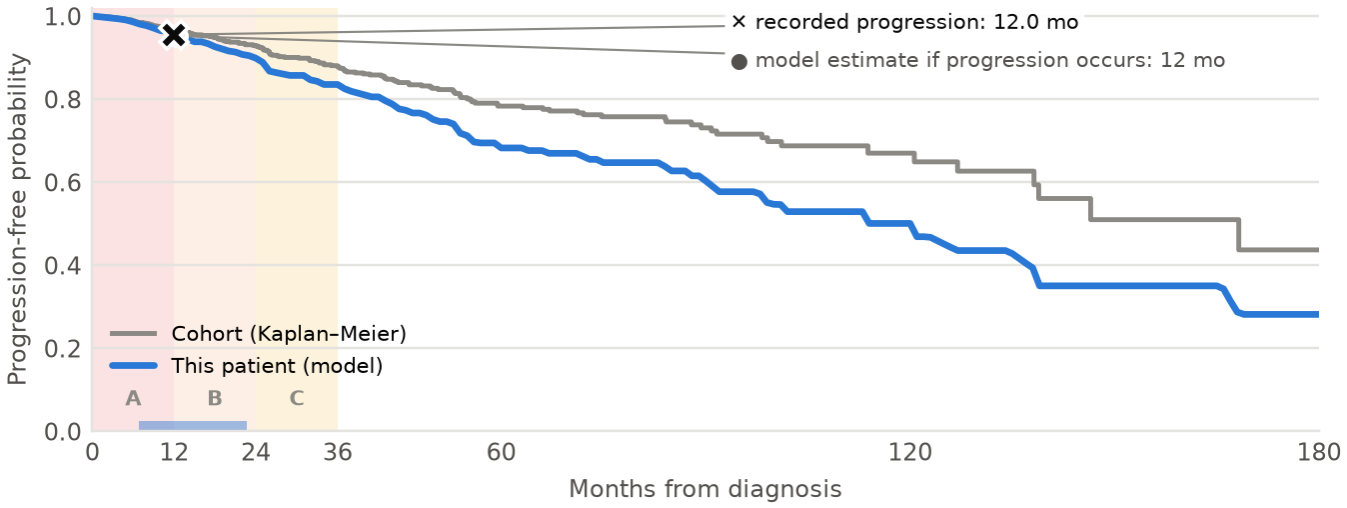
≈ 12 months

Window 12–24 months · IQR 7–23 mo if progression occurs

Recorded progression

12.0 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 0 mo



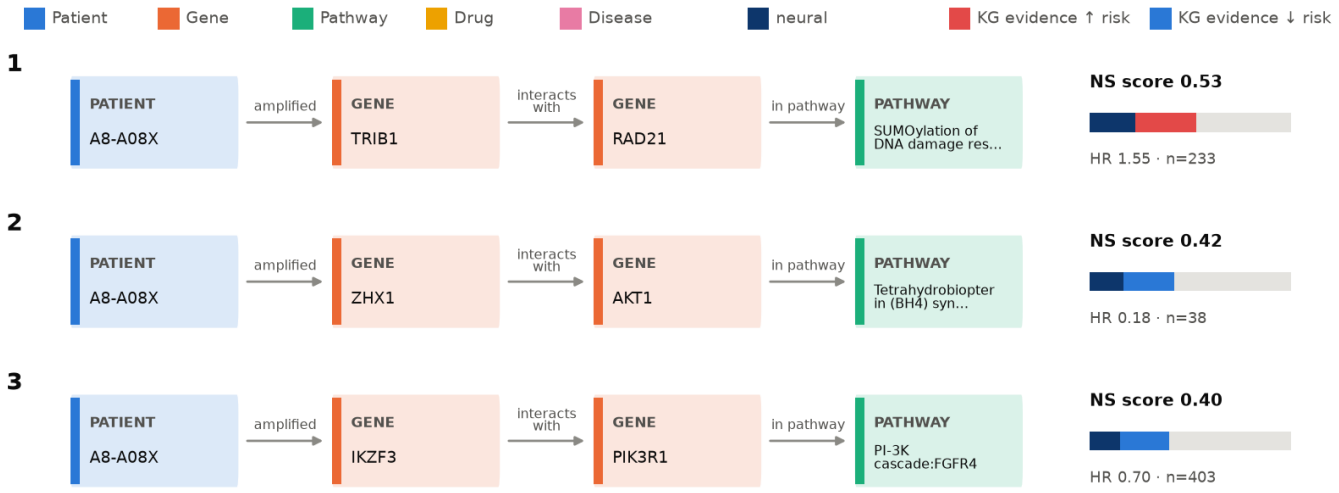
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 43-year-old patient has a Her2 tumour, AJCC stage III (T4, N3, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TRIB1** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.55). Progression was recorded at 12.0 months; the model's estimate given progression was 12 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

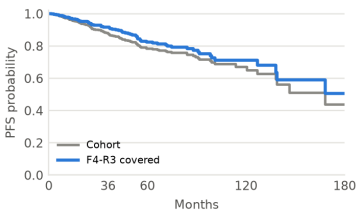


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A8-A08X** amplified → **TRIB1** interacts with → **RAD21** in pathway → **SUMOylation of DNA damage response and repair proteins**	38	↑ **faster progression** HR 1.55, z +2.3, n = 233 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.53**
2	**TCGA-A8-A08X** amplified → **ZHX1** interacts with → **AKT1** in pathway → **Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation**	28	↓ **slower progression** HR 0.18, z -1.9, n = 38 carriers of “Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation”	**0.42**
3	**TCGA-A8-A08X** amplified → **IKZF3** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR4**	25	↓ **slower progression** HR 0.70, z -1.8, n = 403 carriers of “PI-3K cascade:FGFR4”	**0.40**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

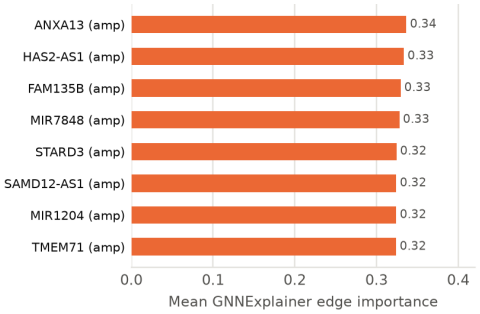
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.C176F	57	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
ANXA13	High-level amplification	GISTIC ±2	0	0.34
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.33
FAM135B	High-level amplification	GISTIC ±2	0	0.33
MIR7848	High-level amplification	GISTIC ±2	0	0.33
STARD3	High-level amplification	GISTIC ±2	1	0.32
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.32
MIR1204	High-level amplification	GISTIC ±2	0	0.32
TMEM71	High-level amplification	GISTIC ±2	0	0.32
ZHX1-C8ORF76	High-level amplification	GISTIC ±2	0	0.32



110 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 45 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CX	52	II	Her2	0.75	no progression, followed 56.8 mo
TCGA-AO-A0JE	53	III	Her2	0.71	no progression, followed 76.8 mo
TCGA-E2-A1B0	50	III	Her2	0.71	no progression, followed 53.6 mo
TCGA-BH-A1F2	53	III	LumB	0.70	no progression, followed 31.5 mo
TCGA-AN-A04C	51	II	Her2	0.69	no progression, followed 1.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**85**	Tumour type	**Breast Invasive Mixed Mucinous Carcinoma**
Sex	**Female**	OncoTree code	**IMMC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8480/3**

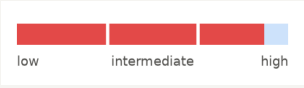
Sample: **TCGA-D8-A1XC-01** · Aneuploidy score: **2** · TMB (nonsyn., /Mb): **0.87** · MSIsensor score: **0.29** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

19% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 31%

Model-estimated progression time

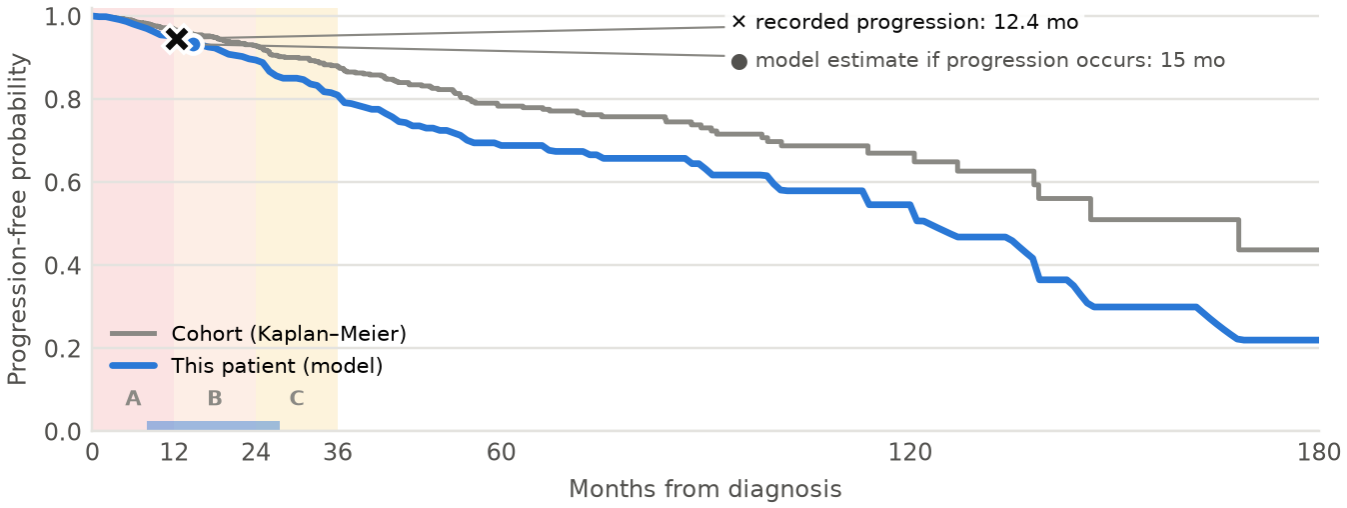
≈ 15 months

Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

12.4 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 2 mo



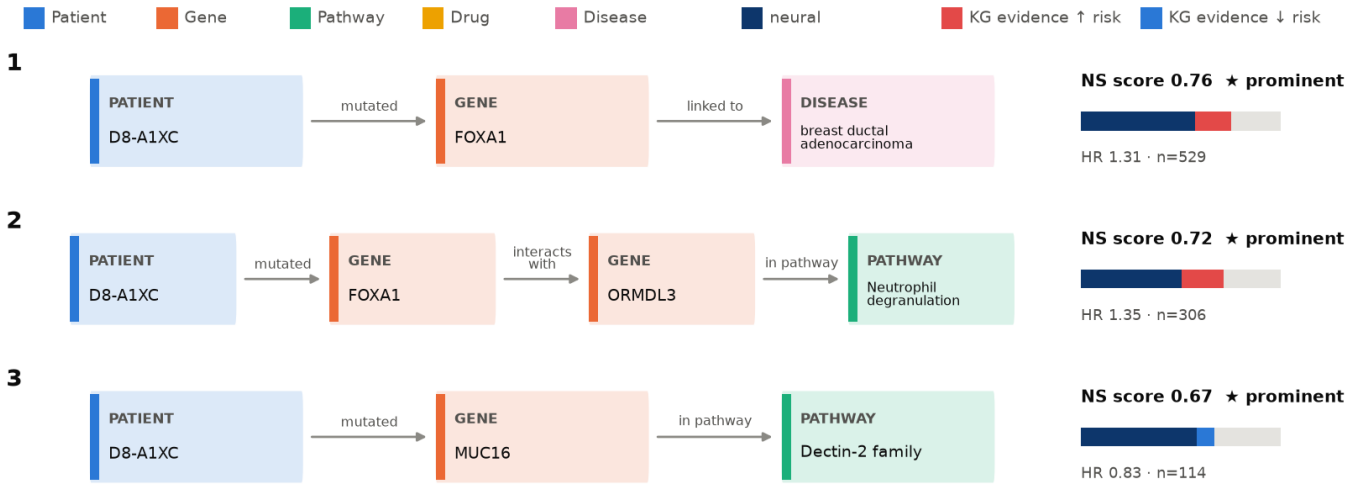
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 85-year-old patient has a LumA tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FOXA1** to **breast ductal adenocarcinoma**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.31). Progression was recorded at 12.4 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

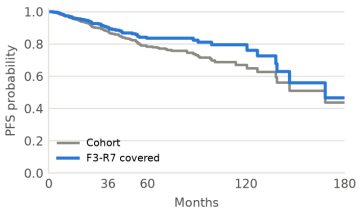


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-D8-A1XC** mutated → **FOXA1** linked to → **breast ductal adenocarcinoma**	95	↑ faster progression HR 1.31, z +1.4, n = 529 carriers of “breast ductal adenocarcinoma”	**0.76**
2 ★	**TCGA-D8-A1XC** mutated → **FOXA1** interacts with → **ORMDL3** in pathway → **Neutrophil degranulation**	84	↑ faster progression HR 1.35, z +1.6, n = 306 carriers of “Neutrophil degranulation”	**0.72**
3 ★	**TCGA-D8-A1XC** mutated → **MUC16** in pathway → **Dectin-2 family**	97	↓ slower progression HR 0.83, z -0.7, n = 114 carriers of “Dectin-2 family”	**0.67**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

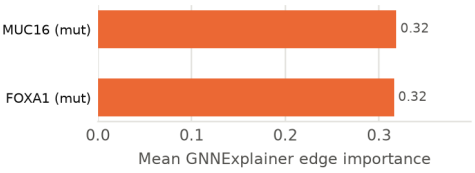
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.E410Afs*2	9	0.32
MUC16	Mutation	p.E6944D	8	0.32



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MUC16 → Oregovomab (DB04964). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1FR	73	III	LumA	0.94	progressed at 54.0 mo (group D)
TCGA-A2-A3KD	47	III	LumA	0.66	no progression, followed 39.6 mo
TCGA-D8-A73W	79	III	LumA	0.64	progressed at 0.7 mo (group A)

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A133	65	III	LumA	0.63	no progression, followed 0.0 mo
TCGA-B6-A0RO	71	III	LumA	0.62	no progression, followed 162.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

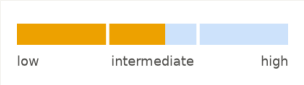
Sample: **TCGA-A2-A0CM-01** · Aneuploidy score: **24** · TMB (nonsyn., /Mb): **1.73** · MSIsensor score: **0.39** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

55th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

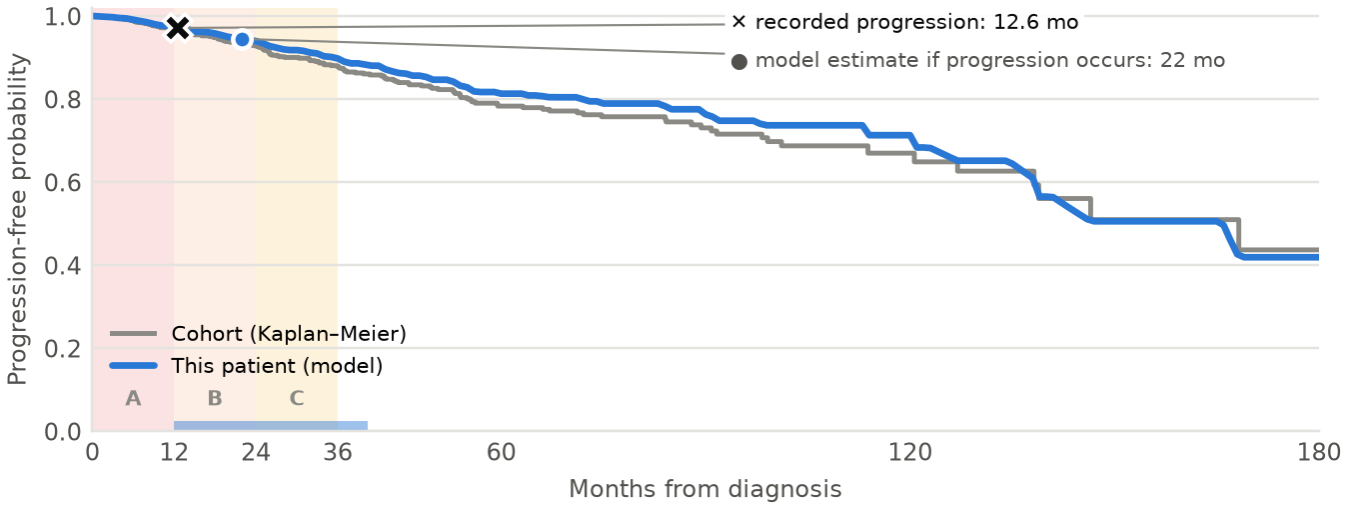
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

12.6 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 9 mo



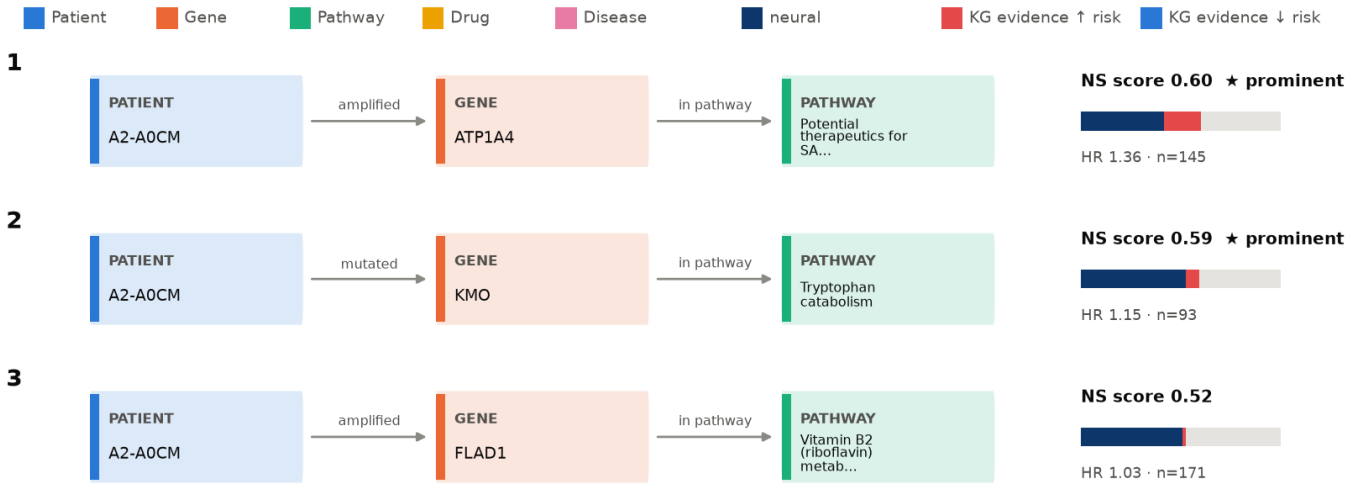
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **55th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ATP1A4** to **Potential therapeutics for SARS**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.36). Progression was recorded at 12.6 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

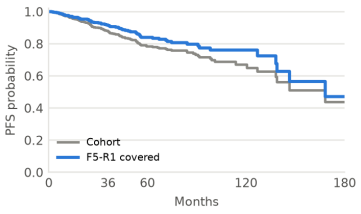


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A0CM** amplified → **ATP1A4** in pathway → **Potential therapeutics for SARS**	70	↑ **faster progression** HR 1.36, z +1.4, n = 145 carriers of “Potential therapeutics for SARS”	**0.60**
2 ★	**TCGA-A2-A0CM** mutated → **KMO** in pathway → **Tryptophan catabolism**	88	↑ **faster progression** HR 1.15, z +0.5, n = 93 carriers of “Tryptophan catabolism”	**0.59**
3	**TCGA-A2-A0CM** amplified → **FLAD1** in pathway → **Vitamin B2 (riboflavin) metabolism**	85	↑ **faster progression** HR 1.03, z +0.1, n = 171 carriers of “Vitamin B2 (riboflavin) metabolism”	**0.52**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

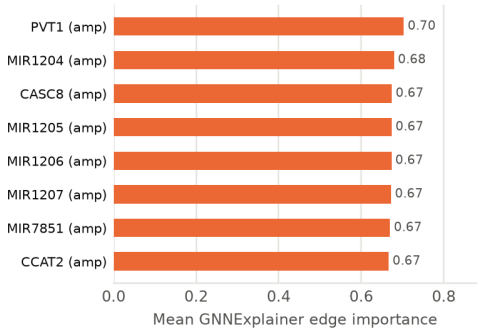
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TP53 · driver	Mutation	p.E204Sfs*43	57	0.31
KMT2C · driver	Homozygous deletion	GISTIC ±2	4	0.31
PVT1	High-level amplification	GISTIC ±2	0	0.70
MIR1204	High-level amplification	GISTIC ±2	0	0.68
CASC8	High-level amplification	GISTIC ±2	0	0.67
MIR1205	High-level amplification	GISTIC ±2	0	0.67
MIR1206	High-level amplification	GISTIC ±2	0	0.67
MIR1207	High-level amplification	GISTIC ±2	0	0.67
MIR7851	High-level amplification	GISTIC ±2	0	0.67
CCAT2	High-level amplification	GISTIC ±2	0	0.67
POU5F1B	High-level amplification	GISTIC ±2	0	0.66



22 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1N8	48	II	Basal	0.82	no progression, followed 34.2 mo
TCGA-AR-A1AQ	49	II	Basal	0.80	no progression, followed 99.3 mo
TCGA-E2-A1B6	44	II	Basal	0.79	no progression, followed 28.5 mo
TCGA-A2-A3XX	49	II	Basal	0.79	no progression, followed 47.3 mo
TCGA-AR-A1AI	47	II	Basal	0.78	progressed at 19.3 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**—**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-C8-A12Q-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.35** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

97th

High risk tier · fused neuro-symbolic model



Probability of progression

27% by 36 mo

12 mo 8% · 24 mo 15% · 60 mo 42%

Model-estimated progression time

≈ 15 months

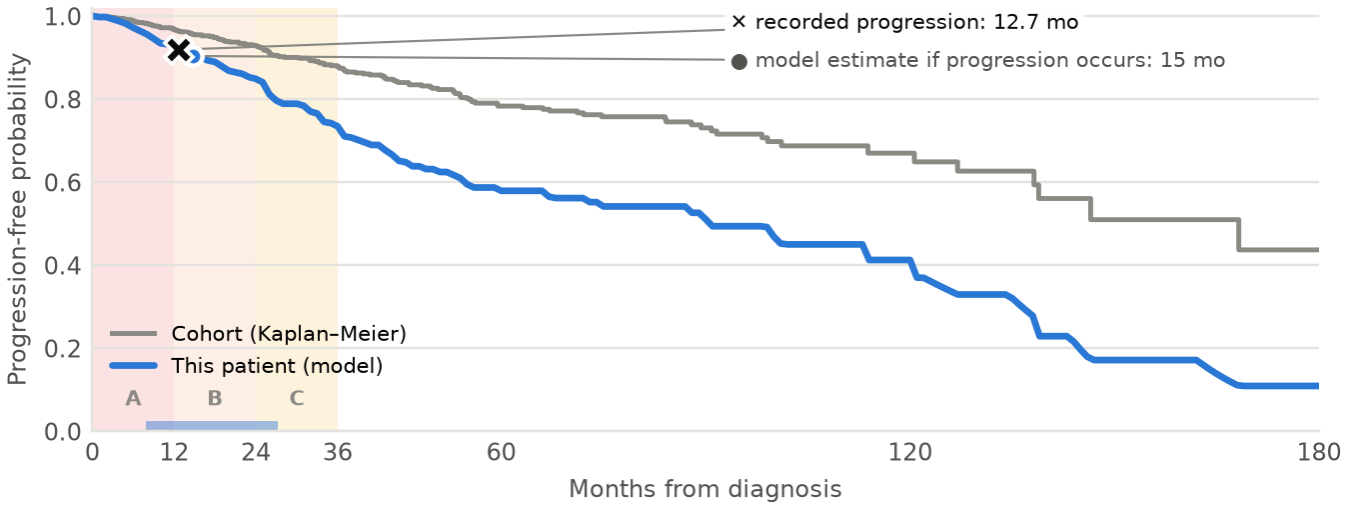
Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

12.7 months

■ **Group B** · Progression in year 2

✓ **estimate in recorded window** · |error| 2 mo



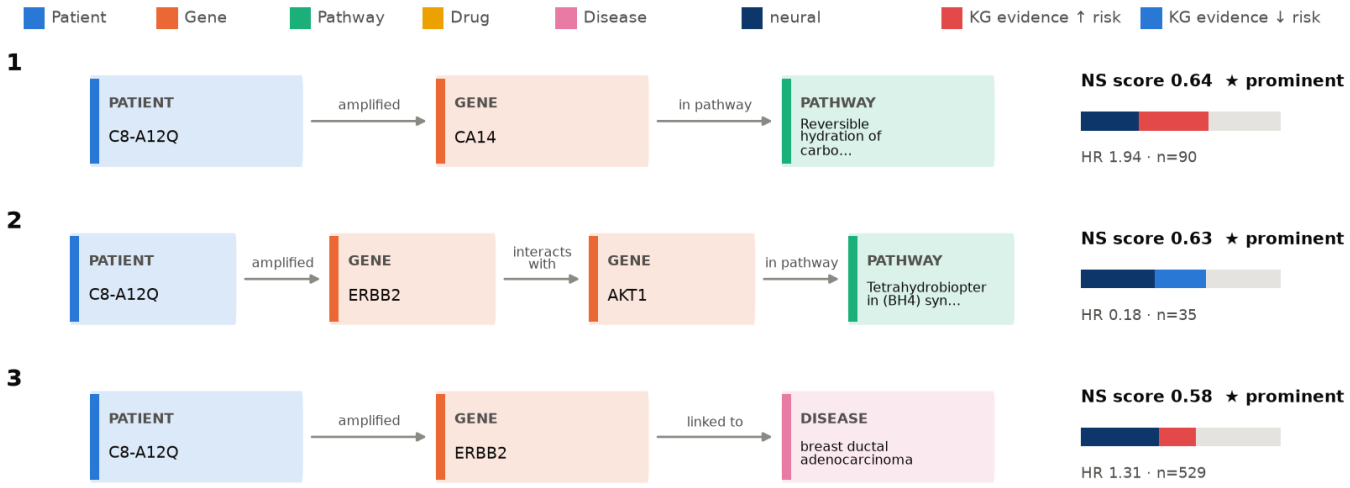
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a Her2 tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **97th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CA14** to **Reversible hydration of carbon dioxide**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.94). Progression was recorded at 12.7 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

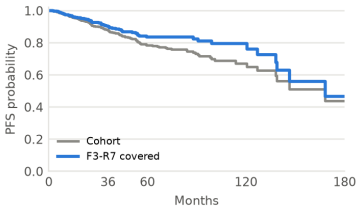


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-C8-A12Q** amplified → **CA14** in pathway → **Reversible hydration of carbon dioxide**	48	↑ **faster progression** HR 1.94, z +2.6, n = 90 carriers of “Reversible hydration of carbon dioxide”	**0.64**
2 ★	**TCGA-C8-A12Q** amplified → **ERBB2** interacts with → **AKT1** in pathway → **Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation**	62	↓ **slower progression** HR 0.18, z -1.9, n = 35 carriers of “Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation”	**0.63**
3 ★	**TCGA-C8-A12Q** amplified → **ERBB2** linked to → **breast ductal adenocarcinoma**	65	↑ **faster progression** HR 1.31, z +1.4, n = 529 carriers of “breast ductal adenocarcinoma”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

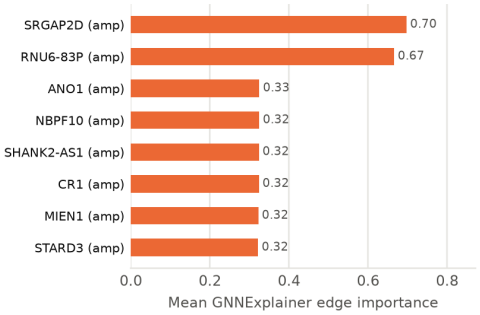
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ **lower risk**	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
KMT2C · driver	Mutation	p.I323=	4	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
TP53 · driver	Mutation	p.Y205N	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.70
RNU6-83P	High-level amplification	GISTIC ±2	0	0.67
ANO1	High-level amplification	GISTIC ±2	2	0.33
NBPF10	High-level amplification	GISTIC ±2	0	0.32
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
CR1	High-level amplification	GISTIC ±2	3	0.32
MIEN1	High-level amplification	GISTIC ±2	0	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32



67 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021) … and 55 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0HY	60	I	LumB	0.71	no progression, followed 50.8 mo
TCGA-C8-A138	54	III	Her2	0.67	no progression, followed 12.5 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0I9	62	IV	Her2	0.66	progressed at 11.9 mo (group A)
TCGA-AR-A254	50	III	Her2	0.66	no progression, followed 85.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

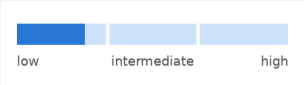
Sample: **TCGA-BH-A1EY-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.23** · MSIsensor score: **0.05** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

25th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

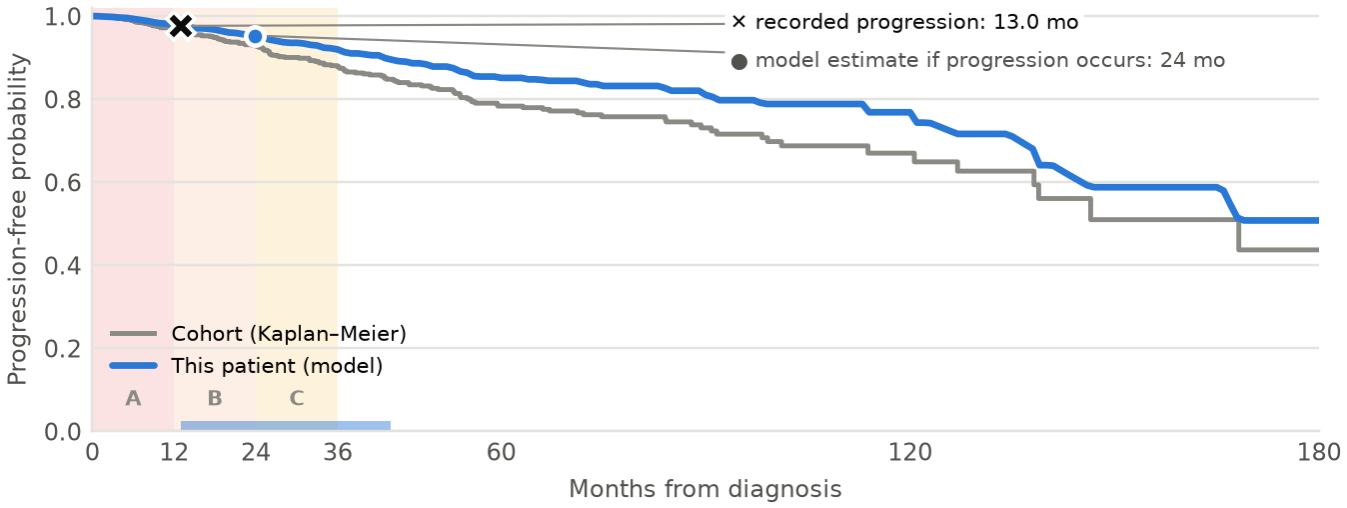
≈ 24 months

Window 12–24 months · IQR 13–44 mo
if progression occurs

Recorded progression

13.0 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 11 mo



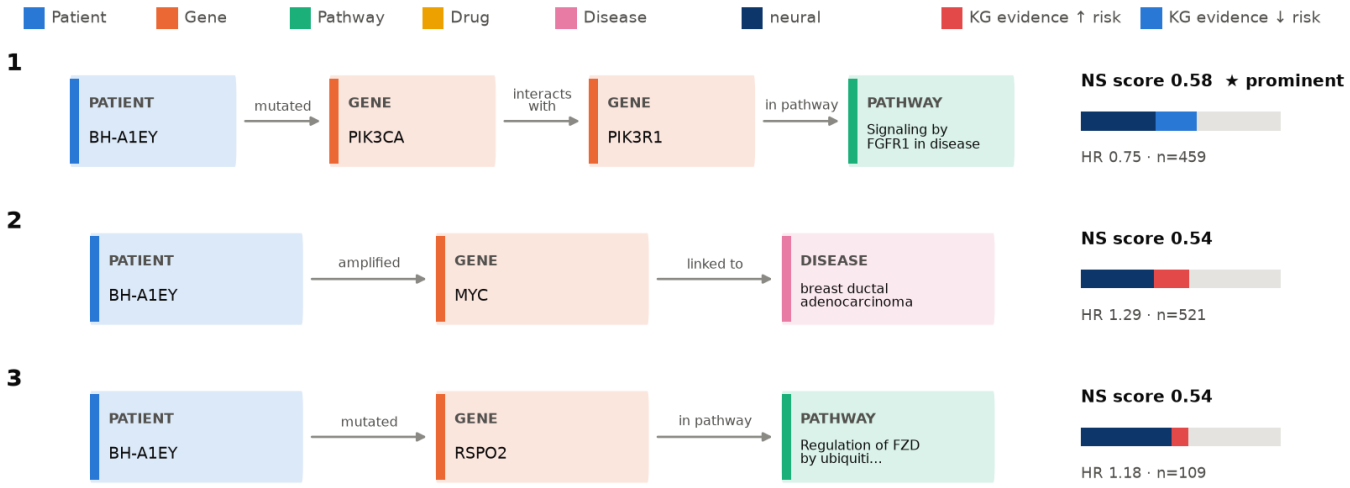
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **25th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **Signaling by FGFR1 in disease**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.75). Progression was recorded at 13.0 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

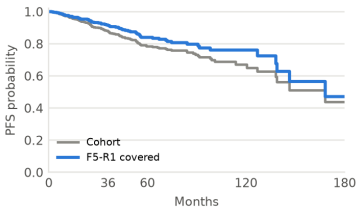


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1EY** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Signaling by FGFR1 in disease**	63	↓ **slower progression** HR 0.75, z -1.5, n = 459 carriers of “Signaling by FGFR1 in disease”	**0.58**
2	**TCGA-BH-A1EY** amplified → **MYC** linked to → **breast ductal adenocarcinoma**	61	↑ **faster progression** HR 1.29, z +1.3, n = 521 carriers of “breast ductal adenocarcinoma”	**0.54**
3	**TCGA-BH-A1EY** mutated → **RSPO2** in pathway → **Regulation of FZD by ubiquitination**	76	↑ **faster progression** HR 1.18, z +0.6, n = 109 carriers of “Regulation of FZD by ubiquitination”	**0.54**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

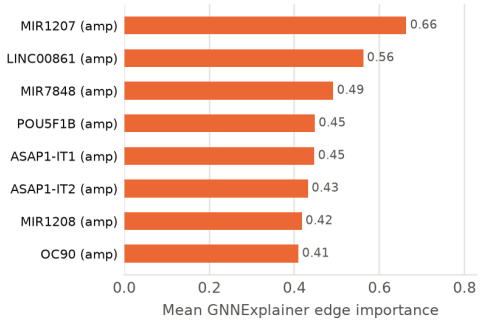
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.Y126C	57	0.32
KMT2C · driver	Mutation	p.S3394N	4	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
MAP2K4 · driver	Mutation	p.X363_splice	12	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
MIR1207	High-level amplification	GISTIC ±2	0	0.66
LINC00861	High-level amplification	GISTIC ±2	0	0.56
MIR7848	High-level amplification	GISTIC ±2	0	0.49
POU5F1B	High-level amplification	GISTIC ±2	0	0.45
ASAP1-IT1	High-level amplification	GISTIC ±2	0	0.45
ASAP1-IT2	High-level amplification	GISTIC ±2	0	0.43
MIR1208	High-level amplification	GISTIC ±2	0	0.42



49 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813) … and 18 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1RH	63	II	Her2	0.79	no progression, followed 46.6 mo
TCGA-BH-A0H9	69	II	LumA	0.78	no progression, followed 41.0 mo
TCGA-BH-A0GY	67	II	LumA	0.78	no progression, followed 30.3 mo
TCGA-A8-A07Z	85	II	LumA	0.76	no progression, followed 45.1 mo
TCGA-S3-AA10	65	II	Basal	0.76	no progression, followed 19.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**70**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

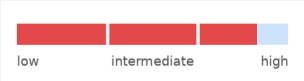
Sample: **TCGA-A7-A425-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **0.93** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

89th

High risk tier · fused neuro-symbolic model



Probability of progression

18% by 36 mo

12 mo 5% · 24 mo 10% · 60 mo 29%

Model-estimated progression time

≈ 14 months

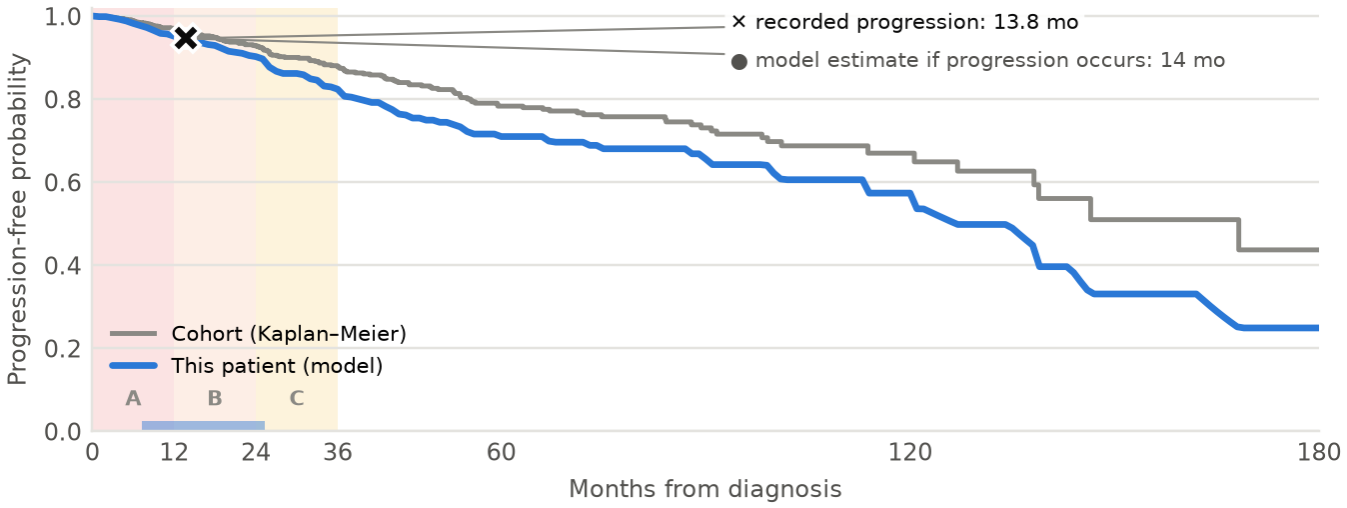
Window 12–24 months · IQR 7–25 mo if progression occurs

Recorded progression

13.8 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 0 mo



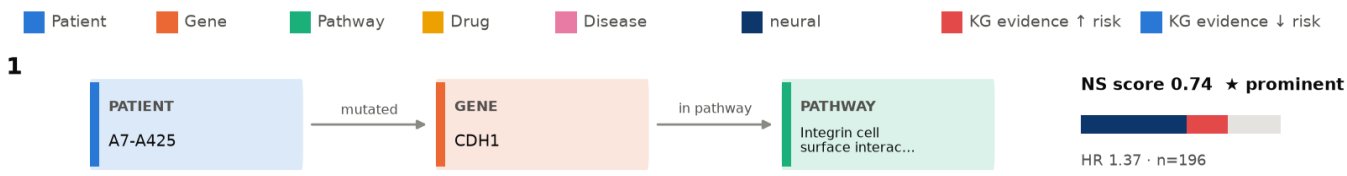
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 70-year-old patient has a LumA tumour, AJCC stage III (T3, N3, MX). The fused neuro-symbolic model places them in the **89th risk percentile** (high tier). The most significant rule covering them (*N stage = N3 AND Aneuploidy score ≥ 2.5 AND TMB (nonsynonymous) < 5.58*) marks a group with a progression hazard 4.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CDH1** to **Integrin cell surface interactions**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.37). Progression was recorded at 13.8 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

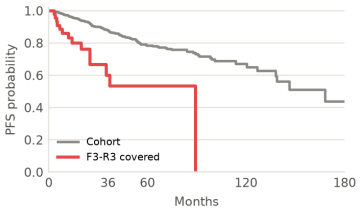


1 of 17 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A7-A425** mutated → **CDH1** in pathway → **Integrin cell surface interactions**	89	↑ **faster progression** HR 1.37, z +1.5, n = 196 carriers of “Integrin cell surface interactions”	**0.74**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

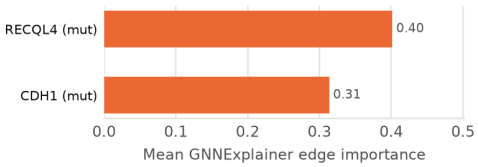
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R3 ↑ **higher risk**	N stage = N3 AND Aneuploidy score ≥ 2.5 AND TMB (nonsynonymous) < 5.58	47 pts 14 events	**4.25** p 1.1e-06	89 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.K397Mfs*17	17	0.31
RECQL4	Mutation	p.P160S	0	0.40



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-Z7-A8R5	61	III	LumA	0.79	progressed at 6.0 mo (group A)
TCGA-GM-A4E0	67	III	LumA	0.78	no progression, followed 72.0 mo
TCGA-E9-A6HE	45	III	LumA	0.77	no progression, followed 27.8 mo
TCGA-D8-A3Z6	56	III	LumA	0.70	no progression, followed 18.5 mo
TCGA-E9-A5FK	60	III	LumA	0.70	no progression, followed 26.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**—**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

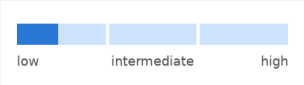
Sample: **TCGA-D8-A1JA-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **26.6** · MSIsensor score: **0.38** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

15th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

≈ 10 months

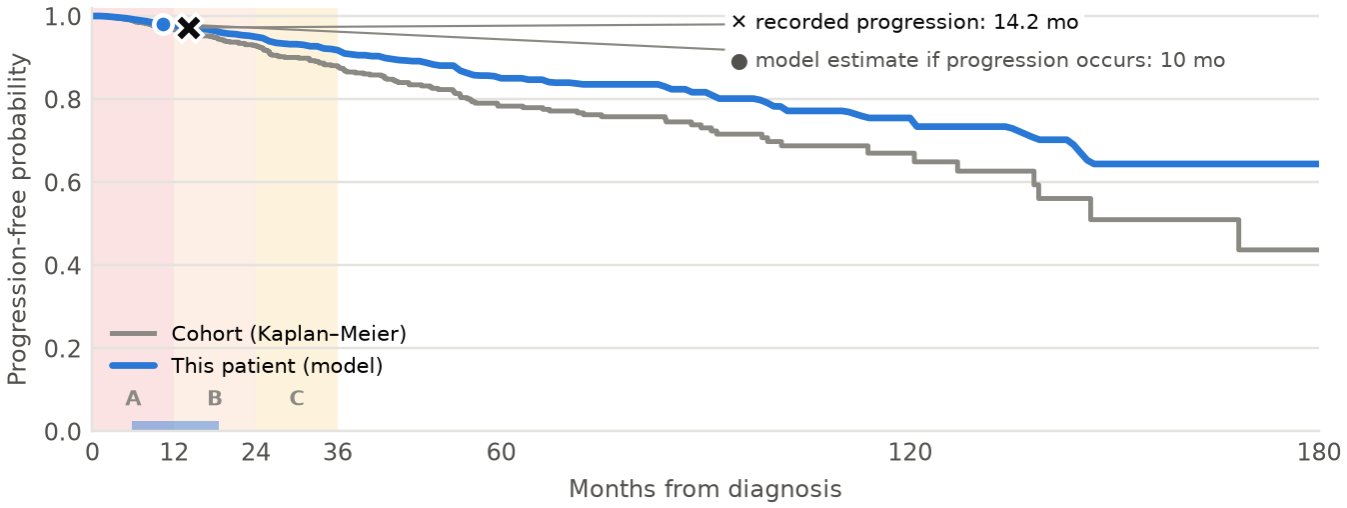
Window ≤ 12 months · IQR 6–19 mo if progression occurs

Recorded progression

14.2 months

Group B · Progression in year 2

✗ recorded window differs · |error| 4 mo



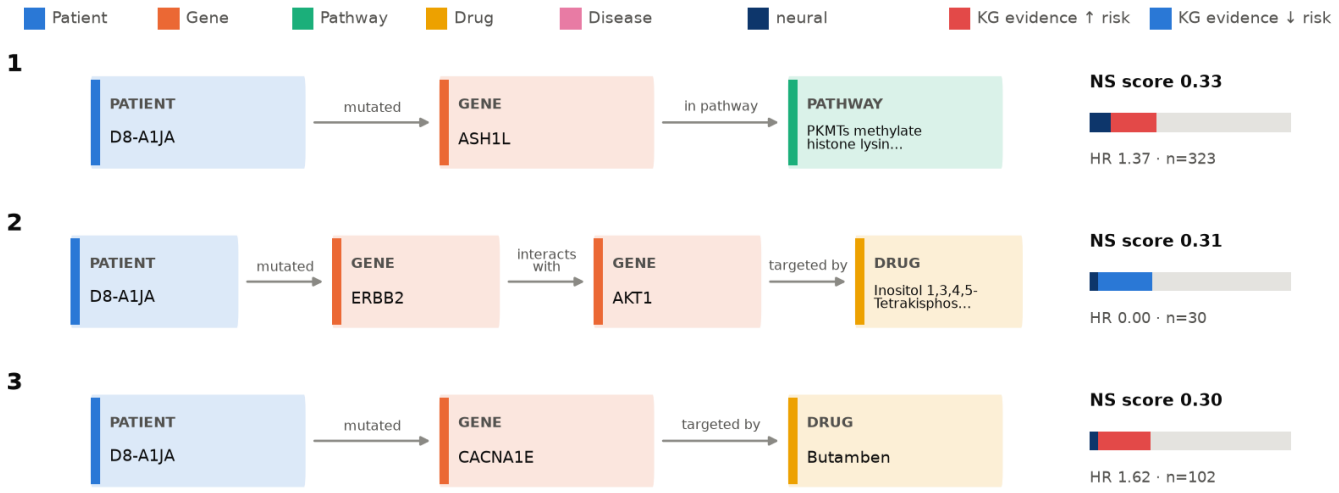
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a Her2 tumour, AJCC stage missing (T4, missing, M0). The fused neuro-symbolic model places them in the **15th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ASH1L** to **PKMTs methylate histone lysines**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.37). Progression was recorded at 14.2 months; the model's estimate given progression was 10 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

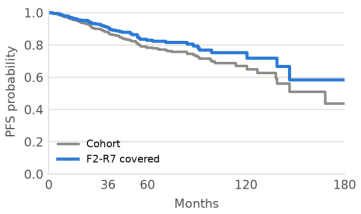


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-D8-A1JA** mutated → **ASH1L** in pathway → **PKMTs methylate histone lysines**	18	↑ **faster progression** HR 1.37, z +1.7, n = 323 carriers of “PKMTs methylate histone lysines”	**0.33**
2	**TCGA-D8-A1JA** mutated → **ERBB2** interacts with → **AKT1** targeted by → **Inositol 1,3,4,5-Tetrakisphosphate**	7	↓ **slower progression** HR 0.00, z -2.0, n = 30 carriers of “Inositol 1,3,4,5-Tetrakisphosphate”	**0.31**
3	**TCGA-D8-A1JA** mutated → **CACNA1E** targeted by → **Butamben**	7	↑ **faster progression** HR 1.62, z +2.0, n = 102 carriers of “Butamben”	**0.30**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

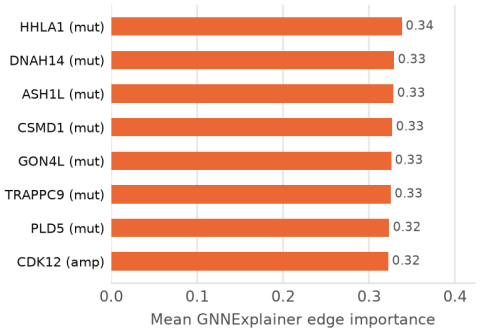
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
ERBB2 · driver	Mutation	p.V91=	55	0.31
FGFR1 · driver	Mutation	p.L675=	43	0.31
RB1 · driver	Mutation	p.V714=	25	0.31
HHLA1	Mutation	p.W420C	0	0.34
DNAH14	Mutation	p.*742*	0	0.33
ASH1L	Mutation	p.E2148Q, p.E2569Q, p.K2155N	1	0.33
CSMD1	Mutation	p.R2141K, p.S1027L	0	0.33
GON4L	Mutation	p.E2209K	0	0.33
TRAPPC9	Mutation	p.E600K	2	0.33
PLD5	Mutation	p.Q6=	0	0.32
CDK12	High-level amplification	GISTIC ±2	2	0.32



56 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 90 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AC-A23H	90	II	Her2	0.55	progressed at 5.7 mo (group A)
TCGA-BH-A1EN	78	II	Her2	0.54	no progression, followed 69.9 mo
TCGA-A8-A09G	79	III	Her2	0.53	no progression, followed 0.0 mo
TCGA-C8-A1HK	53	II	Her2	0.52	no progression, followed 12.0 mo
TCGA-A2-A04X	34	II	Her2	0.51	no progression, followed 55.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE III**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-AR-A2LH-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **2** · MSIsensor score: **0.01** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

86th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 3% · 24 mo 9% · 60 mo 29%

Model-estimated progression time

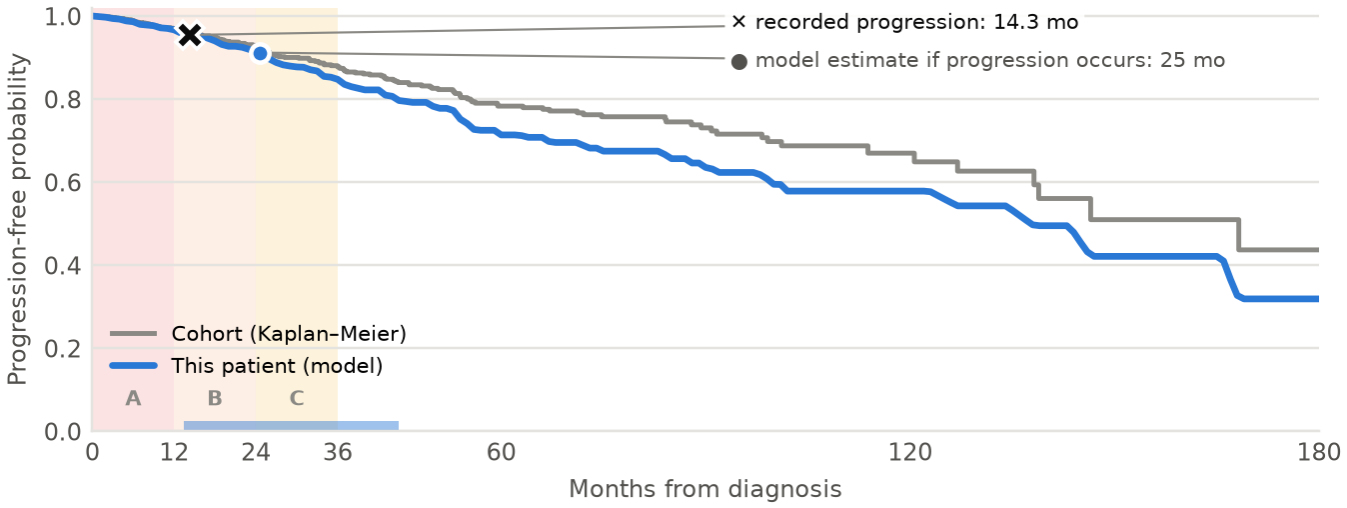
≈ 25 months

Window 24–36 months · IQR 13–45 mo
if progression occurs

Recorded progression

14.3 months

Group B · Progression in year 2
✗ recorded window differs · |error| 10 mo



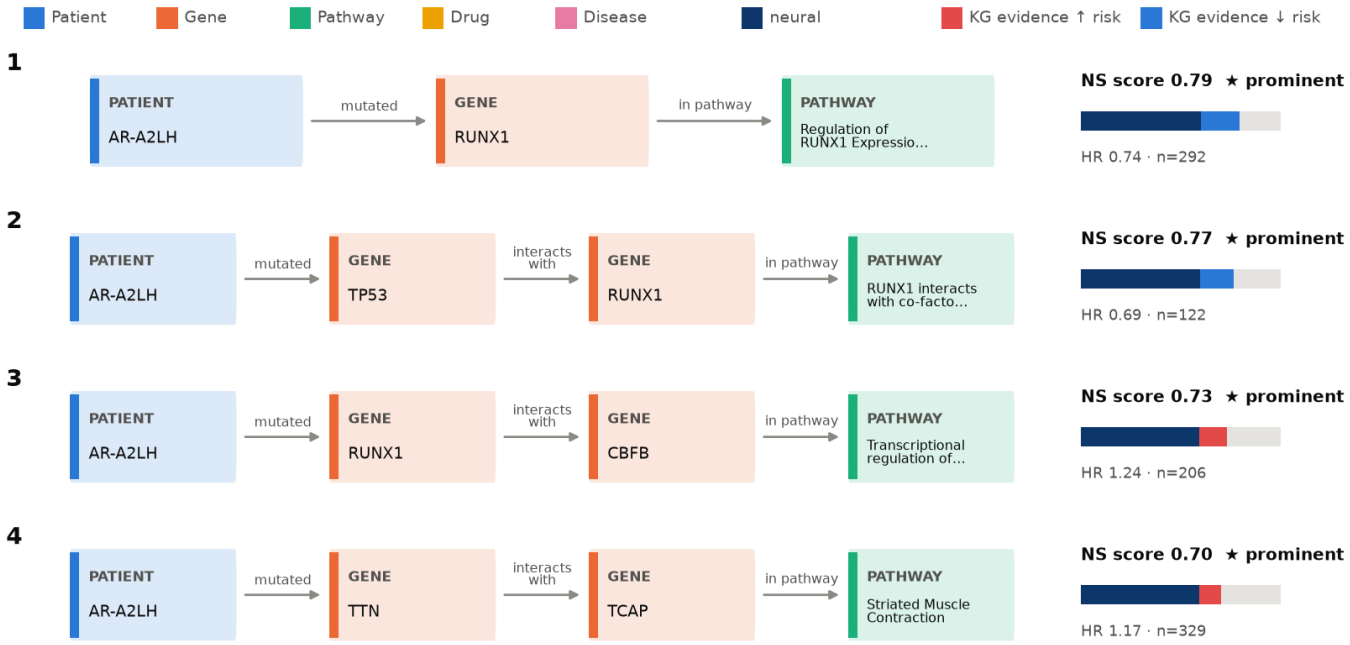
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a Normal tumour, AJCC stage III (T3, N3, M0). The fused neuro-symbolic model places them in the **86th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **RUNX1** to **Regulation of RUNX1 Expression and Activity**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.74). Progression was recorded at 14.3 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

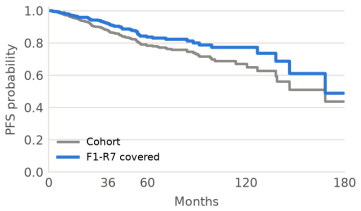


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AR-A2LH** mutated → **RUNX1** in pathway → **Regulation of RUNX1 Expression and Activity**	100	↓ **slower progression** HR 0.74, z -1.5, n = 292 carriers of “Regulation of RUNX1 Expression and Activity”	**0.79**
2 ★	**TCGA-AR-A2LH** mutated → **TP53** interacts with → **RUNX1** in pathway → **RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known**	100	↓ **slower progression** HR 0.69, z -1.3, n = 122 carriers of “RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known”	**0.77**
3 ★	**TCGA-AR-A2LH** mutated → **RUNX1** interacts with → **CBFB** in pathway → **Transcriptional regulation of granulopoiesis**	99	↑ **faster progression** HR 1.24, z +1.0, n = 206 carriers of “Transcriptional regulation of granulopoiesis”	**0.73**
4 ★	**TCGA-AR-A2LH** mutated → **TTN** interacts with → **TCAP** in pathway → **Striated Muscle Contraction**	99	↑ **faster progression** HR 1.17, z +0.8, n = 329 carriers of “Striated Muscle Contraction”	**0.70**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

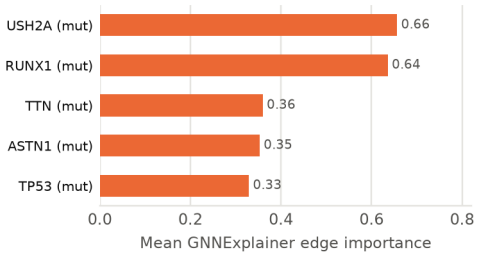
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R7 ↓ **lower risk**	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RUNX1 · driver	Mutation	p.Q264*	18	0.64
TP53 · driver	Mutation	p.R273H	57	0.33
USH2A	Mutation	p.E1739Q	0	0.66
TTN	Mutation	p.G12310E	2	0.36
ASTN1	Mutation	p.I825=	0	0.35



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A0JB	50	III	Normal	0.64	no progression, followed 50.7 mo
TCGA-BH-A0HP	65	III	LumA	0.63	no progression, followed 13.6 mo
TCGA-A8-A09X	62	III	Her2	0.62	no progression, followed 14.0 mo
TCGA-LD-A74U	79	III	LumA	0.62	no progression, followed 13.2 mo
TCGA-E9-A6HE	45	III	LumA	0.60	no progression, followed 27.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**52**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E9-A1N6-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **—** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

70th

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 24%

Model-estimated progression time

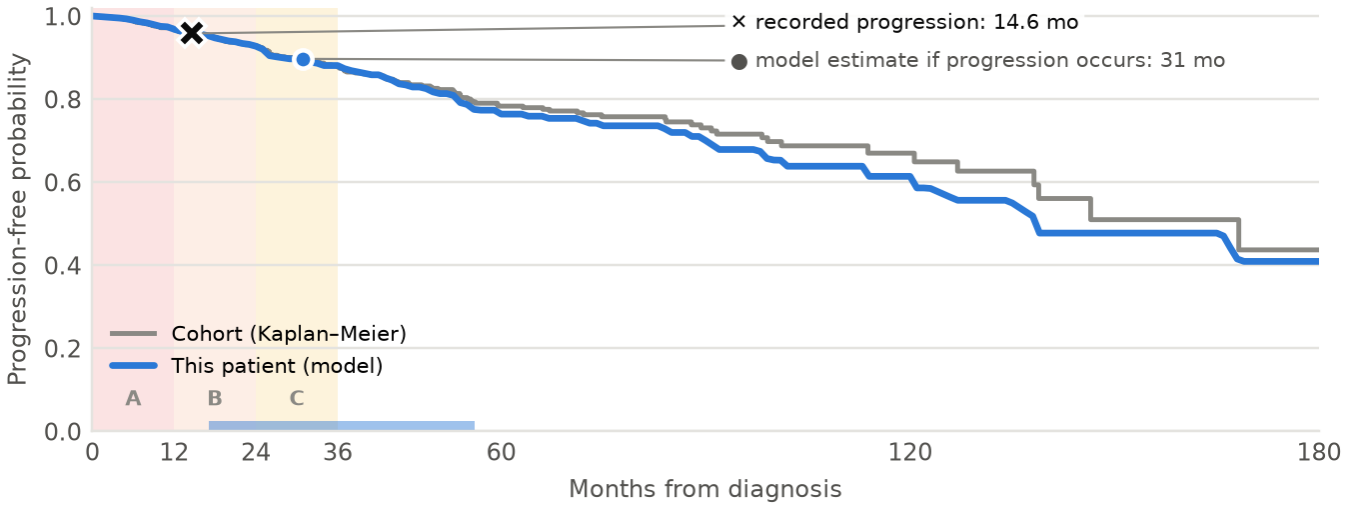
≈ 31 months

Window 24–36 months · IQR 17–56 mo
if progression occurs

Recorded progression

14.6 months

Group B · Progression in year 2
✗ recorded window differs · |error| 16 mo



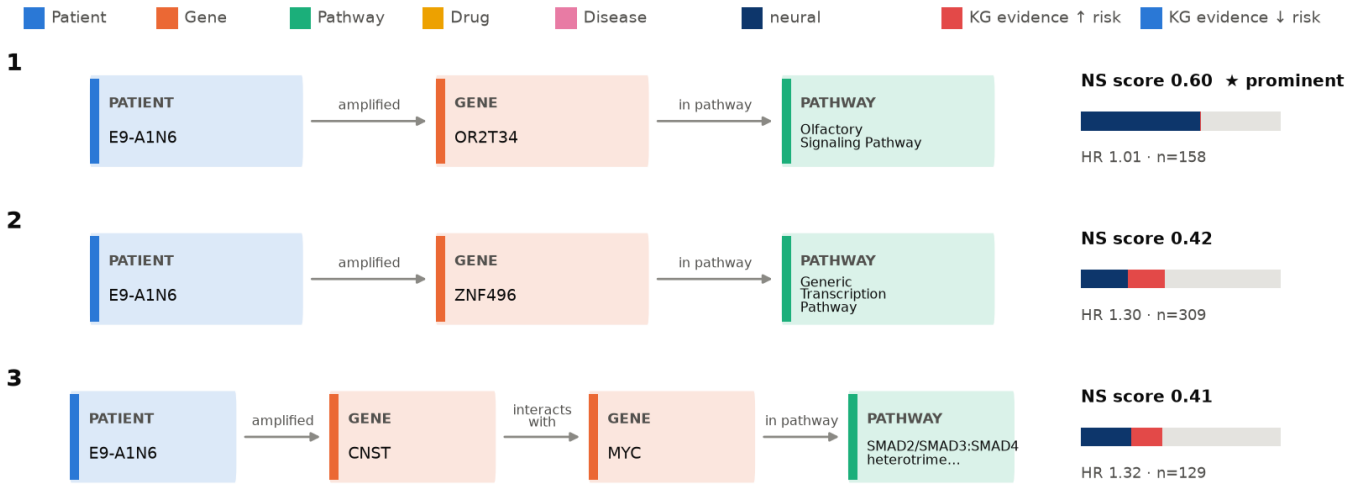
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 52-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **70th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND Age ≥ 34.5*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **OR2T34** to **Olfactory Signaling Pathway**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.01). Progression was recorded at 14.6 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

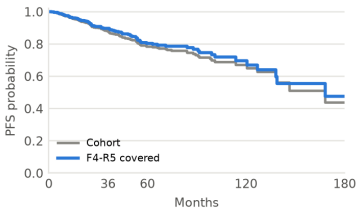


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E9-A1N6** amplified → **OR2T34** in pathway → **Olfactory Signaling Pathway**	99	↑ **faster progression** HR 1.01, z +0.1, n = 158 carriers of “Olfactory Signaling Pathway”	**0.60**
2	**TCGA-E9-A1N6** amplified → **ZNF496** in pathway → **Generic Transcription Pathway**	39	↑ **faster progression** HR 1.30, z +1.4, n = 309 carriers of “Generic Transcription Pathway”	**0.42**
3	**TCGA-E9-A1N6** amplified → **CNST** interacts with → **MYC** in pathway → **SMAD2/SMAD3:SMAD4 heterotrimer regulates transcription**	42	↑ **faster progression** HR 1.32, z +1.2, n = 129 carriers of “SMAD2/SMAD3:SMAD4 heterotrimer regulates transcription”	**0.41**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

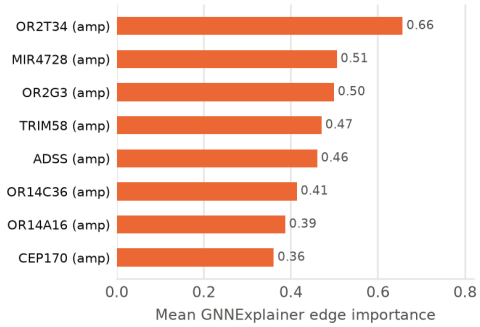
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
OR2T34	High-level amplification	GISTIC ±2	1	0.66
MIR4728	High-level amplification	GISTIC ±2	0	0.51
OR2G3	High-level amplification	GISTIC ±2	1	0.50
TRIM58	High-level amplification	GISTIC ±2	0	0.47
ADSS	High-level amplification	GISTIC ±2	0	0.46
OR14C36	High-level amplification	GISTIC ±2	1	0.41
OR14A16	High-level amplification	GISTIC ±2	1	0.39
CEP170	High-level amplification	GISTIC ±2	0	0.36
ZPBP2	High-level amplification	GISTIC ±2	0	0.35



70 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 60 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A0U1	36	II	missing	0.82	no progression, followed 133.2 mo
TCGA-D8-A1JB	54	II	LumA	0.78	no progression, followed 55.5 mo
TCGA-BH-A0B7	42	II	Her2	0.77	no progression, followed 84.1 mo
TCGA-AO-A03N	59	II	LumB	0.74	progressed at 54.1 mo (group D)
TCGA-AN-A0FT	63	II	LumA	0.73	no progression, followed 7.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**58**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0WW-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **1.13** · MSIsensor score: **0.67** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

93rd

High risk tier · fused neuro-symbolic model



Probability of progression

19% by 36 mo

12 mo 4% · 24 mo 11% · 60 mo 35%

Model-estimated progression time

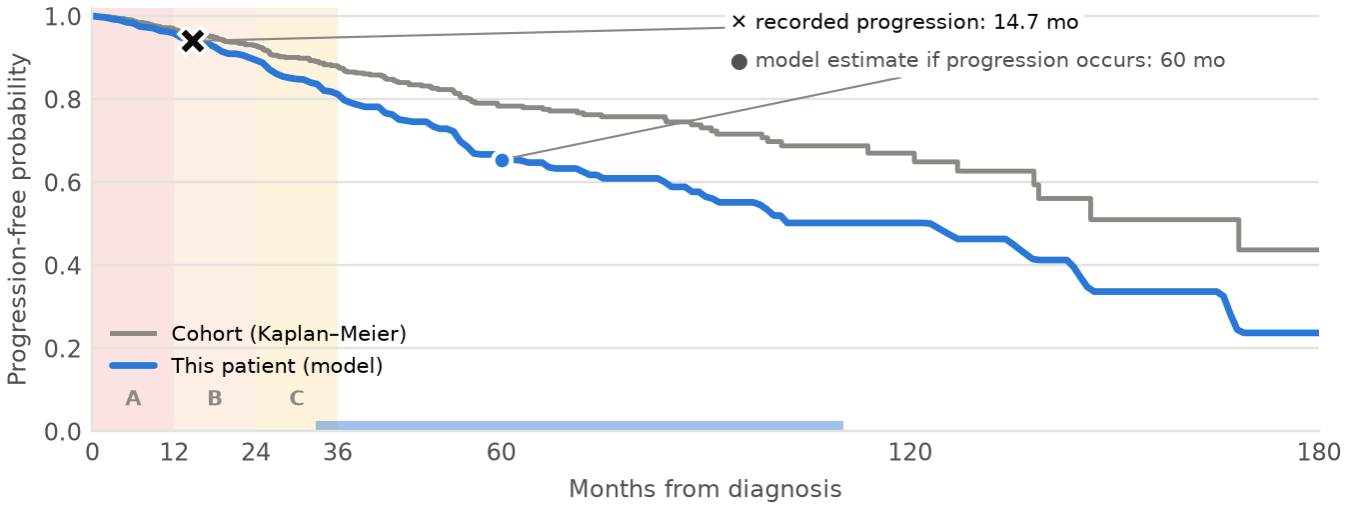
≈ 60 months

Window > 36 months · IQR 33–110 mo
if progression occurs

Recorded progression

14.7 months

Group B · Progression in year 2
✗ recorded window differs · |error| 45 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 58-year-old patient has a LumB tumour, AJCC stage X (T2, N1, MX). The fused neuro-symbolic model places them in the **93rd risk percentile** (high tier). The most significant rule covering them (*AJCC stage = III AND altered gene linked to 'breast ductal adenocarcinoma' AND Aneuploidy score < 26.5*) marks a group with a progression hazard 2.8× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PPFIA1** to **Norepinephrine Neurotransmitter Release Cycle**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.59). Progression was recorded at 14.7 months; the model's estimate given progression was 60 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

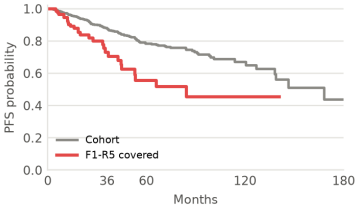


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0WW** amplified → **PPFIA1** in pathway → **Norepinephrine Neurotransmitter Release Cycle**	100	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of “Norepinephrine Neurotransmitter Release Cycle”	**0.82**
2 ★	**TCGA-B6-A0WW** amplified → **ANO1** in pathway → **Stimuli-sensing channels**	98	↓ **slower progression** HR 0.78, z -1.2, n = 261 carriers of “Stimuli-sensing channels”	**0.74**
3 ★	**TCGA-B6-A0WW** amplified → **KCNU1** in pathway → **Sperm Motility And Taxes**	98	↓ **slower progression** HR 0.82, z -0.7, n = 108 carriers of “Sperm Motility And Taxes”	**0.68**
4 ★	**TCGA-B6-A0WW** amplified → **FADD** in pathway → **Regulation of TNFR1 signaling**	72	↓ **slower progression** HR 0.64, z -1.8, n = 208 carriers of “Regulation of TNFR1 signaling”	**0.68**
5 ★	**TCGA-B6-A0WW** mutated → **PTEN** interacts with → **SHARPIN** in pathway → **Neurexins and neuroligins**	78	↓ **slower progression** HR 0.69, z -1.6, n = 210 carriers of “Neurexins and neuroligins”	**0.68**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

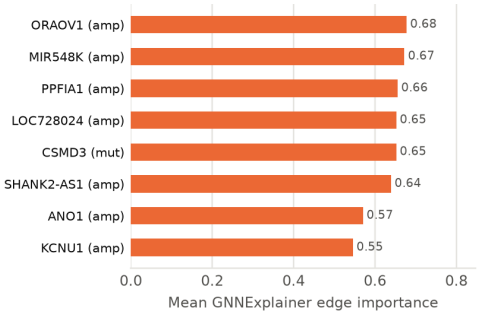
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R5 ↑ higher risk	AJCC stage = III AND altered gene linked to 'breast ductal adenocarcinoma' AND Aneuploidy score < 26.5 [KG]	121 pts 28 events	**2.84** p 1.1e-06	84 mo
F1-R6 ↑ higher risk	AJCC stage = III AND Age ≥ 31.5	198 pts 38 events	**2.05** p 2.0e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
PTEN · driver	Mutation	p.K266*, p.N48Tfs*6	22	0.31
ORAOV1	High-level amplification	GISTIC ±2	0	0.68
MIR548K	High-level amplification	GISTIC ±2	0	0.67
PPFIA1	High-level amplification	GISTIC ±2	6	0.66
LOC728024	High-level amplification	GISTIC ±2	0	0.65
CSMD3	Mutation	p.S3666L	0	0.65
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.64
ANO1	High-level amplification	GISTIC ±2	2	0.57
KCNU1	High-level amplification	GISTIC ±2	1	0.55
ZNF703	High-level amplification	GISTIC ±2	1	0.50



26 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 9 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JI	54	II	LumB	0.72	no progression, followed 19.0 mo
TCGA-AC-A2BM	41	II	LumB	0.70	no progression, followed 99.4 mo
TCGA-E2-A15K	58	II	LumB	0.69	no progression, followed 9.0 mo
TCGA-AN-A03Y	66	II	LumB	0.66	no progression, followed 0.3 mo
TCGA-A1-A0SQ	45	II	LumA	0.66	no progression, followed 18.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**44**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A409-01** · Aneuploidy score: **30** · TMB (nonsyn., /Mb): **0.97** · MSIsensor score: **0.07** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

85th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 28%

Model-estimated progression time

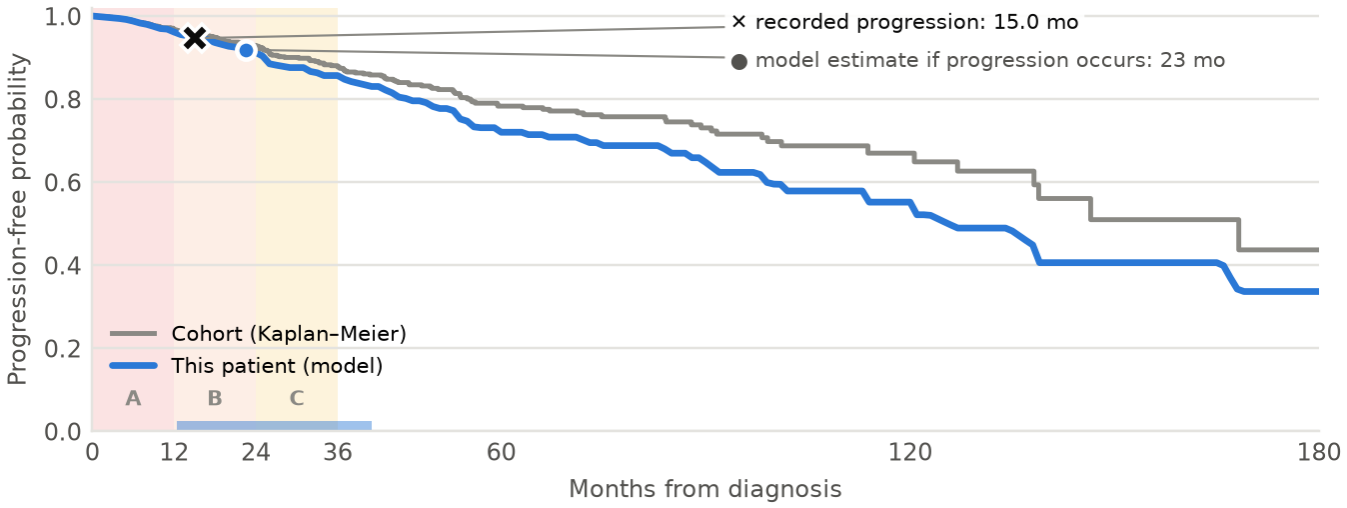
≈ 23 months

Window 12–24 months · IQR 12–41 mo
if progression occurs

Recorded progression

15.0 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 8 mo



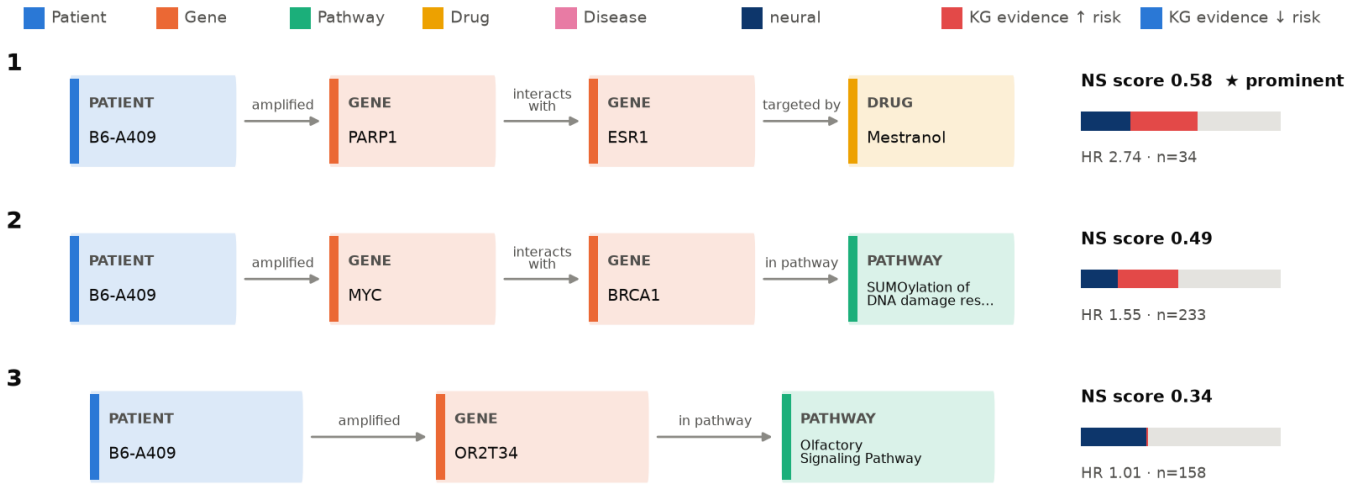
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 44-year-old patient has a Basal tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **85th risk percentile** (high tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PARP1** to **Mestranol**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.74). Progression was recorded at 15.0 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

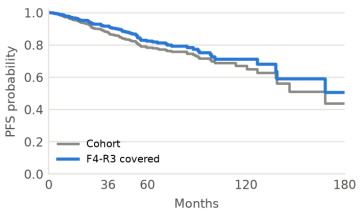


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A409** amplified → **PARP1** interacts with → **ESR1** targeted by → **Mestranol**	42	↑ **faster progression** HR 2.74, z +2.5, n = 34 carriers of “Mestranol”	**0.58**
2	**TCGA-B6-A409** amplified → **MYC** interacts with → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	31	↑ **faster progression** HR 1.55, z +2.3, n = 233 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.49**
3	**TCGA-B6-A409** amplified → **OR2T34** in pathway → **Olfactory Signaling Pathway**	55	↑ **faster progression** HR 1.01, z +0.1, n = 158 carriers of “Olfactory Signaling Pathway”	**0.34**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

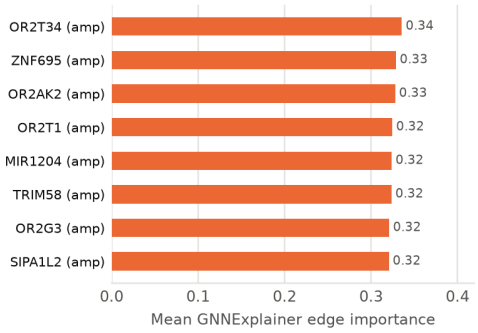
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
TP53 · driver	Mutation	p.T253Hfs*11	57	0.31
OR2T34	High-level amplification	GISTIC ±2	1	0.34
ZNF695	High-level amplification	GISTIC ±2	0	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.33
OR2T1	High-level amplification	GISTIC ±2	1	0.32
MIR1204	High-level amplification	GISTIC ±2	0	0.32
TRIM58	High-level amplification	GISTIC ±2	0	0.32
OR2G3	High-level amplification	GISTIC ±2	1	0.32
SIPA1L2	High-level amplification	GISTIC ±2	0	0.32
HEATR1	High-level amplification	GISTIC ±2	2	0.32
OR11L1	High-level amplification	GISTIC ±2	1	0.32



77 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548) … and 35 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A6VW	48	II	Basal	0.81	no progression, followed 9.4 mo
TCGA-E9-A22G	47	II	Basal	0.79	no progression, followed 40.7 mo
TCGA-A2-A0D0	60	II	Basal	0.78	no progression, followed 67.3 mo
TCGA-A2-A0SX	48	I	Basal	0.76	progressed at 40.7 mo (group D)
TCGA-B6-A0WX	40	III	Basal	0.76	progressed at 18.5 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**63**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**EUR**	Pathologic T	**T4**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

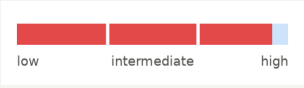
Sample: **TCGA-B6-A0IK-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **3.8** · MSIsensor score: **0.25** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 5% · 24 mo 12% · 60 mo 34%

Model-estimated progression time

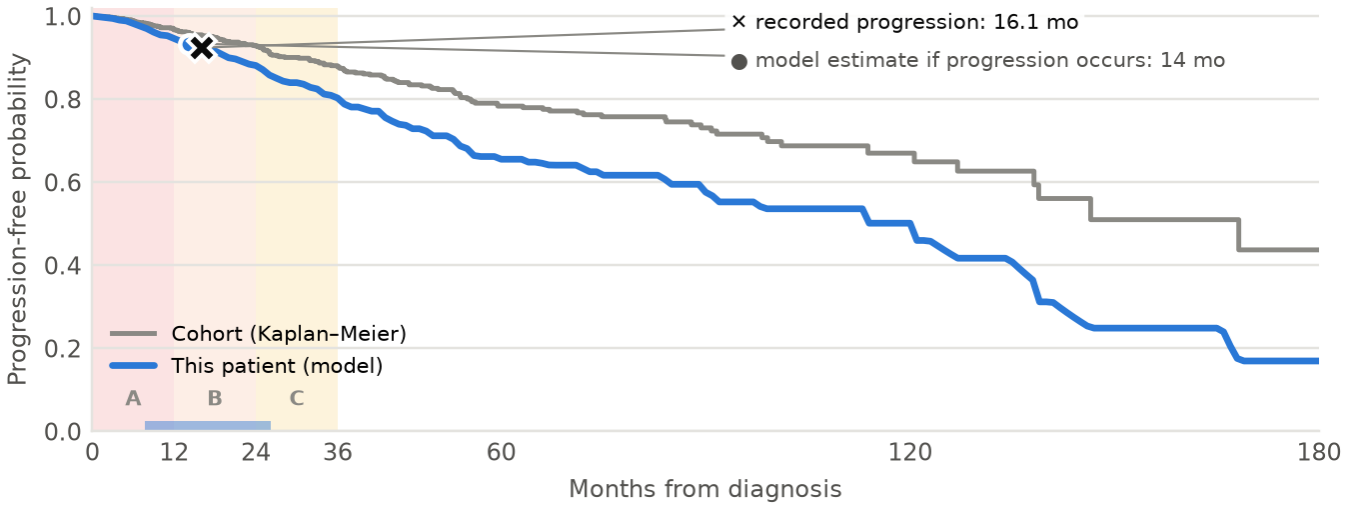
≈ 14 months

Window 12–24 months · IQR 8–26 mo if progression occurs

Recorded progression

16.1 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 2 mo



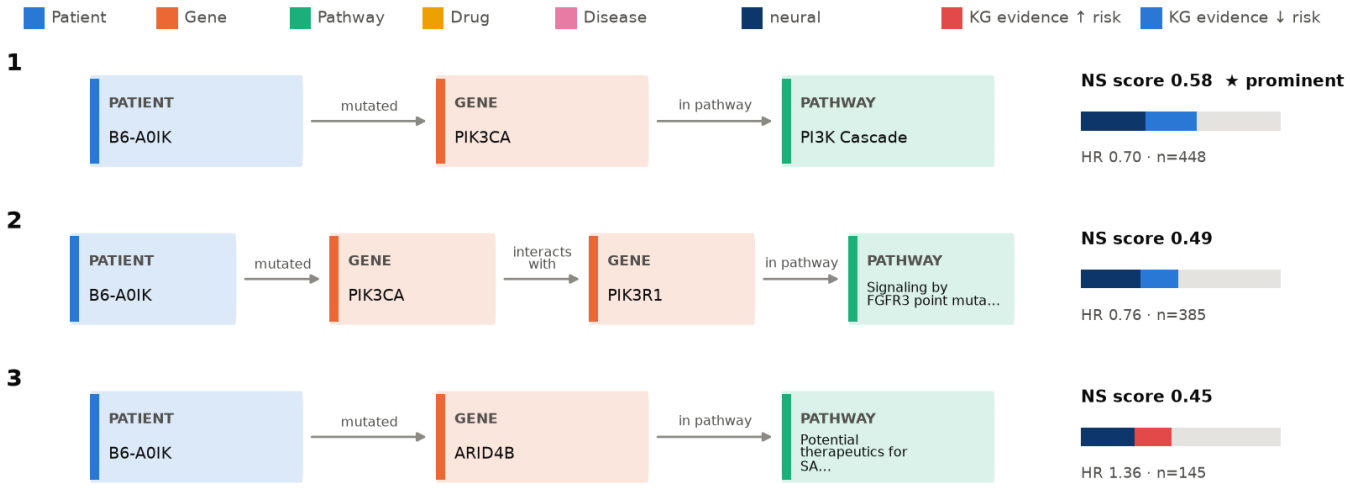
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 63-year-old patient has a Her2 tumour, AJCC stage III (T4, N1, M0). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **PI3K Cascade**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 16.1 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

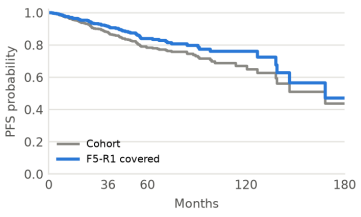


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0IK** mutated → **PIK3CA** in pathway → **PI3K Cascade**	54	↓ **slower progression** HR 0.70, z -1.9, n = 448 carriers of “PI3K Cascade”	**0.58**
2	**TCGA-B6-A0IK** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Signaling by FGFR3 point mutants in cancer**	50	↓ **slower progression** HR 0.76, z -1.4, n = 385 carriers of “Signaling by FGFR3 point mutants in cancer”	**0.49**
3	**TCGA-B6-A0IK** mutated → **ARID4B** in pathway → **Potential therapeutics for SARS**	45	↑ **faster progression** HR 1.36, z +1.4, n = 145 carriers of “Potential therapeutics for SARS”	**0.45**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

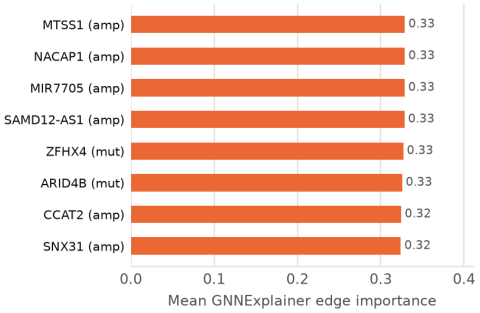
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R4 ↓ **lower risk**	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.32
NF1 · driver	Mutation	p.P2696Tfs*14	6	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
BAP1 · driver	Homozygous deletion	GISTIC ±2	9	0.31
TP53 · driver	Mutation	p.L194R	57	0.31
MTSS1	High-level amplification	GISTIC ±2	0	0.33
NACAP1	High-level amplification	GISTIC ±2	0	0.33
MIR7705	High-level amplification	GISTIC ±2	0	0.33
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.33
ZFHX4	Mutation	p.L2845V	0	0.33
ARID4B	Mutation	p.S1018*	3	0.33
CCAT2	High-level amplification	GISTIC ±2	0	0.32



81 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); TP53 → Acetylsalicylic acid (DB00945) … and 16 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1B0	50	III	Her2	0.73	no progression, followed 53.6 mo
TCGA-A8-A09Q	83	III	LumB	0.69	no progression, followed 25.0 mo
TCGA-BH-A1F2	53	III	LumB	0.69	no progression, followed 31.5 mo
TCGA-A8-A08R	52	II	Basal	0.69	no progression, followed 1.0 mo
TCGA-C8-A12P	55	II	Her2	0.69	no progression, followed 11.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0X1-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **2.27** · MSIsensor score: **1.54** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

34% by 36 mo

12 mo 10% · 24 mo 22% · 60 mo 55%

Model-estimated progression time

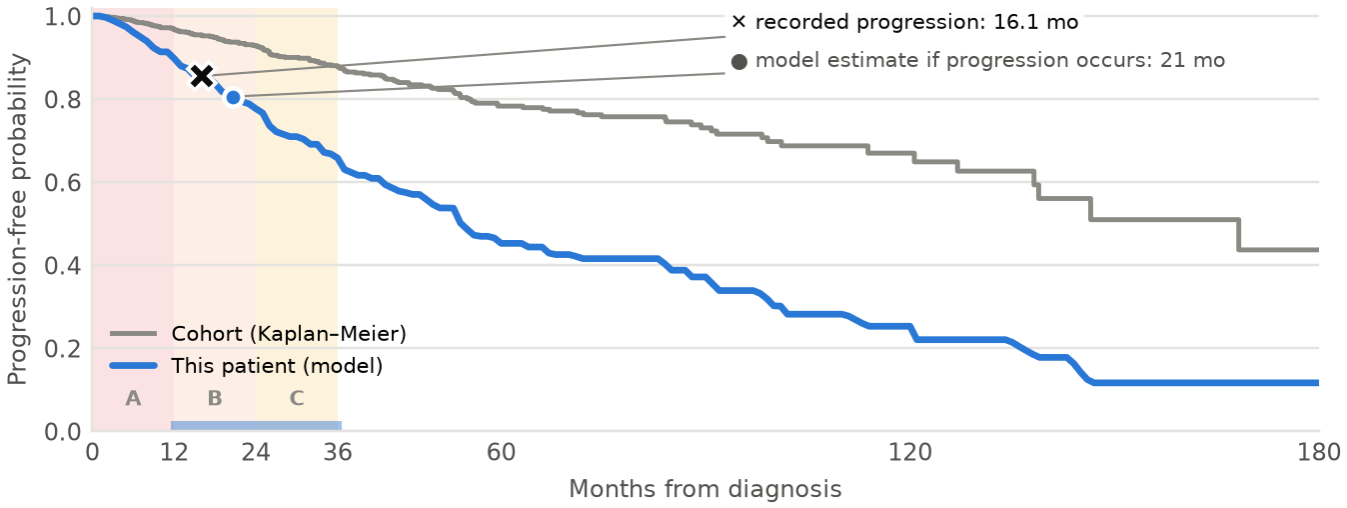
≈ 21 months

Window 12–24 months · IQR 12–37 mo
if progression occurs

Recorded progression

16.1 months

■ **Group B** · Progression in year 2
✓ **estimate in recorded window** · |error| 4 mo



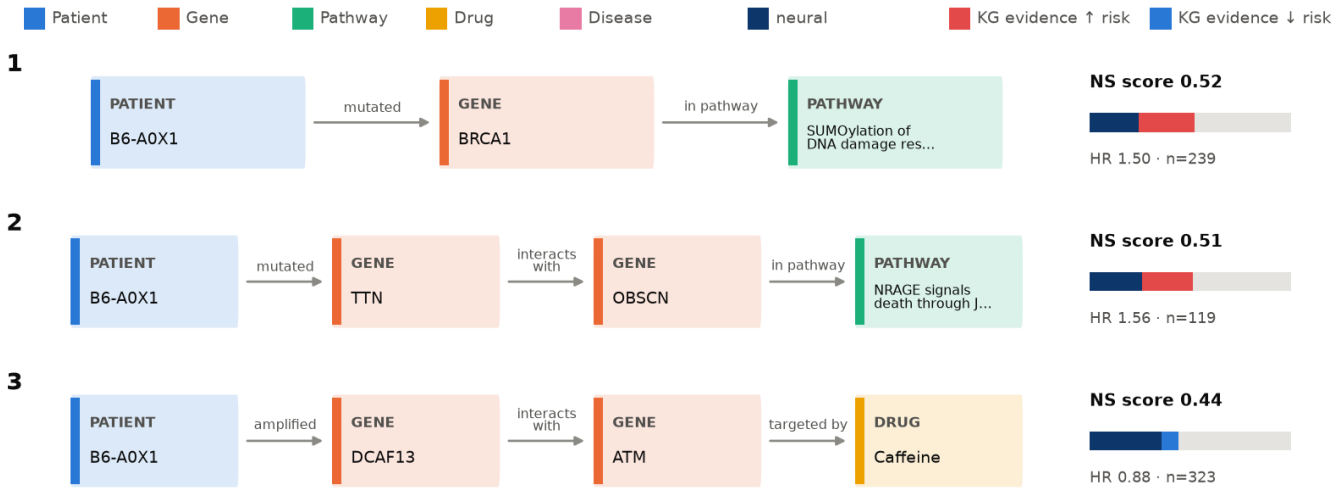
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a Basal tumour, AJCC stage II (T2, N1, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The most significant rule covering them (*40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **BRCA1** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.50). Progression was recorded at 16.1 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

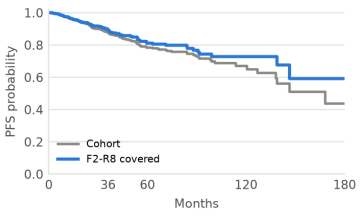


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0X1** mutated → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	40	↑ **faster progression** HR 1.50, z +2.1, n = 239 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.52**
2	**TCGA-B6-A0X1** mutated → **TTN** interacts with → **OBSCN** in pathway → **NRAGE signals death through JNK**	43	↑ **faster progression** HR 1.56, z +1.9, n = 119 carriers of “NRAGE signals death through JNK”	**0.51**
3	**TCGA-B6-A0X1** amplified → **DCAF13** interacts with → **ATM** targeted by → **Caffeine**	59	↓ **slower progression** HR 0.88, z -0.6, n = 323 carriers of “Caffeine”	**0.44**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

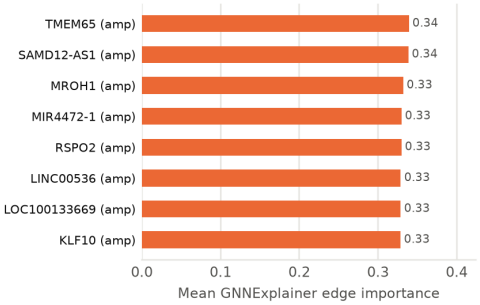
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R8 ↓ **lower risk**	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA1 · driver	Mutation	p.E23Sfs*8	29	0.31
TP53 · driver	Mutation	p.X331_splice	57	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
TMEM65	High-level amplification	GISTIC ±2	0	0.34
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.34
MROH1	High-level amplification	GISTIC ±2	0	0.33
MIR4472-1	High-level amplification	GISTIC ±2	0	0.33
RSPO2	High-level amplification	GISTIC ±2	1	0.33
LINC00536	High-level amplification	GISTIC ±2	0	0.33
LOC100133669	High-level amplification	GISTIC ±2	0	0.33
KLF10	High-level amplification	GISTIC ±2	0	0.33
MROH5	High-level amplification	GISTIC ±2	0	0.33



280 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813) … and 92 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A0TS	46	II	Basal	0.89	no progression, followed 84.1 mo
TCGA-C8-A12K	80	II	Basal	0.88	no progression, followed 0.0 mo
TCGA-A2-A3Y0	57	II	Basal	0.87	no progression, followed 50.8 mo
TCGA-A2-A0T0	59	II	Basal	0.86	no progression, followed 17.5 mo
TCGA-AN-A0FL	62	II	Basal	0.85	no progression, followed 7.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**72**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-E9-A2JS-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **1.93** · MSIsensor score: **0.81** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

37th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

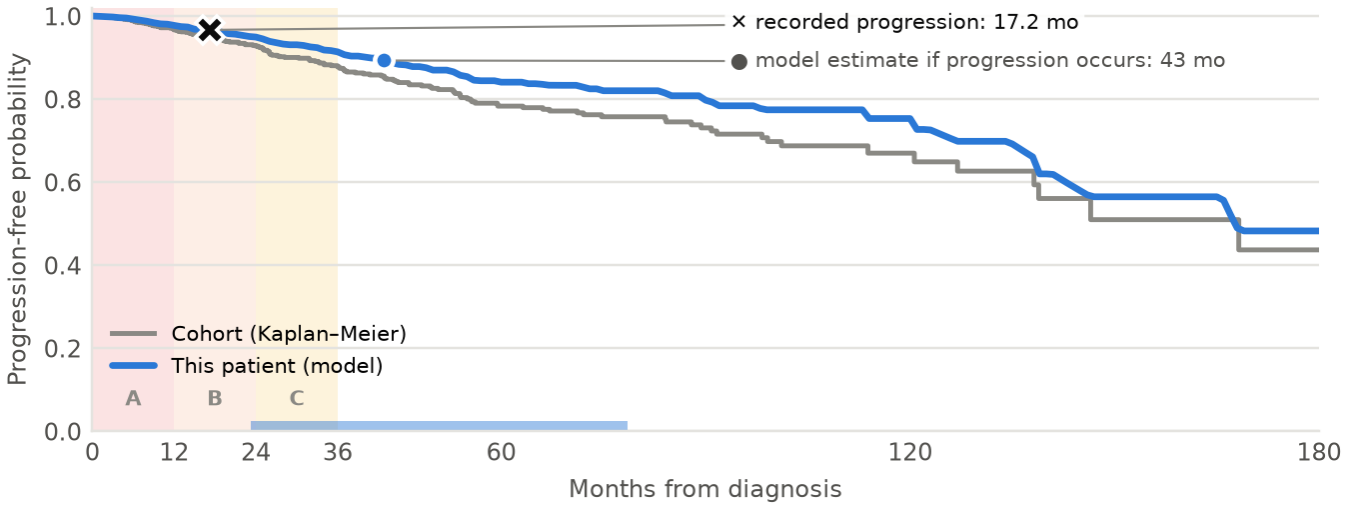
≈ 43 months

Window > 36 months · IQR 23–79 mo if progression occurs

Recorded progression

17.2 months

Group B · Progression in year 2
✗ recorded window differs · |error| 26 mo



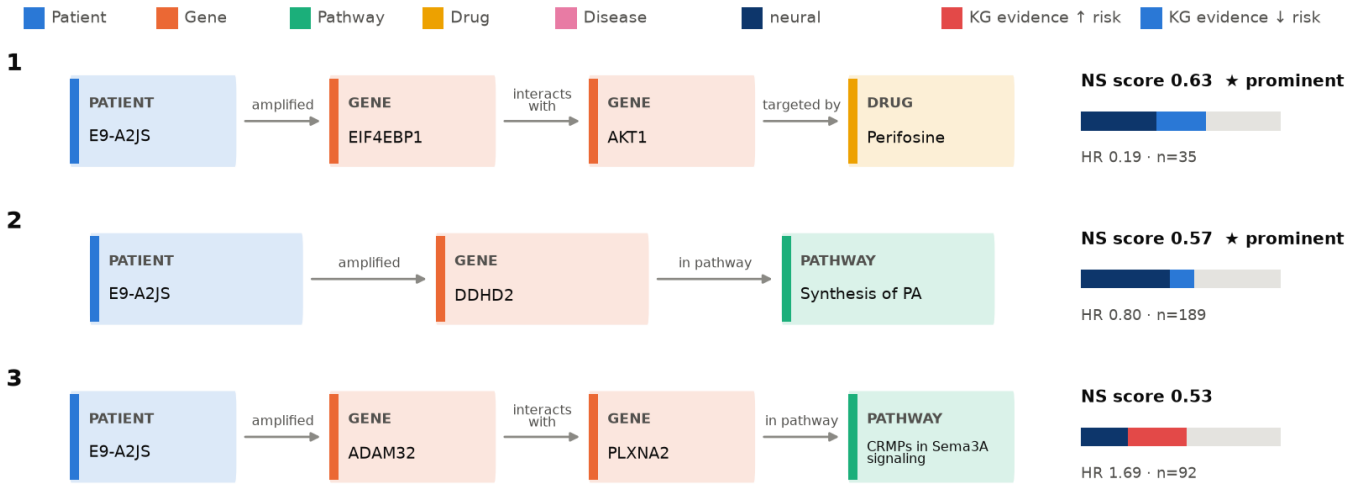
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 72-year-old patient has a LumB tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **37th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **EIF4EBP1** to **Perifosine**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.19). Progression was recorded at 17.2 months; the model's estimate given progression was 43 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

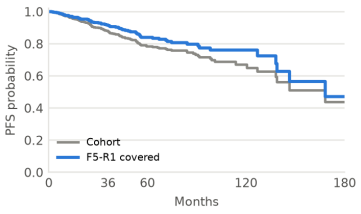


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E9-A2JS** amplified → **EIF4EBP1** interacts with → **AKT1** targeted by → **Perifosine**	63	↓ **slower progression** HR 0.19, z -1.9, n = 35 carriers of “Perifosine”	**0.63**
2 ★	**TCGA-E9-A2JS** amplified → **DDHD2** in pathway → **Synthesis of PA**	74	↓ **slower progression** HR 0.80, z -0.9, n = 189 carriers of “Synthesis of PA”	**0.57**
3	**TCGA-E9-A2JS** amplified → **ADAM32** interacts with → **PLXNA2** in pathway → **CRMPs in Sema3A signaling**	39	↑ **faster progression** HR 1.69, z +2.2, n = 92 carriers of “CRMPs in Sema3A signaling”	**0.53**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

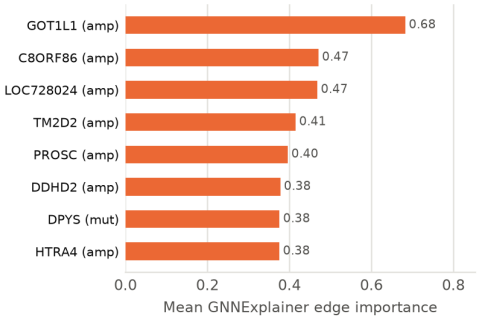
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
GOT1L1	High-level amplification	GISTIC ±2	0	0.68
C8ORF86	High-level amplification	GISTIC ±2	0	0.47
LOC728024	High-level amplification	GISTIC ±2	0	0.47
TM2D2	High-level amplification	GISTIC ±2	0	0.41
PROSC	High-level amplification	GISTIC ±2	0	0.40
DDHD2	High-level amplification	GISTIC ±2	1	0.38
DPYS	Mutation	Splice_Region	1	0.38
HTRA4	High-level amplification	GISTIC ±2	0	0.38
LETM2	High-level amplification	GISTIC ±2	0	0.35
STAR	High-level amplification	GISTIC ±2	1	0.33
PLEKHA2	High-level amplification	GISTIC ±2	1	0.33



40 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 45 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A204	80	II	LumB	0.84	no progression, followed 83.3 mo
TCGA-E2-A1IE	61	II	LumA	0.78	progressed at 59.5 mo (group D)
TCGA-A2-A0ES	52	II	LumA	0.78	no progression, followed 72.0 mo
TCGA-E9-A2JT	63	II	LumA	0.73	no progression, followed 9.5 mo
TCGA-HN-A2OB	45	II	LumA	0.73	progressed at 50.7 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**69**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

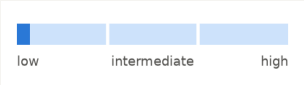
Sample: **TCGA-GM-A2D9-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **14.57** · MSIsensor score: **0.07** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

5th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

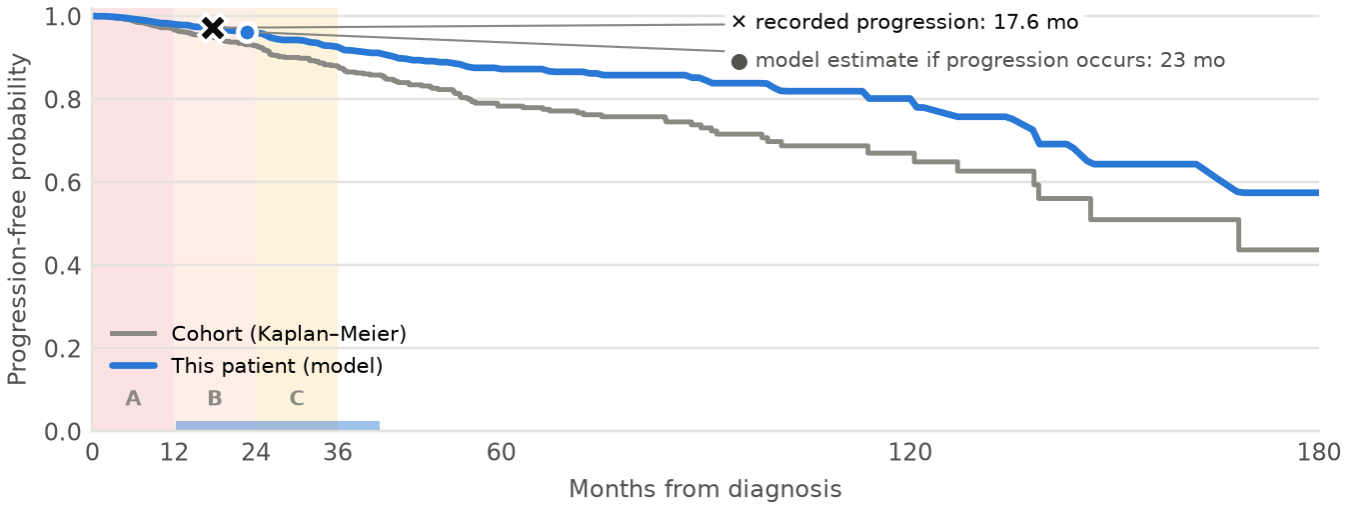
≈ 23 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

17.6 months

■ **Group B** · Progression in year 2
✓ **estimate in recorded window** · |error| 5 mo



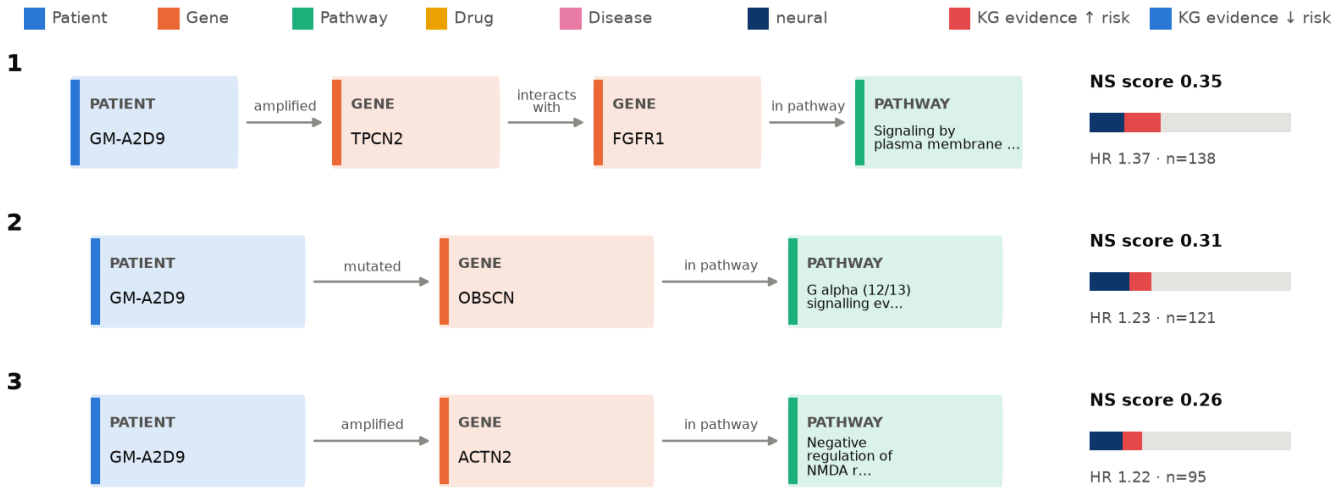
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 69-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **5th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TPCN2** to **Signaling by plasma membrane FGFR1 fusions**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.37). Progression was recorded at 17.6 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

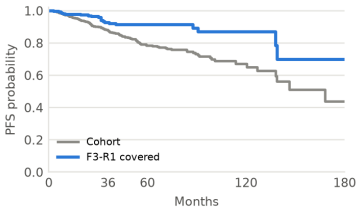


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-GM-A2D9** amplified → **TPCN2** interacts with → **FGFR1** in pathway → **Signaling by plasma membrane FGFR1 fusions**	29	↑ **faster progression** HR 1.37, z +1.4, n = 138 carriers of “Signaling by plasma membrane FGFR1 fusions”	**0.35**
2	**TCGA-GM-A2D9** mutated → **OBSCN** in pathway → **G alpha (12/13) signalling events**	33	↑ **faster progression** HR 1.23, z +0.8, n = 121 carriers of “G alpha (12/13) signalling events”	**0.31**
3	**TCGA-GM-A2D9** amplified → **ACTN2** in pathway → **Negative regulation of NMDA receptor-mediated neuronal transmission**	27	↑ **faster progression** HR 1.22, z +0.7, n = 95 carriers of “Negative regulation of NMDA receptor-mediated neuronal transmission”	**0.26**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

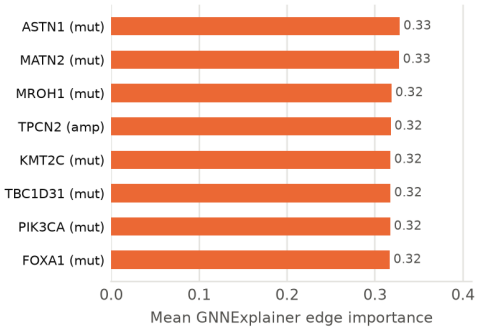
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q982K	4	0.32
PIK3CA · driver	Mutation	p.E453K, p.H1047R	70	0.32
FOXA1 · driver	Mutation	p.K237E, p.K237N, p.K237R	9	0.32
RUNX1 · driver	Mutation	p.E316K	18	0.32
CBFB · driver	Mutation	p.G123C	29	0.32
BRCA1 · driver	Mutation	p.S915C	29	0.32
MAP3K1 · driver	Mutation	p.Q345*	13	0.31
ASTN1	Mutation	p.D1287N	0	0.33
MATN2	Mutation	p.E804Q	0	0.33
MROH1	Mutation	p.X239_splice	0	0.32
TPCN2	High-level amplification	GISTIC ±2	1	0.32
TBC1D31	Mutation	p.R374K	0	0.32



35 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0EO	54	I	LumA	0.79	no progression, followed 80.3 mo
TCGA-D8-A1JP	73	I	LumA	0.75	no progression, followed 21.0 mo
TCGA-B6-A0RN	60	I	LumA	0.74	no progression, followed 263.3 mo
TCGA-E9-A1R5	63	I	LumA	0.73	no progression, followed 3.0 mo
TCGA-BH-A0H3	46	I	LumA	0.70	no progression, followed 63.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**54**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

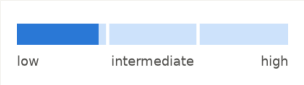
Sample: **TCGA-AR-A0TY-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **2.7** · MSIsensor score: **0.75** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

30th

Low risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

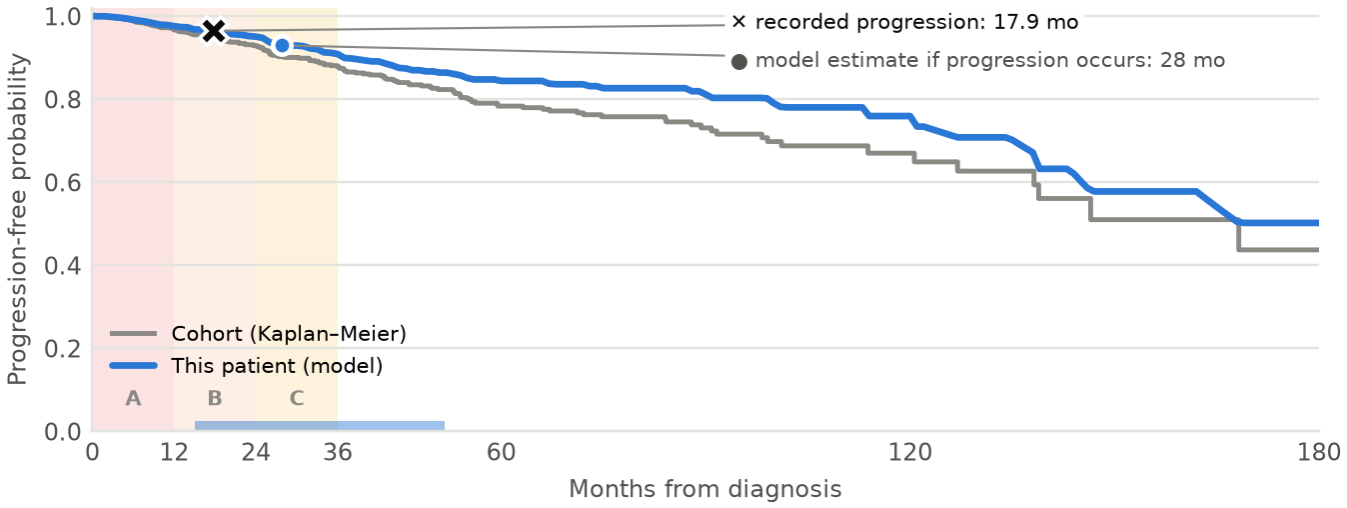
≈ 28 months

Window 24–36 months · IQR 15–52 mo
if progression occurs

Recorded progression

17.9 months

Group B · Progression in year 2
✗ recorded window differs · |error| 10 mo



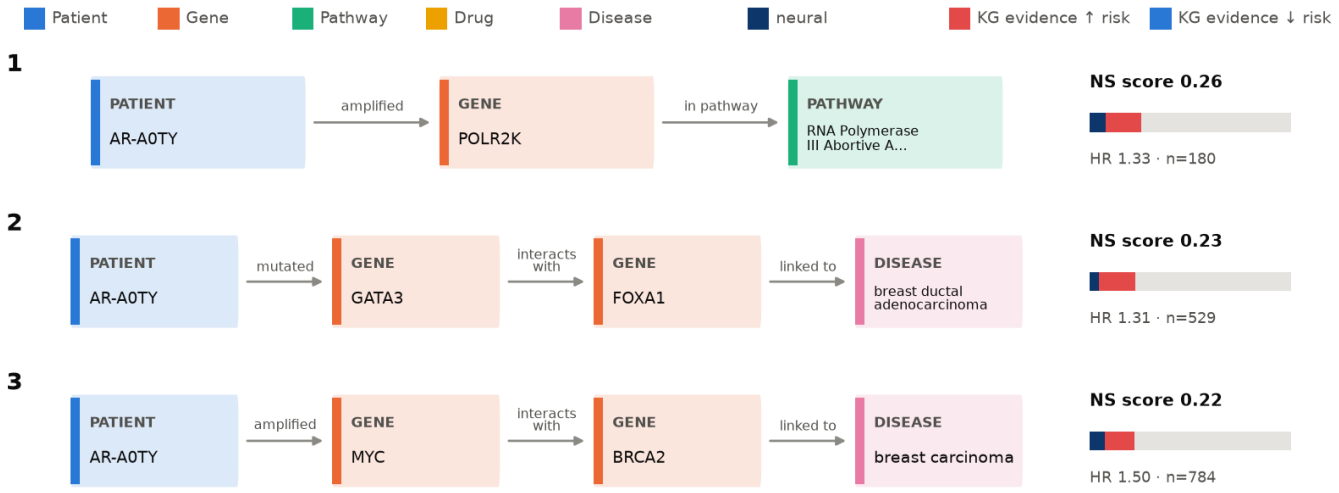
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 54-year-old patient has a LumB tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **30th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **POLR2K** to **RNA Polymerase III Abortive And Retractive Initiation**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.33). Progression was recorded at 17.9 months; the model's estimate given progression was 28 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

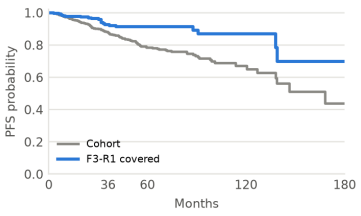


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A0TY** amplified → **POLR2K** in pathway → **RNA Polymerase III Abortive And Retractive Initiation**	13	↑ **faster progression** HR 1.33, z +1.3, n = 180 carriers of “RNA Polymerase III Abortive And Retractive Initiation”	**0.26**
2	**TCGA-AR-A0TY** mutated → **GATA3** interacts with → **FOXA1** linked to → **breast ductal adenocarcinoma**	7	↑ **faster progression** HR 1.31, z +1.4, n = 529 carriers of “breast ductal adenocarcinoma”	**0.23**
3	**TCGA-AR-A0TY** amplified → **MYC** interacts with → **BRCA2** linked to → **breast carcinoma**	12	↑ **faster progression** HR 1.50, z +1.1, n = 784 carriers of “breast carcinoma”	**0.22**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

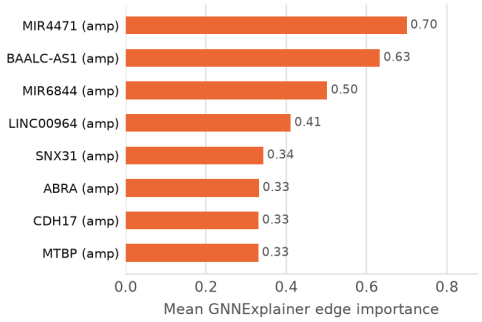
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.H434Pfs*42	13	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
MAP2K4 · driver	Homozygous deletion	GISTIC ±2	12	0.31
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.31
MIR4471	High-level amplification	GISTIC ±2	0	0.70
BAALC-AS1	High-level amplification	GISTIC ±2	0	0.63
MIR6844	High-level amplification	GISTIC ±2	0	0.50
LINC00964	High-level amplification	GISTIC ±2	0	0.41
SNX31	High-level amplification	GISTIC ±2	0	0.34
ABRA	High-level amplification	GISTIC ±2	0	0.33
CDH17	High-level amplification	GISTIC ±2	1	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33



136 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); STK3 → Fostamatinib (DB12010); NDUFB9 → NADH (DB00157); SQLE → Naftifine (DB00735); SQLE → Terbinafine (DB00857); SQLE → Butenafine (DB01091); SQLE → Ellagic acid (DB08846); YWHAZ → Phenethyl Isothiocyanate (DB12695); ATP6V1C1 → Tiludronic acid (DB01133) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A14T	52	II	LumA	0.78	no progression, followed 76.0 mo
TCGA-AR-A0U2	47	III	LumB	0.77	progressed at 52.9 mo (group D)
TCGA-BH-A0AY	62	II	LumA	0.76	no progression, followed 25.5 mo
TCGA-C8-A12N	58	II	LumA	0.76	no progression, followed 11.8 mo
TCGA-A2-A0CU	73	II	LumA	0.76	no progression, followed 5.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A7-A13E-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **3.1** · MSIsensor score: **4.87** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

51st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 18%

Model-estimated progression time

≈ 15 months

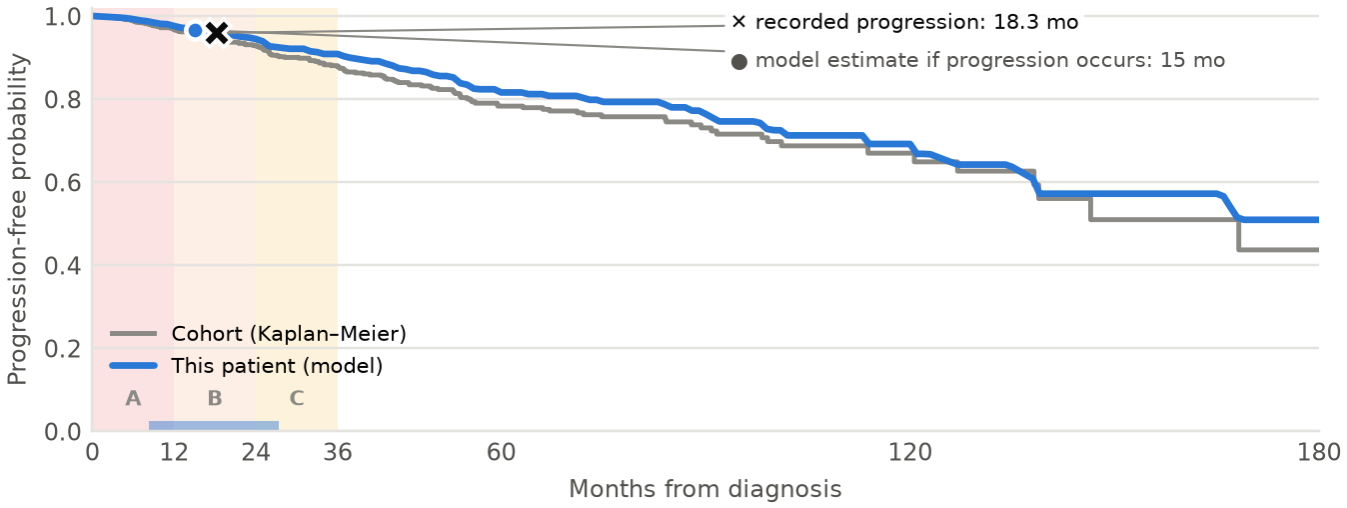
Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

18.3 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 3 mo



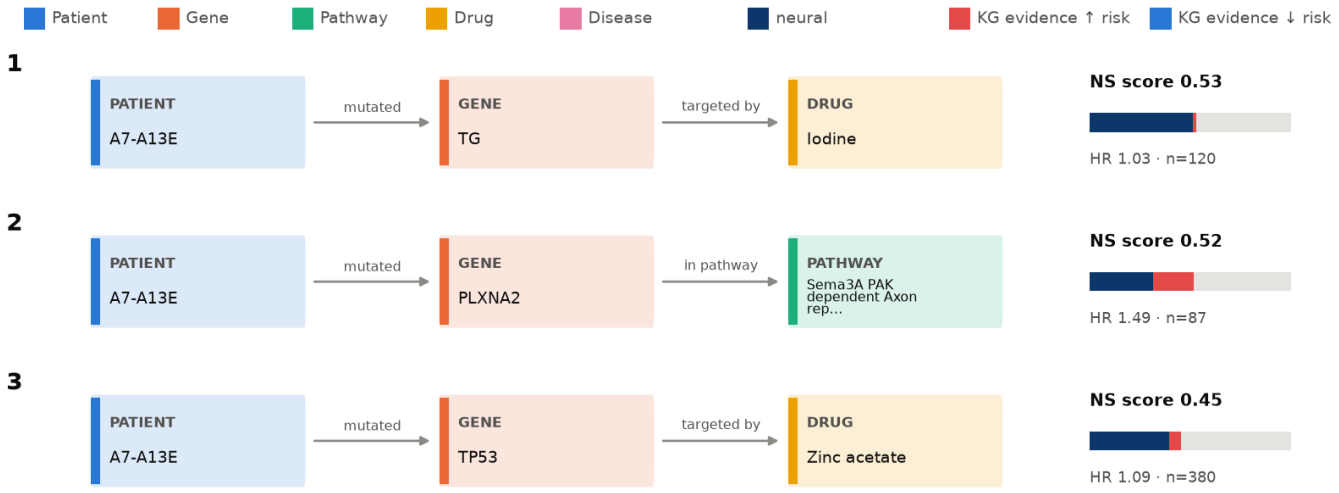
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a Basal tumour, AJCC stage II (T2, N1, MX). The fused neuro-symbolic model places them in the **51st risk percentile** (intermediate tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TG** to **Iodine**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.03). Progression was recorded at 18.3 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

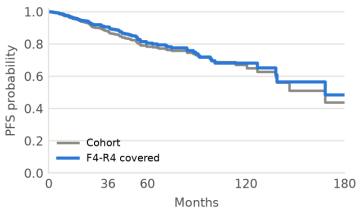


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A7-A13E** mutated → **TG** targeted by → **Iodine**	85	↑ faster progression HR 1.03, z +0.1, n = 120 carriers of “Iodine”	**0.53**
2	**TCGA-A7-A13E** mutated → **PLXNA2** in pathway → **Sema3A PAK dependent Axon repulsion**	53	↑ faster progression HR 1.49, z +1.5, n = 87 carriers of “Sema3A PAK dependent Axon repulsion”	**0.52**
3	**TCGA-A7-A13E** mutated → **TP53** targeted by → **Zinc acetate**	66	↑ faster progression HR 1.09, z +0.4, n = 380 carriers of “Zinc acetate”	**0.45**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

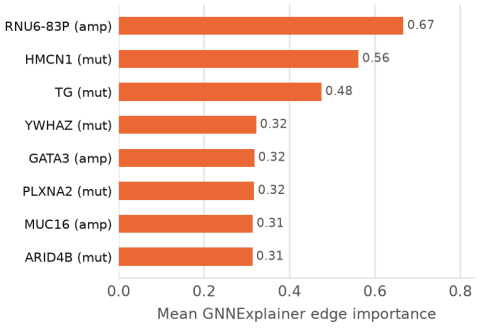
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.32
TP53 · driver	Mutation	p.P77Gfs*65, p.R175H	57	0.31
RNU6-83P	High-level amplification	GISTIC ±2	0	0.67
HMCN1	Mutation	p.I4988=	0	0.56
TG	Mutation	p.R1549W	1	0.48
YWHAZ	Mutation	3'UTR	15	0.32
PLXNA2	Mutation	p.S1617F	4	0.32
MUC16	High-level amplification	GISTIC ±2	8	0.31
ARID4B	Mutation	p.L83Qfs*8	3	0.31
TP53BP2	Mutation	p.P544=	12	0.31
NFASC	Mutation	3'UTR	3	0.31



11 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); TG → Iodine (DB05382); YWHAZ → Phenethyl Isothiocyanate (DB12695) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1AZ	63	II	Basal	0.74	no progression, followed 76.6 mo
TCGA-BH-A5IZ	51	II	Basal	0.73	no progression, followed 18.6 mo
TCGA-LL-A6FR	50	II	Basal	0.72	no progression, followed 16.1 mo
TCGA-LL-A7SZ	49	II	LumA	0.72	no progression, followed 19.5 mo
TCGA-A2-A3XT	45	II	Basal	0.71	no progression, followed 91.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0YJ-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **2.5** · MSIsensor score: **0.47** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 10% · 60 mo 31%

Model-estimated progression time

≈ 16 months

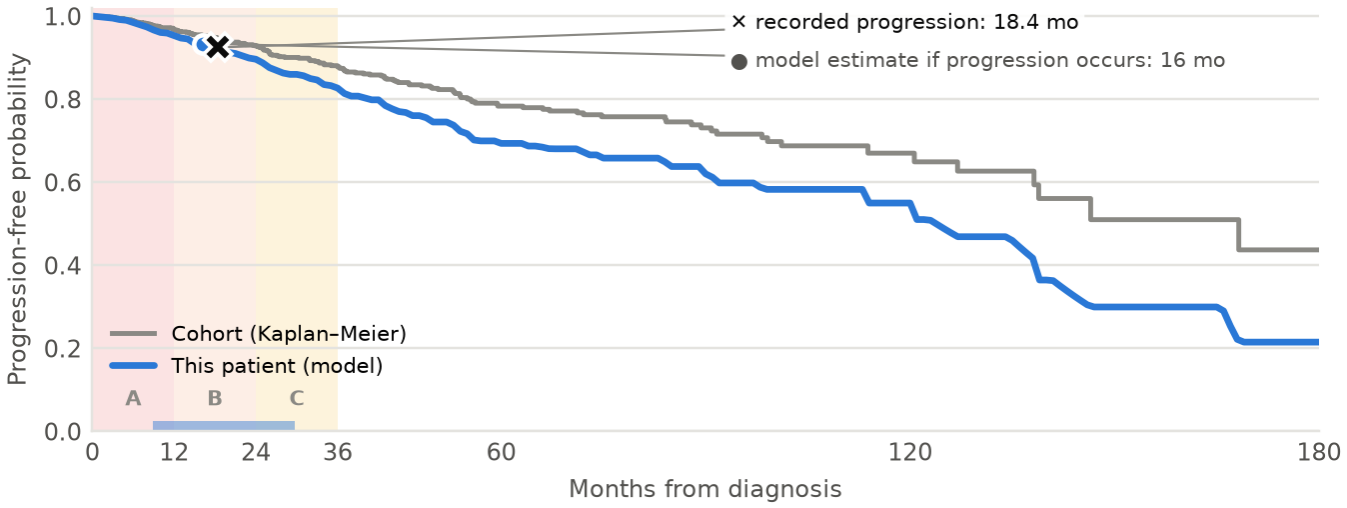
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

18.4 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 2 mo



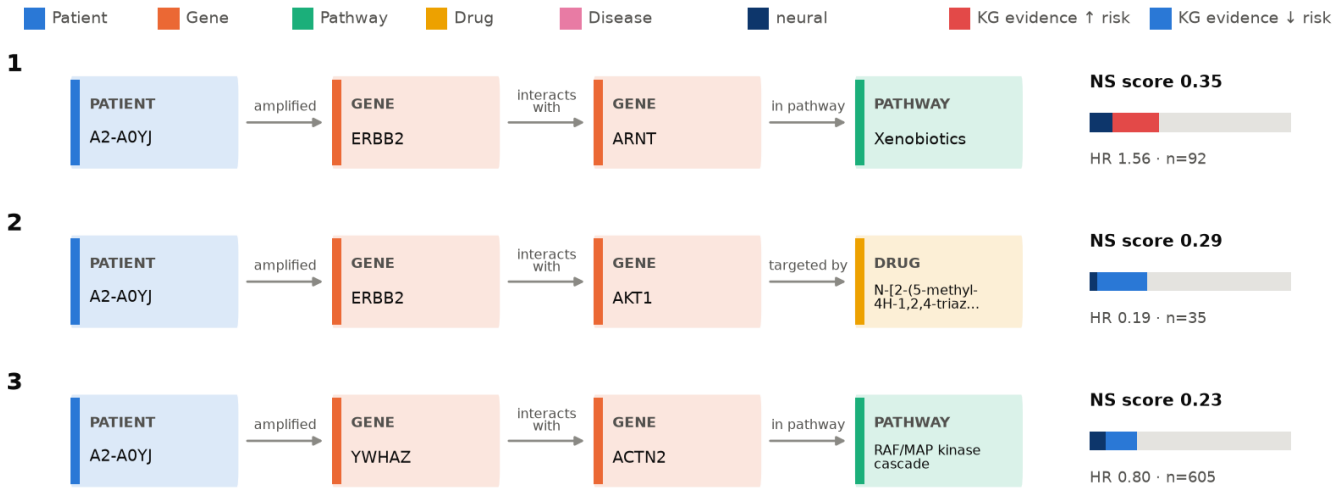
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a Basal tumour, AJCC stage III (T3, N2, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ERBB2** to **Xenobiotics**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.56). Progression was recorded at 18.4 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

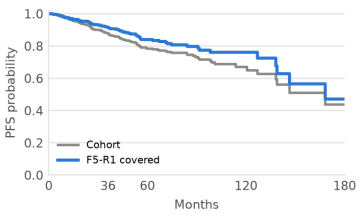


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A2-A0YJ** amplified → **ERBB2** interacts with → **ARNT** in pathway → **Xenobiotics**	19	↑ **faster progression** HR 1.56, z +1.8, n = 92 carriers of “Xenobiotics”	**0.35**
2	**TCGA-A2-A0YJ** amplified → **ERBB2** interacts with → **AKT1** targeted by → **N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine**	6	↓ **slower progression** HR 0.19, z -1.9, n = 35 carriers of “N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine”	**0.29**
3	**TCGA-A2-A0YJ** amplified → **YWHAZ** interacts with → **ACTN2** in pathway → **RAF/MAP kinase cascade**	13	↓ **slower progression** HR 0.80, z -1.2, n = 605 carriers of “RAF/MAP kinase cascade”	**0.23**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

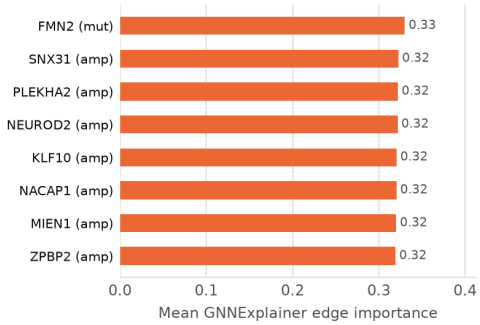
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R342*	57	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
FMN2	Mutation	p.V1487L	0	0.33
SNX31	High-level amplification	GISTIC ±2	0	0.32
PLEKHA2	High-level amplification	GISTIC ±2	1	0.32
NEUROD2	High-level amplification	GISTIC ±2	0	0.32
KLF10	High-level amplification	GISTIC ±2	0	0.32
NACAP1	High-level amplification	GISTIC ±2	0	0.32
MIEN1	High-level amplification	GISTIC ±2	0	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
AZIN1	High-level amplification	GISTIC ±2	1	0.32



66 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1B0	50	III	Her2	0.73	no progression, followed 53.6 mo
TCGA-B6-A1KN	57	III	LumB	0.70	no progression, followed 139.2 mo
TCGA-C8-A27A	48	II	LumB	0.68	no progression, followed 24.6 mo
TCGA-E2-A14Y	35	II	Basal	0.64	no progression, followed 69.3 mo
TCGA-C8-A130	52	III	LumB	0.64	no progression, followed 12.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**40**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-B6-A0WX-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **0.9** · MSIsensor score: **0.18** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

93rd

High risk tier · fused neuro-symbolic model



Probability of progression

21% by 36 mo

12 mo 6% · 24 mo 12% · 60 mo 34%

Model-estimated progression time

≈ 15 months

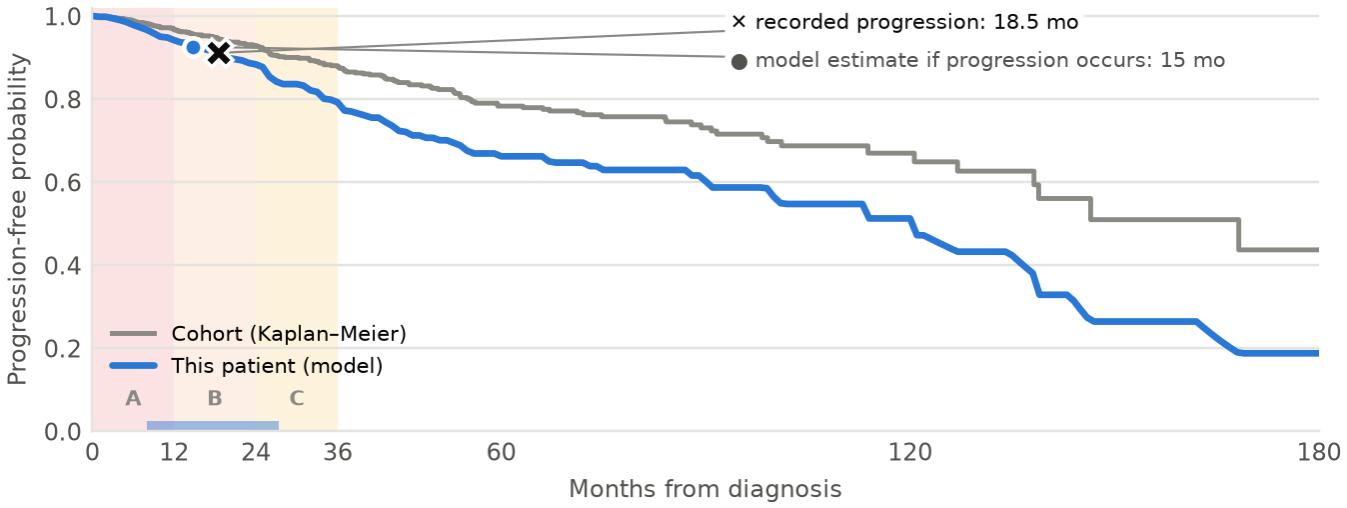
Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

18.5 months

Group B · Progression in year 2

✓ **estimate in recorded window** · |error| 4 mo



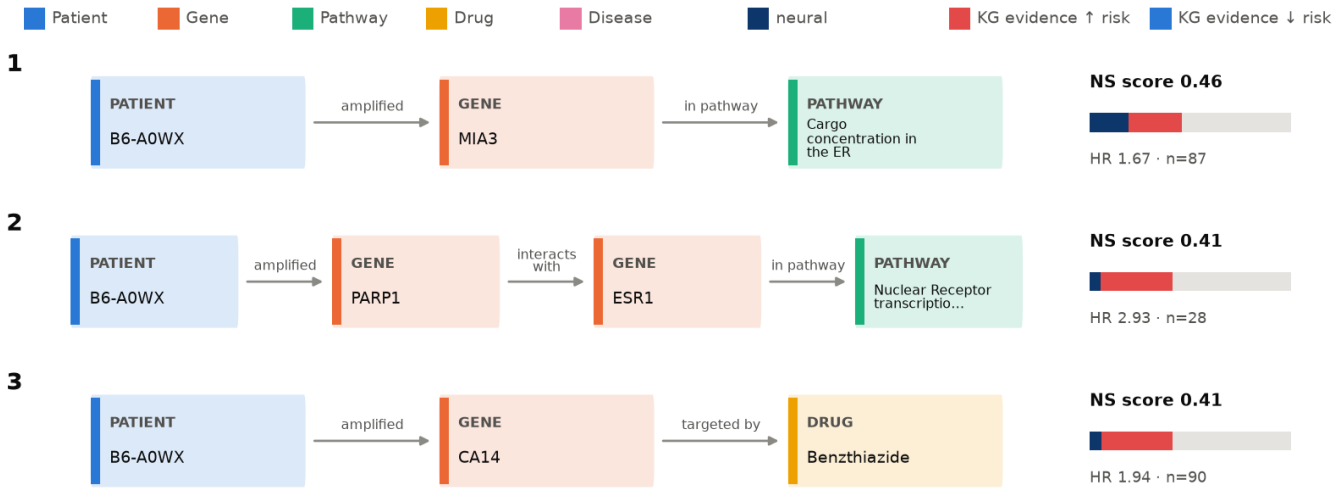
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 40-year-old patient has a Basal tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **93rd risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **MIA3** to **Cargo concentration in the ER**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.67). Progression was recorded at 18.5 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

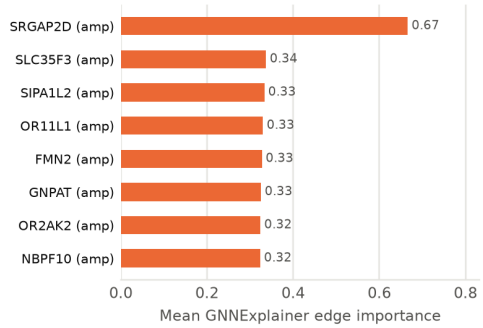
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0WX** amplified → **MIA3** in pathway → **Cargo concentration in the ER**	33	↑ **faster progression** HR 1.67, z +2.0, n = 87 carriers of “Cargo concentration in the ER”	**0.46**
2	**TCGA-B6-A0WX** amplified → **PARP1** interacts with → **ESR1** in pathway → **Nuclear Receptor transcription pathway**	9	↑ **faster progression** HR 2.93, z +2.7, n = 28 carriers of “Nuclear Receptor transcription pathway”	**0.41**
3	**TCGA-B6-A0WX** amplified → **CA14** targeted by → **Benzthiazide**	10	↑ **faster progression** HR 1.94, z +2.6, n = 90 carriers of “Benzthiazide”	**0.41**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.L114*, p.R248W	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.67
SLC35F3	High-level amplification	GISTIC ±2	0	0.34
SIPA1L2	High-level amplification	GISTIC ±2	0	0.33
OR11L1	High-level amplification	GISTIC ±2	1	0.33
FMN2	High-level amplification	GISTIC ±2	0	0.33
GNPAT	High-level amplification	GISTIC ±2	3	0.33
OR2AK2	High-level amplification	GISTIC ±2	1	0.32
NBPF10	High-level amplification	GISTIC ±2	0	0.32
DISP1	High-level amplification	GISTIC ±2	0	0.32
OR2L13	High-level amplification	GISTIC ±2	1	0.32
ZNF496	High-level amplification	GISTIC ±2	1	0.32



110 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); NID1 → Urokinase (DB00013); SV2A → Levetiracetam (DB01202) … and 62 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IQ	40	III	Basal	0.86	no progression, followed 140.9 mo
TCGA-AR-A1AH	51	II	Basal	0.84	progressed at 37.3 mo (group D)
TCGA-BH-A0BA	51	III	LumA	0.83	no progression, followed 37.2 mo
TCGA-A7-A0CE	57	II	Basal	0.82	no progression, followed 35.3 mo
TCGA-AR-A24T	46	III	LumA	0.82	no progression, followed 105.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**35**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1MI**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A3XU-01** · Aneuploidy score: **10** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.6** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

52nd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

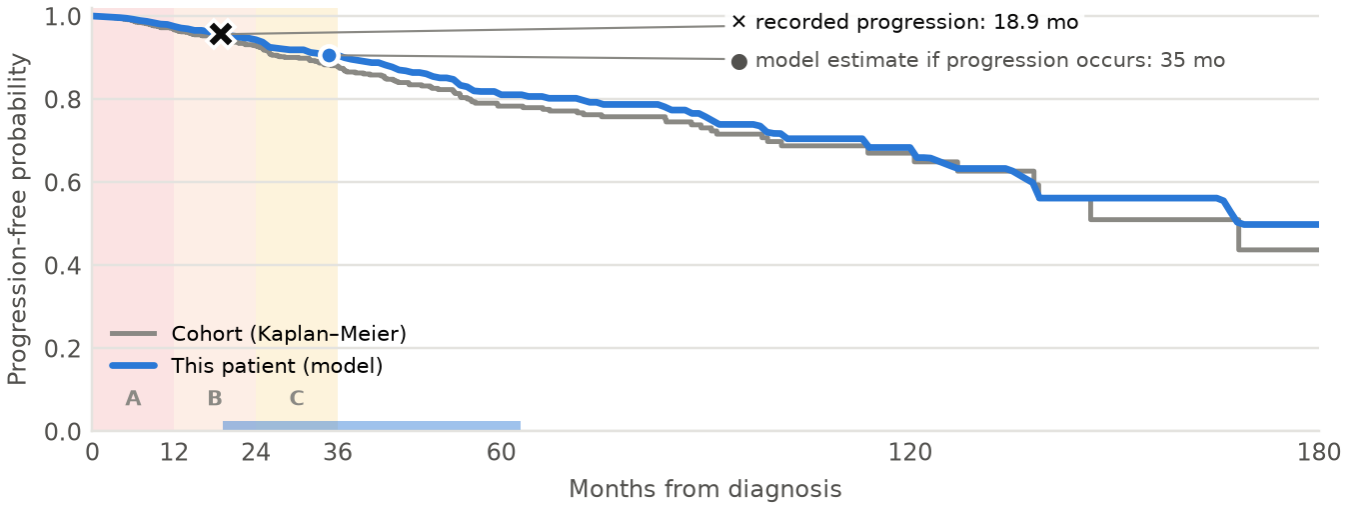
≈ 35 months

Window 24–36 months · IQR 19–63 mo
if progression occurs

Recorded progression

18.9 months

Group B · Progression in year 2
✗ recorded window differs · |error| 16 mo



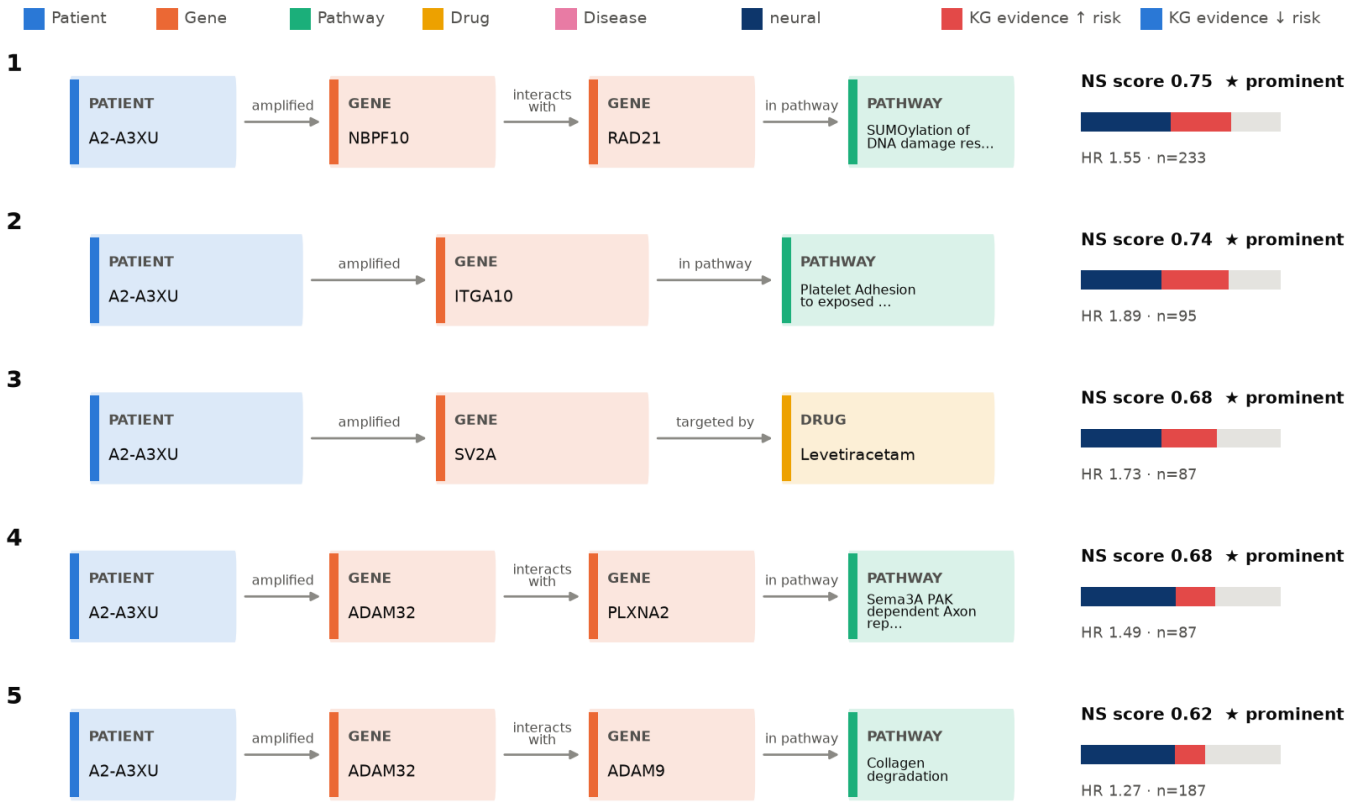
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 35-year-old patient has a Basal tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **52nd risk percentile** (intermediate tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **NBPF10** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.55). Progression was recorded at 18.9 months; the model's estimate given progression was 35 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

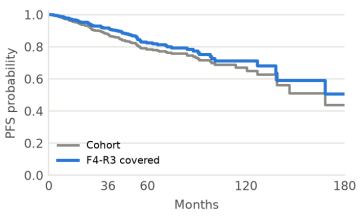


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A3XU** amplified → **NBPF10** interacts with → **RAD21** in pathway → **SUMOylation of DNA damage response and repair proteins**	75	↑ **faster progression** HR 1.55, z +2.3, n = 233 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.75**
2 ★	**TCGA-A2-A3XU** amplified → **ITGA10** in pathway → **Platelet Adhesion to exposed collagen**	67	↑ **faster progression** HR 1.89, z +2.5, n = 95 carriers of “Platelet Adhesion to exposed collagen”	**0.74**
3 ★	**TCGA-A2-A3XU** amplified → **SV2A** targeted by → **Levetiracetam**	68	↑ **faster progression** HR 1.73, z +2.1, n = 87 carriers of “Levetiracetam”	**0.68**
4 ★	**TCGA-A2-A3XU** amplified → **ADAM32** interacts with → **PLXNA2** in pathway → **Sema3A PAK dependent Axon repulsion**	79	↑ **faster progression** HR 1.49, z +1.5, n = 87 carriers of “Sema3A PAK dependent Axon repulsion”	**0.68**
5 ★	**TCGA-A2-A3XU** amplified → **ADAM32** interacts with → **ADAM9** in pathway → **Collagen degradation**	79	↑ **faster progression** HR 1.27, z +1.1, n = 187 carriers of “Collagen degradation”	**0.62**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

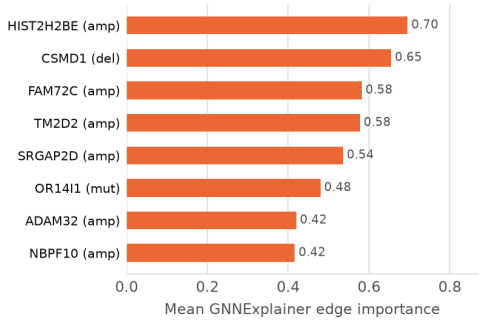
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.70
CSMD1	Homozygous deletion	GISTIC ±2	0	0.65
FAM72C	High-level amplification	GISTIC ±2	0	0.58
TM2D2	High-level amplification	GISTIC ±2	0	0.58
SRGAP2D	High-level amplification	GISTIC ±2	0	0.54
OR14I1	Mutation	p.A26T	1	0.48
ADAM32	High-level amplification	GISTIC ±2	0	0.42
NBPF10	High-level amplification	GISTIC ±2	0	0.42
HAS2	Mutation	p.L407I	1	0.37
HTRA4	High-level amplification	GISTIC ±2	0	0.34
ITGA10	High-level amplification	GISTIC ±2	3	0.34
PDE4DIP	High-level amplification	GISTIC ±2	0	0.32



21 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

SV2A → Levetiracetam (DB01202); SV2A → Brivaracetam (DB05541); SV2A → Seletracetam (DB05885); ADAM9 → INCB7839 (DB05033); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593); PARP1 → Carba-nicotinamide-adenine-dinucleotide (DB02498); PARP1 → NU1025 (DB02690); PARP1 → Nicotinamide (DB02701); PARP1 → 2-{3-[4-(4-Fluorophenyl)-3,6-Dihydro-1(2h)-Pyridinyl]Propyl}-8-Methyl-4(3h)-Quinazolinone (DB03072) … and 16 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1JM	59	II	Basal	0.66	no progression, followed 19.4 mo
TCGA-S3-AA15	51	II	Basal	0.66	no progression, followed 17.3 mo
TCGA-E2-A1LI	57	II	Basal	0.63	no progression, followed 102.6 mo
TCGA-A1-A0SO	67	II	Basal	0.63	no progression, followed 28.0 mo
TCGA-BH-A1EW	38	II	LumA	0.63	progressed at 55.7 mo (group D)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

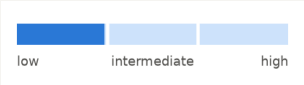
Sample: **TCGA-A7-A13G-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **0.18** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

32nd

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

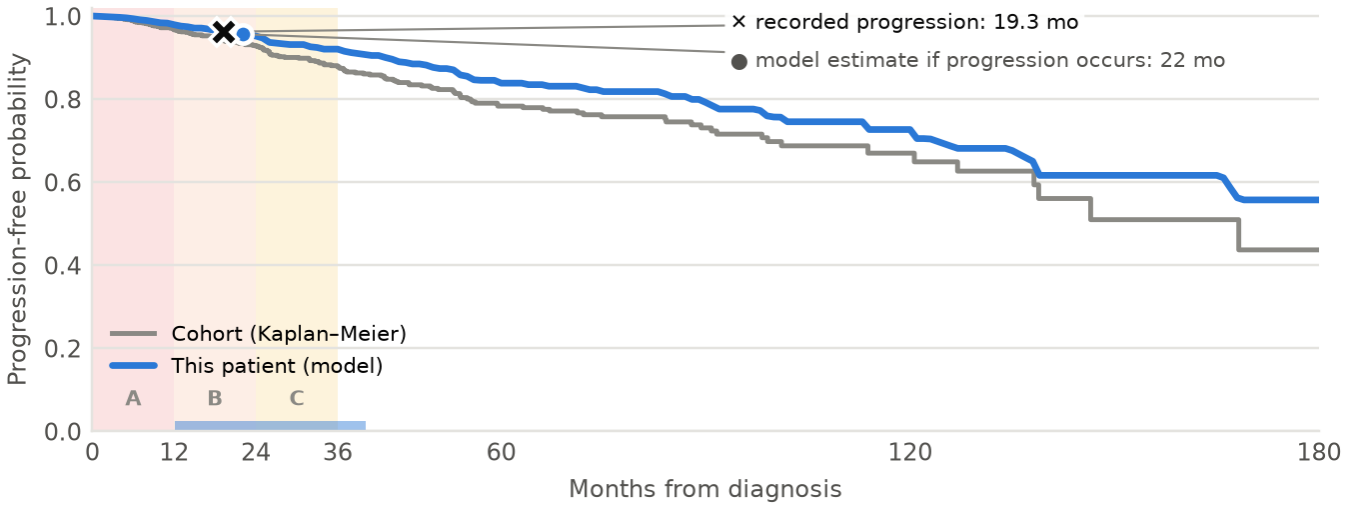
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

19.3 months

■ **Group B** · Progression in year 2
✓ **estimate in recorded window** · |error| 3 mo



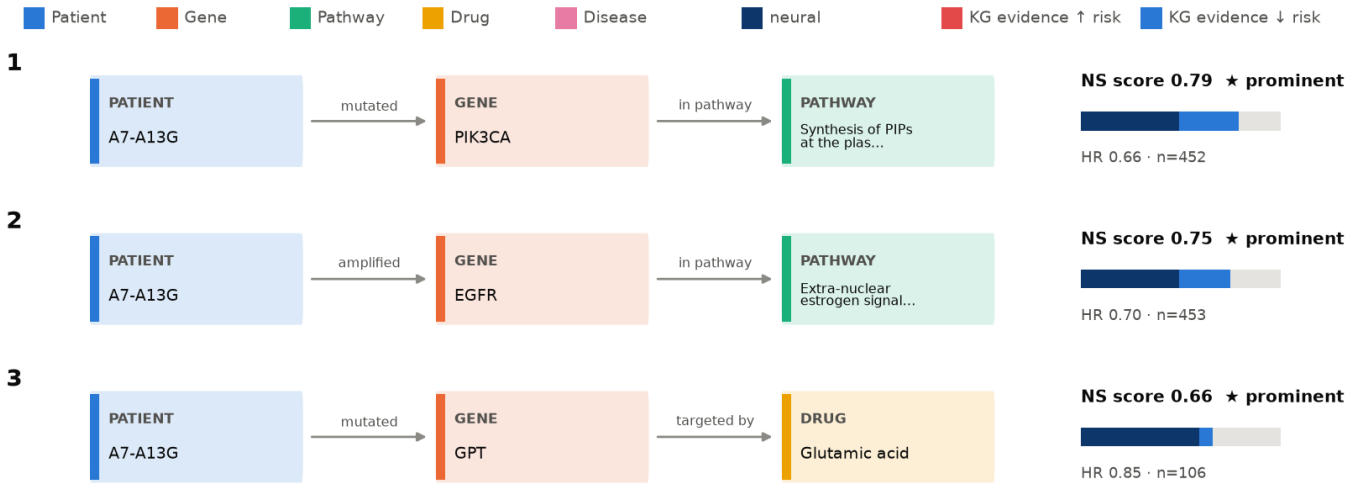
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage II (T2, N0, MX). The fused neuro-symbolic model places them in the **32nd risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND TMB (nonsynonymous) ≥ 0.38 AND Age ≥ 40.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **Synthesis of PIPs at the plasma membrane**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.66). Progression was recorded at 19.3 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

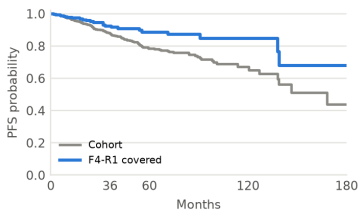


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A7-A13G** mutated → **PIK3CA** in pathway → **Synthesis of PIPs at the plasma membrane**	82	↓ slower progression HR 0.66, z -2.2, n = 452 carriers of “Synthesis of PIPs at the plasma membrane”	**0.79**
2 ★	**TCGA-A7-A13G** amplified → **EGFR** in pathway → **Extra-nuclear estrogen signaling**	82	↓ slower progression HR 0.70, z -1.9, n = 453 carriers of “Extra-nuclear estrogen signaling”	**0.75**
3 ★	**TCGA-A7-A13G** mutated → **GPT** targeted by → **Glutamic acid**	99	↓ slower progression HR 0.85, z -0.5, n = 106 carriers of “Glutamic acid”	**0.66**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

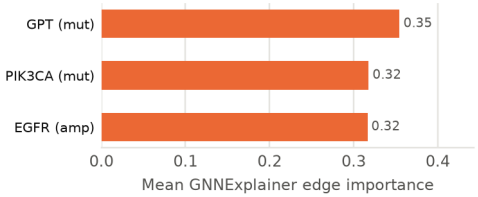
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R1 ↓ lower risk	N stage = N0 AND TMB (nonsynonymous) ≥ 0.38 AND Age ≥ 40.5	355 pts 26 events	**0.39** p 7.7e-06	not reached
F4-R2 ↓ lower risk	N stage = N0 AND 34.5 ≤ Age < 83.5	386 pts 32 events	**0.43** p 2.5e-05	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.A1046=, p.H1047R	70	0.32
EGFR · driver	High-level amplification	GISTIC ±2	73	0.32
GPT	Mutation	p.A114=	5	0.35



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); EGFR → Cetuximab (DB00002); EGFR → Lidocaine (DB00281); EGFR → Gefitinib (DB00317) … and 32 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1RD	67	II	LumA	0.77	no progression, followed 1.1 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XY	74	II	LumA	0.77	no progression, followed 16.5 mo
TCGA-C8-A1HE	59	II	LumA	0.76	no progression, followed 12.3 mo
TCGA-AN-A0XL	61	II	LumA	0.76	no progression, followed 5.4 mo
TCGA-OL-A66P	75	II	Her2	0.76	no progression, followed 14.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**47**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8510/3**

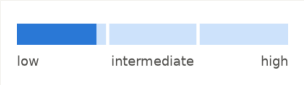
Sample: **TCGA-AR-A1AI-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **2.33** · MSIsensor score: **0.79** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

29th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

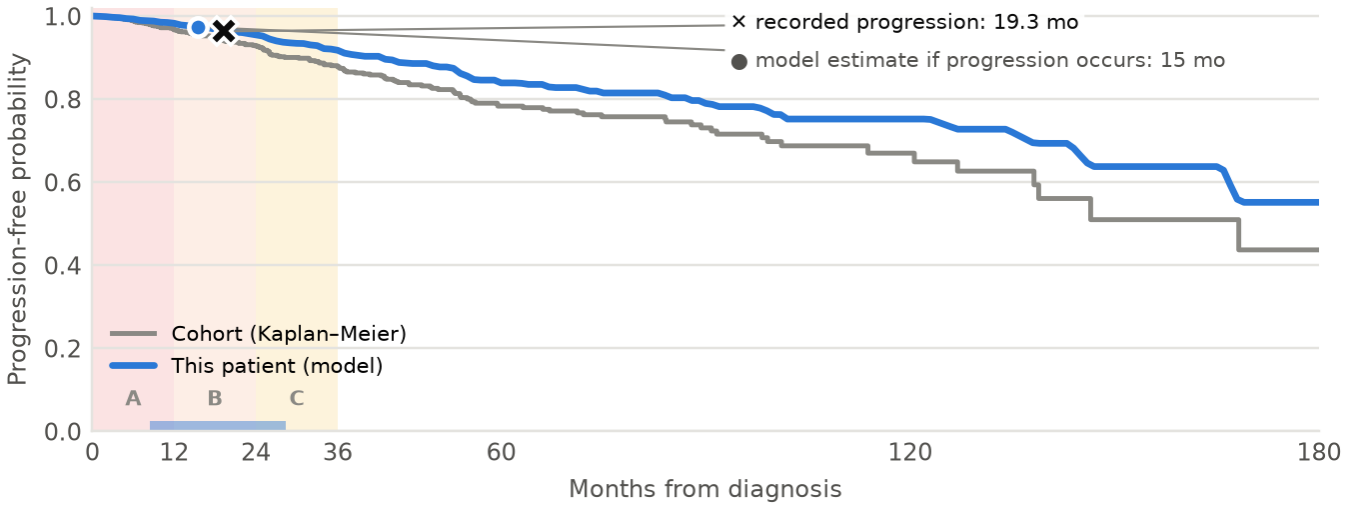
≈ 15 months

Window 12–24 months · IQR 8–28 mo if progression occurs

Recorded progression

19.3 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 4 mo



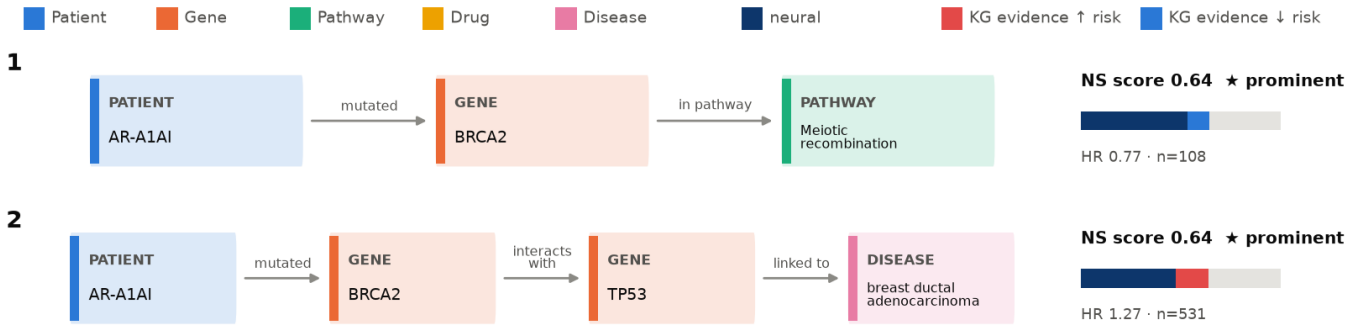
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 47-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **29th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **BRCA2** to **Meiotic recombination**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.77). Progression was recorded at 19.3 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

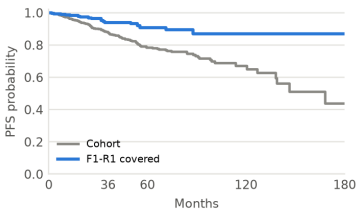


2 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AR-A1AI** mutated → **BRCA2** in pathway → **Meiotic recombination**	89	↓ **slower progression** HR 0.77, z -0.8, n = 108 carriers of “Meiotic recombination”	**0.64**
2 ★	**TCGA-AR-A1AI** mutated → **BRCA2** interacts with → **TP53** linked to → **breast ductal adenocarcinoma**	79	↑ **faster progression** HR 1.27, z +1.2, n = 531 carriers of “breast ductal adenocarcinoma”	**0.64**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

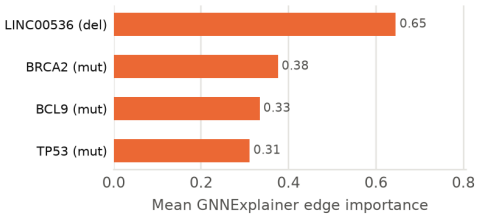
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
BRCA2 · driver	Mutation	p.R2336C	14	0.38
TP53 · driver	Mutation	p.F270I	57	0.31
LINC00536	Homozygous deletion	GISTIC ±2	0	0.65
BCL9	Mutation	p.F1204L	3	0.33



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XX	49	II	Basal	0.94	no progression, followed 47.3 mo
TCGA-E2-A1B6	44	II	Basal	0.92	no progression, followed 28.5 mo
TCGA-C8-A12V	55	II	Basal	0.91	no progression, followed 12.7 mo
TCGA-BH-A0RX	59	II	Basal	0.90	no progression, followed 5.6 mo
TCGA-E9-A1N8	48	II	Basal	0.90	no progression, followed 34.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**ADMIX**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N3**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-E2-A2P5-01** · Aneuploidy score: **26** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

21% by 36 mo

12 mo 6% · 24 mo 13% · 60 mo 36%

Model-estimated progression time

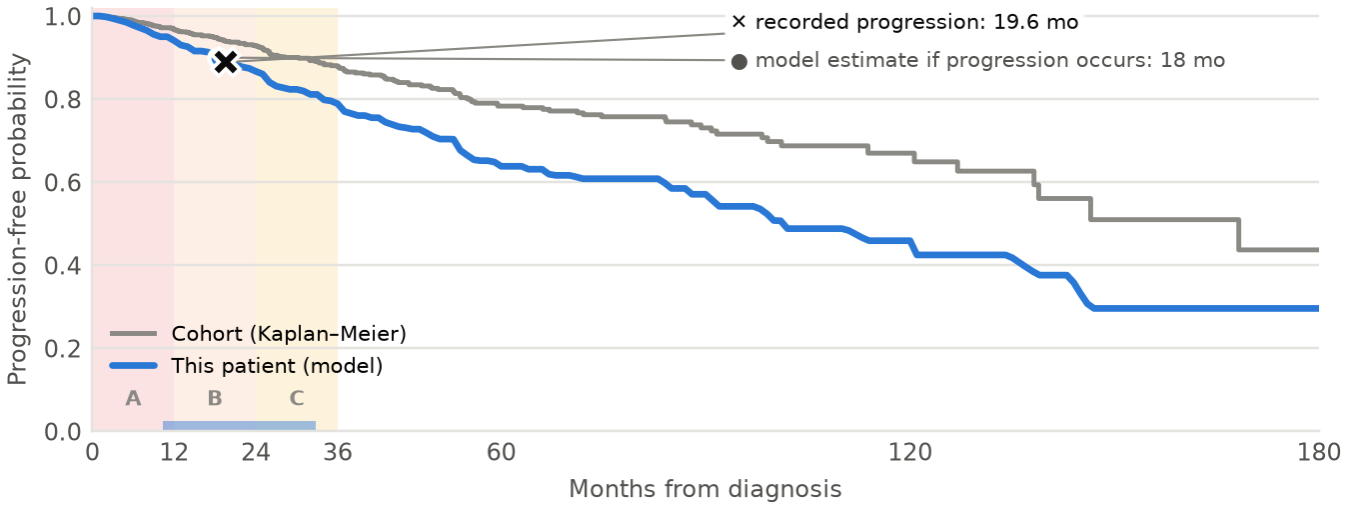
≈ 18 months

Window 12–24 months · IQR 10–33 mo
if progression occurs

Recorded progression

19.6 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 1 mo



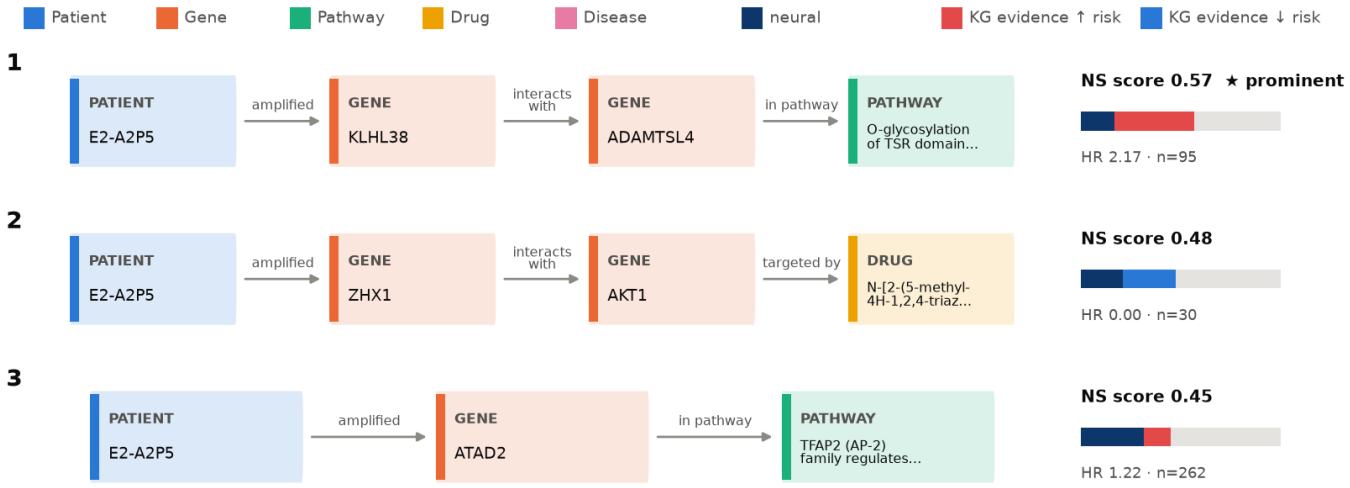
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a subtype-unassigned tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **KLHL38** to **O-glycosylation of TSR domain-containing proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.17). Progression was recorded at 19.6 months; the model's estimate given progression was 18 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

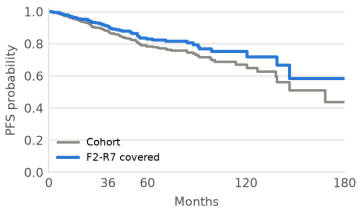


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A2P5** amplified → **KLHL38** interacts with → **ADAMTSL4** in pathway → **O-glycosylation of TSR domain-containing proteins**	28	↑ **faster progression** HR 2.17, z +3.3, n = 95 carriers of “O-glycosylation of TSR domain-containing proteins”	**0.57**
2	**TCGA-E2-A2P5** amplified → **ZHX1** interacts with → **AKT1** targeted by → **N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine**	35	↓ **slower progression** HR 0.00, z -2.0, n = 30 carriers of “N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine”	**0.48**
3	**TCGA-E2-A2P5** amplified → **ATAD2** in pathway → **TFAP2 (AP-2) family regulates transcription of growth factors and their receptors**	53	↑ **faster progression** HR 1.22, z +1.0, n = 262 carriers of “TFAP2 (AP-2) family regulates transcription of growth factors and their receptors”	**0.45**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

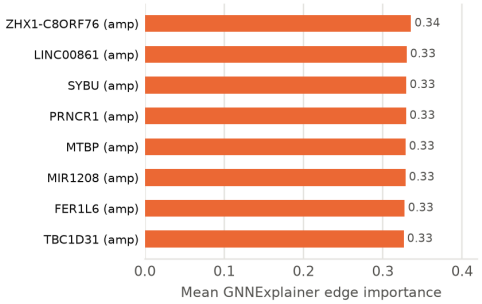
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ **lower risk**	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
ZHX1-C8ORF76	High-level amplification	GISTIC ±2	0	0.34
LINC00861	High-level amplification	GISTIC ±2	0	0.33
SYBU	High-level amplification	GISTIC ±2	0	0.33
PRNCR1	High-level amplification	GISTIC ±2	0	0.33
MTBP	High-level amplification	GISTIC ±2	0	0.33
MIR1208	High-level amplification	GISTIC ±2	0	0.33
FER1L6	High-level amplification	GISTIC ±2	0	0.33
TBC1D31	High-level amplification	GISTIC ±2	0	0.33
EFR3A	High-level amplification	GISTIC ±2	0	0.33
ZNF572	High-level amplification	GISTIC ±2	0	0.32
HAS2-AS1	High-level amplification	GISTIC ±2	0	0.32



89 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); NDUFB9 → NADH (DB00157); TG → Iodine (DB05382); SQLE → Naftifine (DB00735); SQLE → Terbinafine (DB00857); SQLE → Butenafine (DB01091); SQLE → Ellagic acid (DB08846); KCNQ3 → Amitriptyline (DB00321); KCNQ3 → Diclofenac (DB00586) … and 4 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A3Z5	54	III	missing	0.65	no progression, followed 33.4 mo
TCGA-E2-A3DX	43	III	missing	0.60	no progression, followed 43.6 mo
TCGA-A2-A4RW	49	III	missing	0.59	no progression, followed 7.3 mo
TCGA-E9-A3Q9	78	III	missing	0.59	no progression, followed 32.9 mo
TCGA-A8-A09N	57	III	LumB	0.58	no progression, followed 1.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18J · Group B

Report date 14 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**56**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**—**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A18J-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **2.5** · MSIsensor score: **0.08** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

60% by 36 mo

12 mo 17% · 24 mo 39% · 60 mo 85%

Model-estimated progression time

≈ 44 months

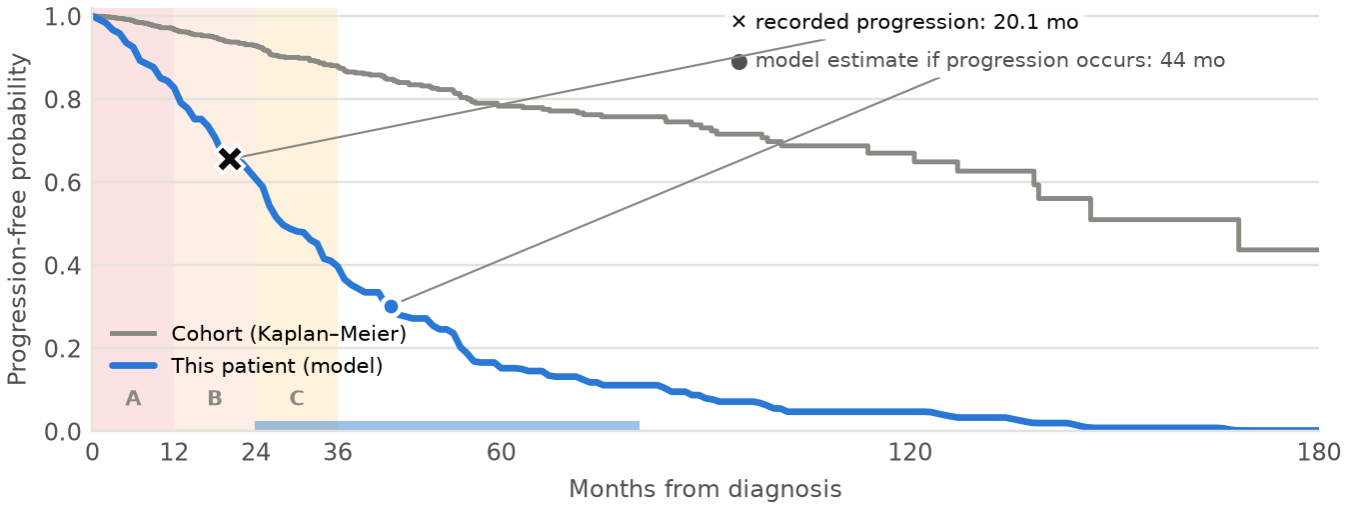
Window > 36 months · IQR 24–80 mo if progression occurs

Recorded progression

20.1 months

Group B · Progression in year 2

✗ recorded window differs · |error| 24 mo



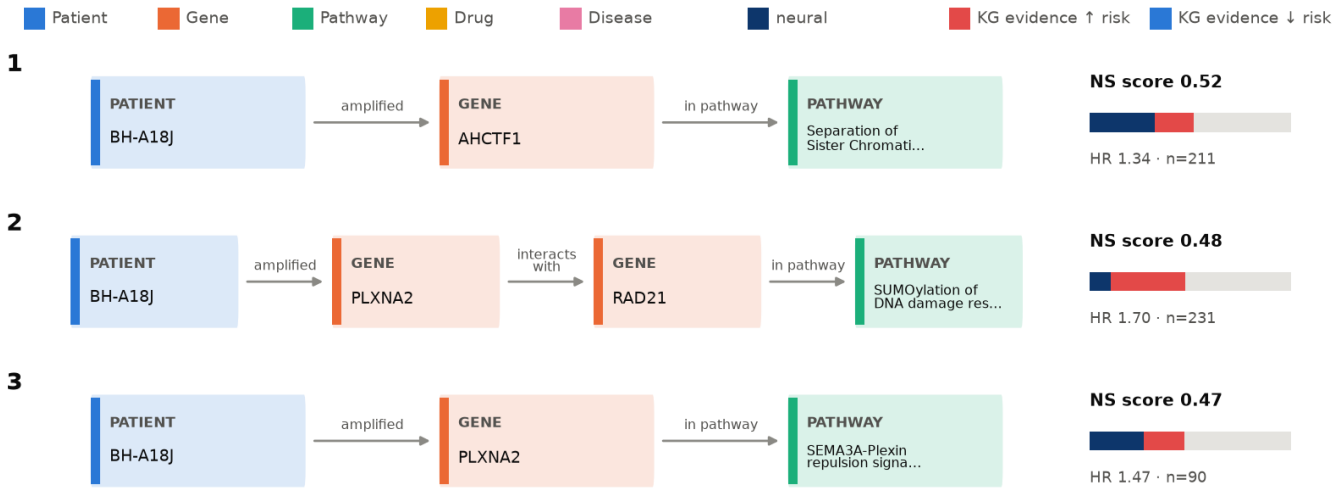
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 56-year-old patient has a LumA tumour, AJCC stage IV (T4, N2, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **AHCTF1** to **Separation of Sister Chromatids**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.34). Progression was recorded at 20.1 months; the model's estimate given progression was 44 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

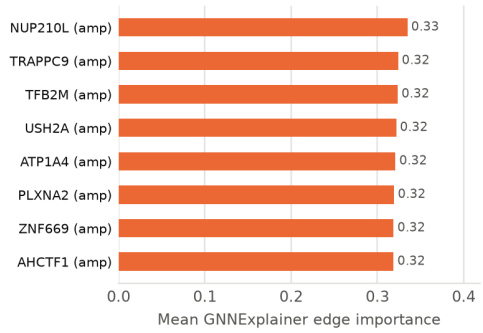
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-BH-A18J** amplified → **AHCTF1** in pathway → **Separation of Sister Chromatids**	54	↑ **faster progression** HR 1.34, z +1.5, n = 211 carriers of “Separation of Sister Chromatids”	**0.52**
2	**TCGA-BH-A18J** amplified → **PLXNA2** interacts with → **RAD21** in pathway → **SUMOylation of DNA damage response and repair proteins**	17	↑ **faster progression** HR 1.70, z +2.8, n = 231 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.48**
3	**TCGA-BH-A18J** amplified → **PLXNA2** in pathway → **SEMA3A-Plexin repulsion signaling by inhibiting Integrin adhesion**	45	↑ **faster progression** HR 1.47, z +1.5, n = 90 carriers of “SEMA3A-Plexin repulsion signaling by inhibiting Integrin adhesion”	**0.47**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q318E	4	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.31
NUP210L	High-level amplification	GISTIC ±2	0	0.33
TRAPPC9	High-level amplification	GISTIC ±2	2	0.32
TFB2M	High-level amplification	GISTIC ±2	2	0.32
USH2A	High-level amplification	GISTIC ±2	0	0.32
ATP1A4	High-level amplification	GISTIC ±2	4	0.32
PLXNA2	High-level amplification	GISTIC ±2	4	0.32
ZNF669	High-level amplification	GISTIC ±2	1	0.32
AHCTF1	High-level amplification	GISTIC ±2	7	0.32
ZNF124	High-level amplification	GISTIC ±2	1	0.32
FAM72C	High-level amplification	GISTIC ±2	0	0.32



40 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); ATP1A4 → Rubidium Rb-82 (DB09479); PARP1 → Theophylline (DB00277); PARP1 → Zinc (DB01593) … and 22 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0CS	73	IV	LumA	0.61	no progression, followed 77.2 mo
TCGA-AN-A0XP	69	III	LumA	0.60	no progression, followed 0.3 mo
TCGA-A8-A09B	58	III	LumA	0.56	no progression, followed 12.0 mo
TCGA-A8-A08T	64	IV	LumA	0.54	progressed at 84.1 mo (group D)
TCGA-AO-A12A	47	II	LumA	0.52	no progression, followed 102.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A2-A0SW · Group B

Report date 14 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**82**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0SW-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **1.73** · MSIsensor score: **1.59** · AJCC edition: —

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

70% by 36 mo

12 mo 27% · 24 mo 47% · 60 mo 88%

Model-estimated progression time

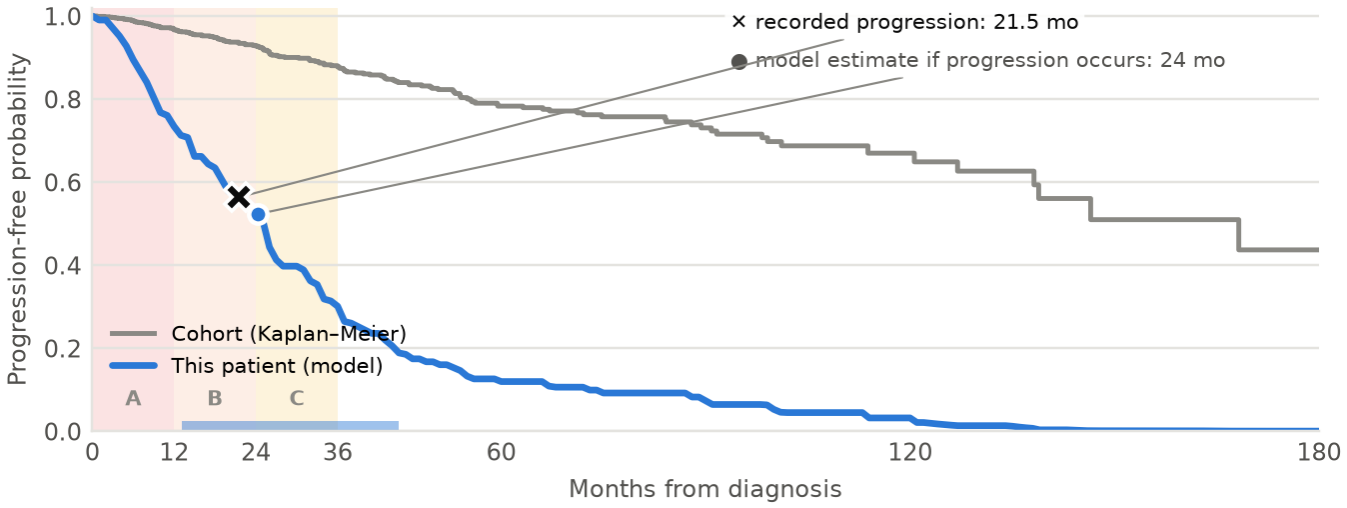
≈ 24 months

Window 24–36 months · IQR 13–45 mo
if progression occurs

Recorded progression

21.5 months

Group B · Progression in year 2
✗ recorded window differs · |error| 3 mo



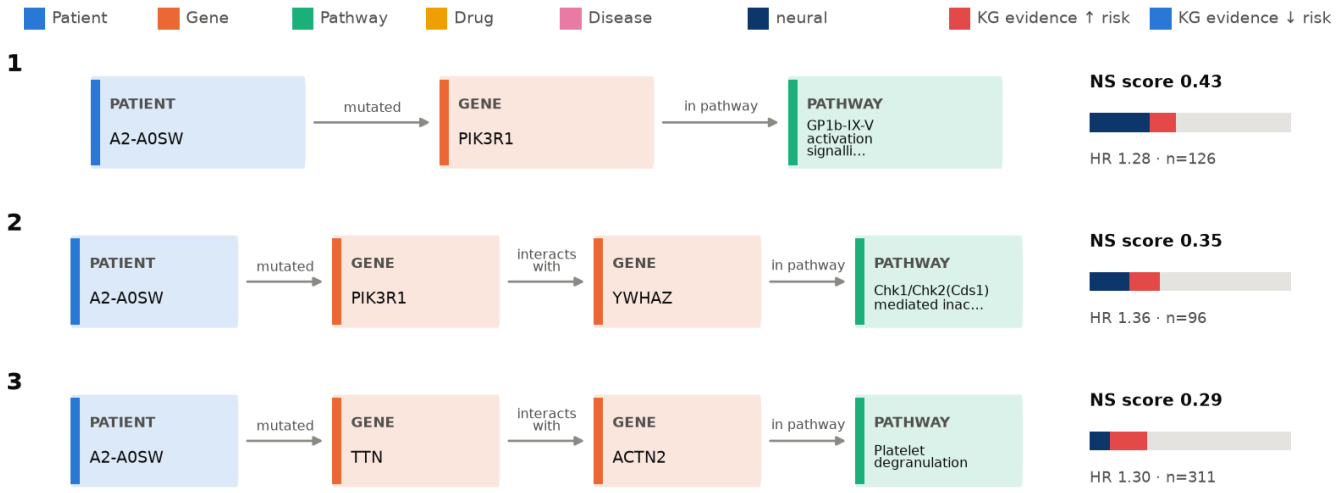
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 82-year-old patient has a LumB tumour, AJCC stage IV (T2, N2, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PIK3R1** to **GP1b-IX-V activation signalling**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.28). Progression was recorded at 21.5 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

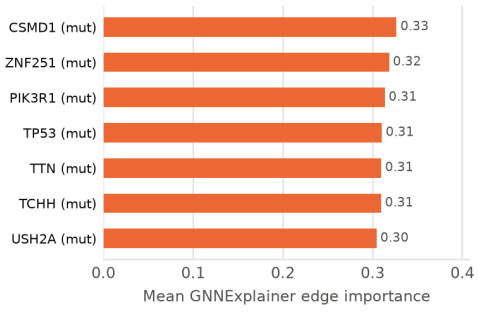
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A2-A0SW** mutated → **PIK3R1** in pathway → **GP1b-IX-V activation signalling**	50	↑ **faster progression** HR 1.28, z +1.0, n = 126 carriers of “GP1b-IX-V activation signalling”	**0.43**
2	**TCGA-A2-A0SW** mutated → **PIK3R1** interacts with → **YWHAZ** in pathway → **Chk1/Chk2(Cds1) mediated inactivation of Cyclin B:Cdk1 complex**	33	↑ **faster progression** HR 1.36, z +1.1, n = 96 carriers of “Chk1/Chk2(Cds1) mediated inactivation of Cyclin B:Cdk1 complex”	**0.35**
3	**TCGA-A2-A0SW** mutated → **TTN** interacts with → **ACTN2** in pathway → **Platelet degranulation**	17	↑ **faster progression** HR 1.30, z +1.4, n = 311 carriers of “Platelet degranulation ”	**0.29**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3R1 · driver	Mutation	p.E468_E469insGLYE	79	0.31
TP53 · driver	Mutation	p.R273C	57	0.31
CSMD1	Mutation	p.V2880=	0	0.33
ZNF251	Mutation	p.F265L	0	0.32
TTN	Mutation	p.*3454*	2	0.31
TCHH	Mutation	p.R68C	1	0.31
USH2A	Mutation	p.Q4382K	0	0.30



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3R1 → Isoprenaline (DB01064); PIK3R1 → SF1126 (DB05210); PIK3R1 → Enzastaurin (DB06486); PIK3R1 → Wortmannin (DB08059); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 2 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A07W	76	IV	LumB	0.73	no progression, followed 10.0 mo
TCGA-A7-A56D	84	II	LumA	0.70	no progression, followed 14.7 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0BF	56	II	LumB	0.64	no progression, followed 43.5 mo
TCGA-E9-A226	45	III	LumB	0.64	progressed at 26.0 mo (group C)
TCGA-AR-A0U0	73	II	Basal	0.64	no progression, followed 65.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**84**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

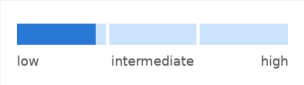
Sample: **TCGA-AC-A2B8-01** · Aneuploidy score: **3** · TMB (nonsyn., /Mb): **2.77** · MSIsensor score: **0.54** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

29th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

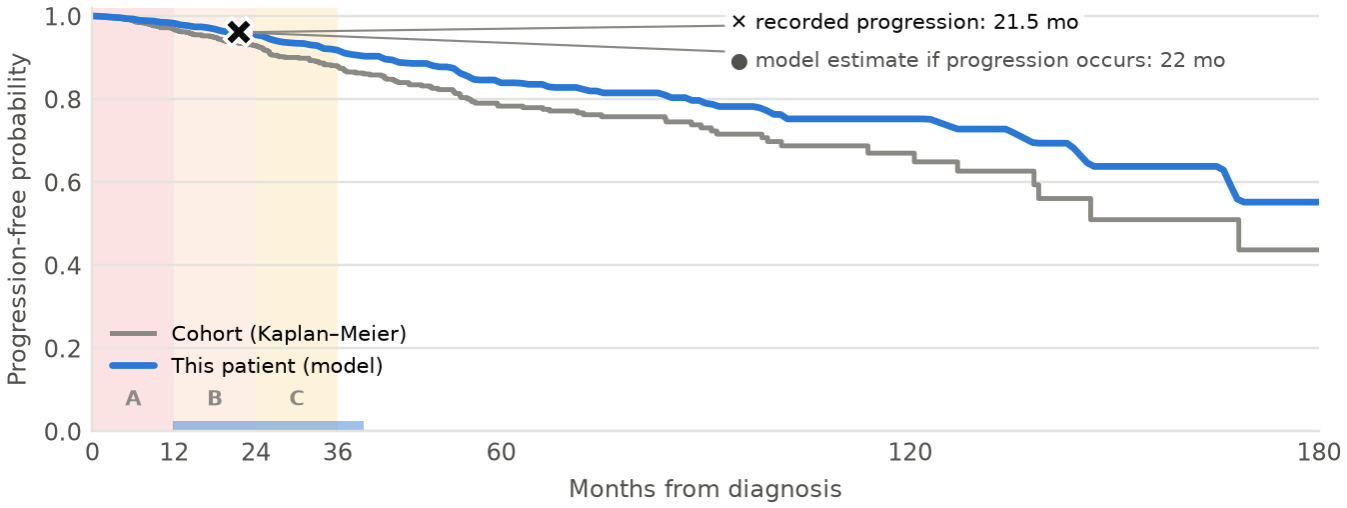
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

21.5 months

■ **Group B** · Progression in year 2
✓ **estimate in recorded window** · |error| 0 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 84-year-old patient has a LumA tumour, AJCC stage II (T3, N0, M0). The fused neuro-symbolic model places them in the **29th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ZP4** to **Interaction With Cumulus Cells And The Zona Pellucida**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.28). Progression was recorded at 21.5 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

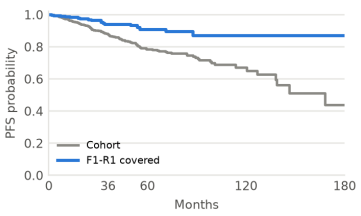


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AC-A2B8** mutated → **ZP4** in pathway → **Interaction With Cumulus Cells And The Zona Pellucida**	99	↑ **faster progression** HR 1.28, z +1.2, n = 183 carriers of “Interaction With Cumulus Cells And The Zona Pellucida”	**0.75**
2 ★	**TCGA-AC-A2B8** mutated → **SF3B1** targeted by → **H3B-8800**	100	↓ **slower progression** HR 0.39, z -1.0, n = 23 carriers of “H3B-8800”	**0.73**
3 ★	**TCGA-AC-A2B8** mutated → **SF3B1** interacts with → **MYC** in pathway → **Constitutive Signaling by NOTCH1 PEST Domain Mutants**	85	↑ **faster progression** HR 1.38, z +1.4, n = 129 carriers of “Constitutive Signaling by NOTCH1 PEST Domain Mutants”	**0.70**
4 ★	**TCGA-AC-A2B8** mutated → **FLG2** in pathway → **Neutrophil degranulation**	76	↑ **faster progression** HR 1.36, z +1.7, n = 316 carriers of “Neutrophil degranulation”	**0.68**
5 ★	**TCGA-AC-A2B8** mutated → **TTN** interacts with → **TCAP** in pathway → **Striated Muscle Contraction**	83	↑ **faster progression** HR 1.17, z +0.8, n = 329 carriers of “Striated Muscle Contraction”	**0.61**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

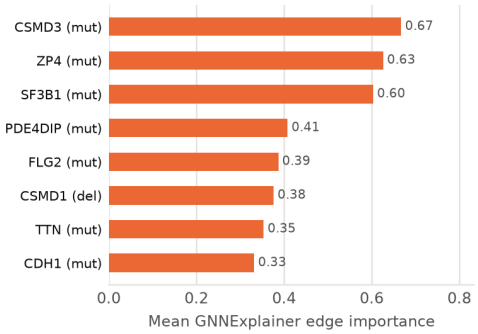
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
SF3B1 · driver	Mutation	p.E622Q	4	0.60
CDH1 · driver	Mutation	p.C603del	17	0.33
PIK3CA · driver	Mutation	p.E545K	70	0.33
CSMD3	Mutation	p.T1007=	0	0.67
ZP4	Mutation	p.R149W	1	0.63
PDE4DIP	Mutation	p.H438=	0	0.41
FLG2	Mutation	p.G965A, p.Q944H	1	0.39
CSMD1	Homozygous deletion	GISTIC ±2	0	0.38
TTN	Mutation	p.R23443P	2	0.35



9 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

SF3B1 → H3B-8800 (DB14017); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-PE-A5DD	64	II	LumA	0.82	no progression, followed 64.2 mo
TCGA-AR-A5QM	62	II	LumA	0.81	progressed at 31.8 mo (group C)
TCGA-A2-A0CR	54	II	LumA	0.78	no progression, followed 107.9 mo
TCGA-E2-A10F	47	II	LumA	0.77	no progression, followed 28.9 mo
TCGA-A2-A0YD	63	II	LumA	0.77	no progression, followed 25.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**73**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-LQ-A4E4-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **—** · MSIsensor score: **0.06** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 10% · 60 mo 32%

Model-estimated progression time

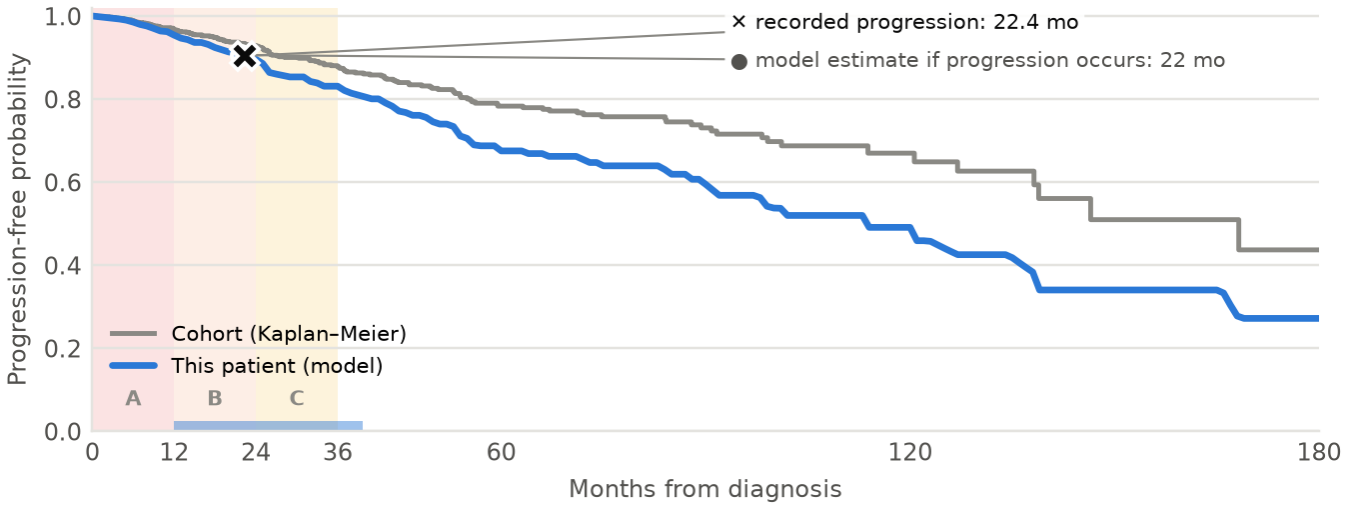
≈ 22 months

Window 12–24 months · IQR 12–40 mo
if progression occurs

Recorded progression

22.4 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 1 mo



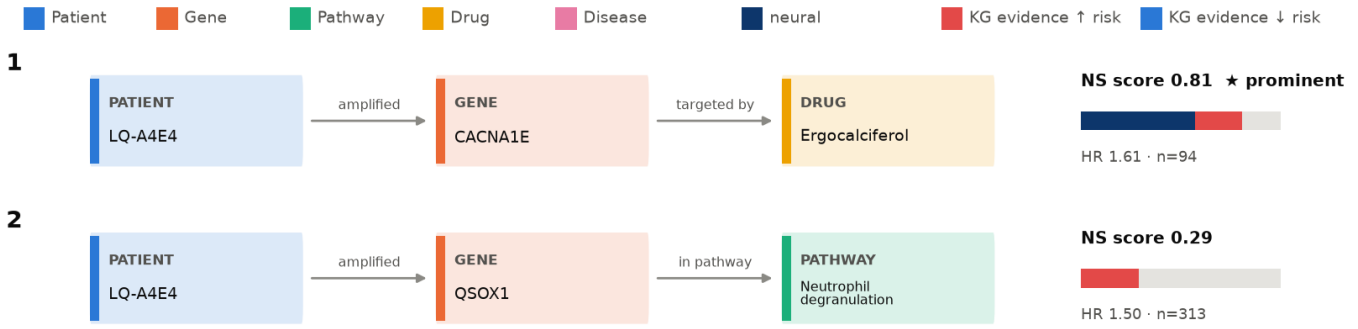
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 73-year-old patient has a subtype-unassigned tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **CACNA1E** to **Ergocalciferol**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.61). Progression was recorded at 22.4 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 16 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

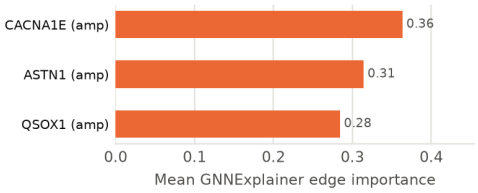
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-LQ-A4E4** amplified → **CACNA1E** targeted by → **Ergocalciferol**	96	↑ **faster progression** HR 1.61, z +1.7, n = 94 carriers of “Ergocalciferol”	**0.81**
2	**TCGA-LQ-A4E4** amplified → **QSOX1** in pathway → **Neutrophil degranulation**	0	↑ **faster progression** HR 1.50, z +2.2, n = 313 carriers of “Neutrophil degranulation”	**0.29**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CACNA1E	High-level amplification	GISTIC ±2	12	0.36
ASTN1	High-level amplification	GISTIC ±2	0	0.31
QSOX1	High-level amplification	GISTIC ±2	4	0.28



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CACNA1E → Ergocalciferol (DB00153); CACNA1E → Enflurane (DB00228); CACNA1E → Topiramate (DB00273); CACNA1E → Spironolactone (DB00421); CACNA1E → Lamotrigine (DB00555); CACNA1E → Calcium citrate (DB11093); CACNA1E → Butamben (DB11148); CACNA1E → Calcium Phosphate (DB11348); CACNA1E → Bioallethrin (DB13746); CACNA1E → Calcium phosphate dihydrate (DB14481). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AC-A2QI	76	III	missing	0.93	no progression, followed 19.3 mo
TCGA-AC-A3W6	90	III	missing	0.90	no progression, followed 19.8 mo
TCGA-EW-A424	51	III	missing	0.88	no progression, followed 23.5 mo
TCGA-AC-A2FE	62	III	missing	0.74	progressed at 25.2 mo (group C)
TCGA-AC-A3TM	50	III	missing	0.72	no progression, followed 25.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A2-A3XS · Group B

Report date 14 Oct 2026

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N2A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A3XS-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.38** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

21% by 36 mo

12 mo 6% · 24 mo 13% · 60 mo 36%

Model-estimated progression time

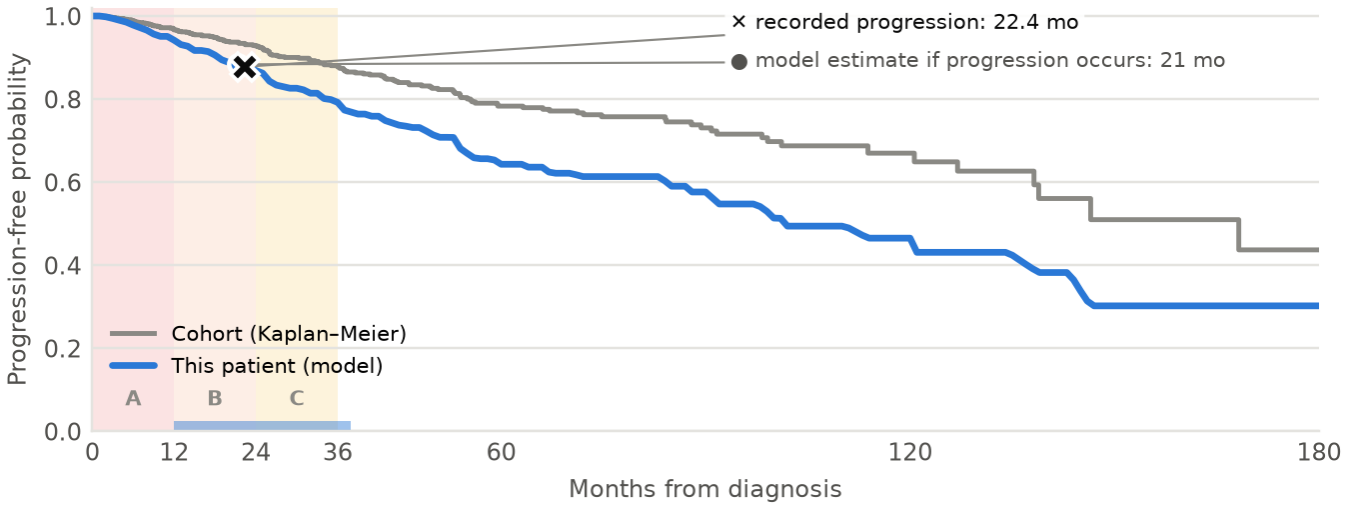
≈ 21 months

Window 12–24 months · IQR 12–38 mo
if progression occurs

Recorded progression

22.4 months

Group B · Progression in year 2
✓ **estimate in recorded window** · |error| 1 mo



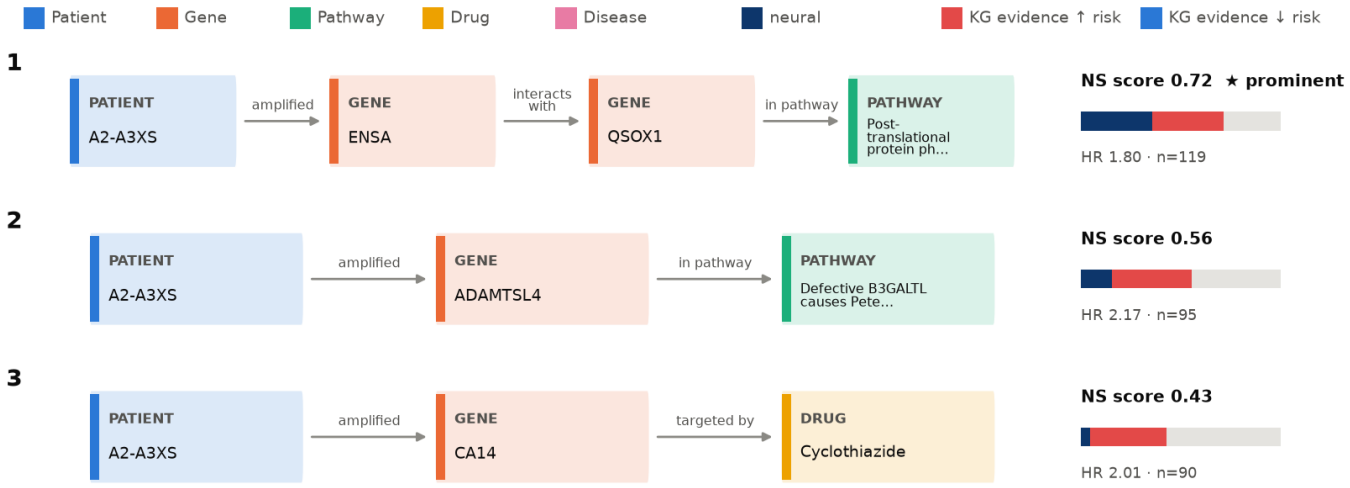
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a subtype-unassigned tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ENSA** to **Post-translational protein phosphorylation**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.80). Progression was recorded at 22.4 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

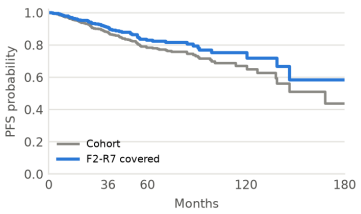


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A3XS** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Post-translational protein phosphorylation**	60	↑ **faster progression** HR 1.80, z +2.7, n = 119 carriers of “Post-translational protein phosphorylation”	**0.72**
2	**TCGA-A2-A3XS** amplified → **ADAMTSL4** in pathway → **Defective B3GALTL causes Peters-plus syndrome (PpS)**	26	↑ **faster progression** HR 2.17, z +3.3, n = 95 carriers of “Defective B3GALTL causes Peters-plus syndrome (PpS)”	**0.56**
3	**TCGA-A2-A3XS** amplified → **CA14** targeted by → **Cyclothiazide**	8	↑ **faster progression** HR 2.01, z +2.8, n = 90 carriers of “Cyclothiazide”	**0.43**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

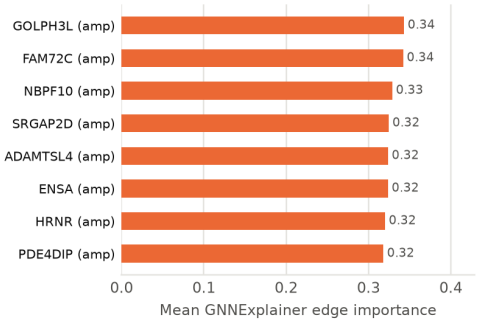
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ **lower risk**	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GOLPH3L	High-level amplification	GISTIC ±2	0	0.34
FAM72C	High-level amplification	GISTIC ±2	0	0.34
NBPF10	High-level amplification	GISTIC ±2	0	0.33
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
ADAMTSL4	High-level amplification	GISTIC ±2	2	0.32
ENSA	High-level amplification	GISTIC ±2	1	0.32
HRNR	High-level amplification	GISTIC ±2	3	0.32
PDE4DIP	High-level amplification	GISTIC ±2	0	0.32
FLG	High-level amplification	GISTIC ±2	1	0.32
BCL9	High-level amplification	GISTIC ±2	3	0.32
ECM1	High-level amplification	GISTIC ±2	1	0.32
SV2A	High-level amplification	GISTIC ±2	7	0.32



24 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

HRNR → Zinc (DB01593); HRNR → Zinc acetate (DB14487); SV2A → Levetiracetam (DB01202); SV2A → Brivaracetam (DB05541); SV2A → Seletracetam (DB05885); TARS2 → Threonine (DB00156); CA14 → Benzthiazide (DB00562); CA14 → Cyclothiazide (DB00606); CA14 → Acetazolamide (DB00819); CA14 → Zonisamide (DB00909) … and 1 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A400	43	III	missing	0.88	no progression, followed 7.1 mo
TCGA-E9-A1RG	62	III	LumB	0.62	no progression, followed 21.3 mo
TCGA-AO-A0JL	59	III	missing	0.62	no progression, followed 55.3 mo
TCGA-GM-A3XN	44	III	missing	0.61	no progression, followed 66.4 mo
TCGA-AR-A1AR	50	III	Basal	0.60	progressed at 8.8 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group B** · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

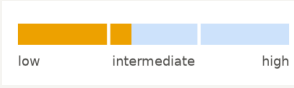
Sample: **TCGA-B6-A0X4-01** · Aneuploidy score: **0** · TMB (nonsyn., /Mb): **0.6** · MSIsensor score: **0** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

42nd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

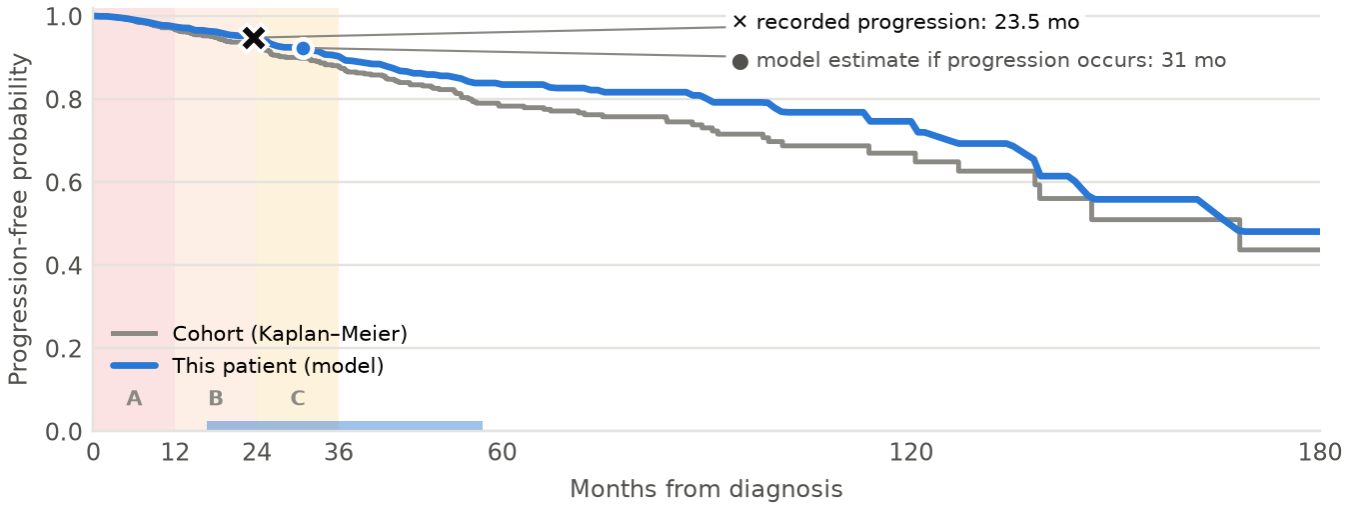
≈ 31 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

23.5 months

■ **Group B** · Progression in year 2
✗ **recorded window differs** · |error| 7 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **42nd risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **KMT2C** to **Activation of anterior HOX genes in hindbrain development during early embryogenesis**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.28). Progression was recorded at 23.5 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

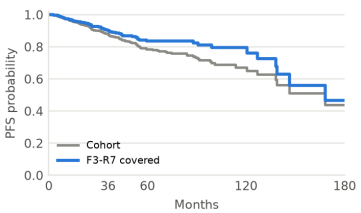


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0X4** mutated → **KMT2C** in pathway → **Activation of anterior HOX genes in hindbrain development during early embryogenesis**	96	↑ **faster progression** HR 1.28, z +1.3, n = 261 carriers of “Activation of anterior HOX genes in hindbrain development during early embryogenesis”	**0.75**
2 ★	**TCGA-B6-A0X4** mutated → **KMT2C** interacts with → **ASH2L** in pathway → **Deactivation of the beta-catenin transactivating complex**	85	↑ **faster progression** HR 1.35, z +1.6, n = 282 carriers of “Deactivation of the beta-catenin transactivating complex”	**0.72**
3 ★	**TCGA-B6-A0X4** mutated → **MAP2K4** linked to → **breast carcinoma**	93	↑ **faster progression** HR 1.50, z +1.1, n = 784 carriers of “breast carcinoma”	**0.71**
4 ★	**TCGA-B6-A0X4** mutated → **GATA3** in pathway → **Estrogen-dependent gene expression**	93	↑ **faster progression** HR 1.23, z +1.1, n = 442 carriers of “Estrogen-dependent gene expression”	**0.71**
5 ★	**TCGA-B6-A0X4** mutated → **ATAD2** in pathway → **TFAP2 (AP-2) family regulates transcription of growth factors and their receptors**	96	↑ **faster progression** HR 1.20, z +0.9, n = 249 carriers of “TFAP2 (AP-2) family regulates transcription of growth factors and their receptors”	**0.69**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

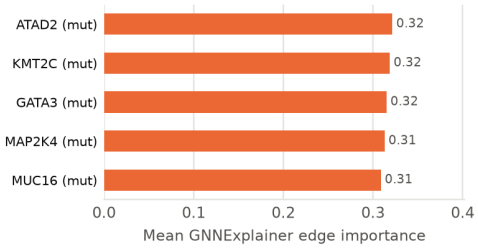
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R8 ↓ lower risk	AJCC stage = II AND Age < 77.5 AND TMB (nonsynonymous) ≥ 0.08	433 pts 40 events	**0.50** p 3.2e-04	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.E640*	4	0.32
GATA3 · driver	Mutation	p.S404Qfs*109	13	0.32
MAP2K4 · driver	Mutation	p.R134W	12	0.31
ATAD2	Mutation	p.I827M	1	0.32
MUC16	Mutation	p.G13230R	8	0.31



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MUC16 → Oregovomab (DB04964). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A18N	88	II	LumA	0.79	no progression, followed 37.7 mo
TCGA-C8-A273	29	II	LumA	0.78	no progression, followed 16.9 mo
TCGA-BH-A1EX	67	II	LumA	0.78	progressed at 49.6 mo (group D)
TCGA-BH-A0E1	52	II	LumA	0.78	no progression, followed 15.7 mo
TCGA-D8-A27K	47	II	LumA	0.78	no progression, followed 48.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**59**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Normal**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**—**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-EW-A1P7-01** · Aneuploidy score: **1** · TMB (nonsyn., /Mb): **0.67** · MSIsensor score: **1.05** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

59th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 19%

Model-estimated progression time

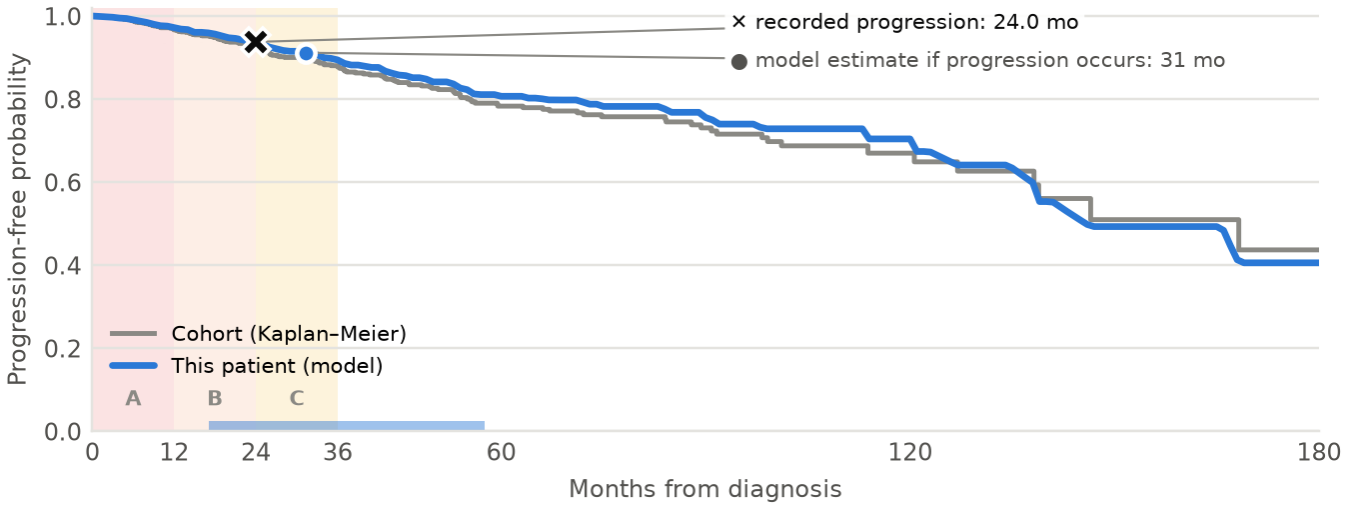
≈ 31 months

Window 24–36 months · IQR 17–58 mo
if progression occurs

Recorded progression

24.0 months

Group B · Progression in year 2
✗ recorded window differs · |error| 7 mo



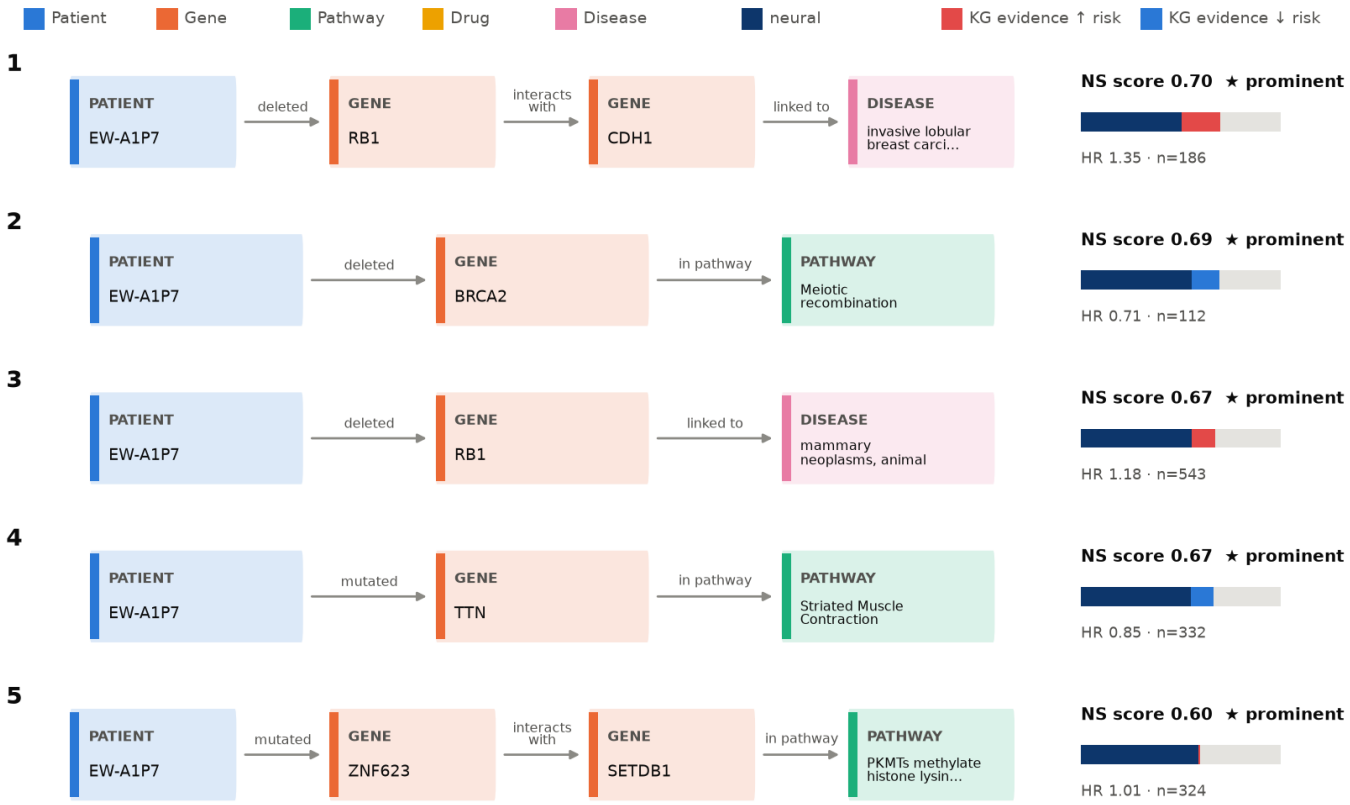
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 59-year-old patient has a Normal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **59th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **RB1** to **invasive lobular breast carcinoma**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.35). Progression was recorded at 24.0 months; the model's estimate given progression was 31 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

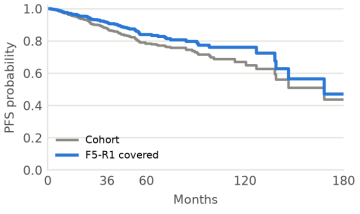


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-EW-A1P7** deleted → **RB1** interacts with → **CDH1** linked to → **invasive lobular breast carcinoma**	84	↑ **faster progression** HR 1.35, z +1.5, n = 186 carriers of “invasive lobular breast carcinoma”	**0.70**
2 ★	**TCGA-EW-A1P7** deleted → **BRCA2** in pathway → **Meiotic recombination**	93	↓ **slower progression** HR 0.71, z -1.0, n = 112 carriers of “Meiotic recombination”	**0.69**
3 ★	**TCGA-EW-A1P7** deleted → **RB1** linked to → **mammary neoplasms, animal**	93	↑ **faster progression** HR 1.18, z +0.9, n = 543 carriers of “mammary neoplasms, animal”	**0.67**
4 ★	**TCGA-EW-A1P7** mutated → **TTN** in pathway → **Striated Muscle Contraction**	92	↓ **slower progression** HR 0.85, z -0.9, n = 332 carriers of “Striated Muscle Contraction”	**0.67**
5 ★	**TCGA-EW-A1P7** mutated → **ZNF623** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	98	↑ **faster progression** HR 1.01, z +0.1, n = 324 carriers of “PKMTs methylate histone lysines”	**0.60**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

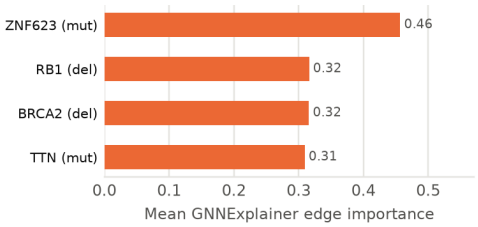
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
RB1 · driver	Homozygous deletion	GISTIC ±2	25	0.32
BRCA2 · driver	Homozygous deletion	GISTIC ±2	14	0.32
ZNF623	Mutation	p.C377Y	0	0.46
TTN	Mutation	p.K2800M	2	0.31



4 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A6R9	61	II	Normal	0.77	no progression, followed 5.3 mo
TCGA-A8-A08C	65	II	missing	0.73	no progression, followed 29.0 mo
TCGA-E2-A14W	78	II	missing	0.73	no progression, followed 32.0 mo
TCGA-HN-A2NL	56	II	Basal	0.71	no progression, followed 2.6 mo
TCGA-D8-A1XW	53	II	Normal	0.71	no progression, followed 43.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.